Weight one forms of level $\Gamma_1(p)$ and local-global compatibility

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Abstract

We give a modular construction of optimal integral structures for weight one modular forms of level $\Gamma_1(p)$ over $\mathbb{Z}_p$, using the coherent cohomology of certain line bundles on the modular curve $\mathcal{X}_0(p)$. Following the ideas of Boxer–Pilloni [BP22, BP26], we develop higher Hida theory for these integral structures and prove their optimality. As an application, we establish local-global compatibility at p for the Galois representations associated to our weight one forms, extending the results of [CG18] at spherical level. In particular, for a fixed nontrivial tame Nebentypus, these representations split as a direct sum of characters at p. We also prove a modularity result that can be viewed as a reinterpretation of Gross's companion forms theorem [Gro90]. Our local–global compatibility result is an ingredient in forthcoming work of the author proving an $R^{\min} = \mathbb{T}$ theorem.

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1 Introduction

1.1 Overview

Let $X \to \mathrm{Spec}(\mathbb{Z}_p)$ be the compactified modular curve of level $\Gamma_1(N)$, where $N \geq 3$ is prime to p, and let ω be the Hodge line bundle associated to the universal generalized elliptic curve $\mathcal{E}/X$. In the 1970s, Katz [Kat73] defined mod p modular forms of weight $k \in \mathbb{Z}_{\geq 0}$ as elements of $\mathrm{H}^0(X_{\mathbb{F}_p}, \omega^{\otimes k})$. From this coherent cohomological perspective, weight one forms are particularly interesting for two reasons:

1. Their cohomology is generally nonzero in both degrees: $\mathrm{H}^0(X_{\mathbb{F}_p}, \omega)$ and $\mathrm{H}^1(X_{\mathbb{F}_p}, \omega)$ may both be nontrivial.

2. Mod p weight one forms need not lift to characteristic zero, in contrast to modular forms of weight $k \geq 2$.

Multiplication by the Hasse invariant, a mod p modular form of weight $p-1$ with q-expansion 1, embeds the space of weight one forms into the space of weight p forms and preserves Hecke eigenvalues away from p. This allows us to attach Galois representations to mod p weight one eigenforms: we multiply by the Hasse invariant, lift the resulting eigensystem to characteristic zero, and apply the classical theory.

This construction naturally raises the question of which additional properties distinguish the Galois representations arising in weight one from those arising in weight p. Serre [Ser87] predicted that the representations attached to mod p weight one forms are unramified at p. In the p-distinguished case, this follows implicitly from the work of Gross [Gro90]. The result was subsequently

established in [Wie14] using the doubling method under a multiplicity-one hypothesis, and by Calegari–Geraghty [CG18] using ordinary deformation rings. Their result also applies to representations valued in Hecke algebras. Roughly speaking, it takes the following form:

Theorem 1.1 ([CG18, Theorem 3.11], informal)**.** *Under the hypotheses of the cited theorem, let $f \in \mathrm{H}^0(X_{\mathbb{F}_p}, \omega)$ be a cuspidal eigenform, and let $\overline{\rho}_f$ be its associated Galois representation. Then $\overline{\rho}_f$ is unramified at p. Moreover, the corresponding Hecke-algebra-valued deformation*

$$\rho : G_{\mathbb{Q}} \longrightarrow \mathrm{GL}_2(\mathbb{T}_{\mathfrak{m}})$$

is also unramified at p.

Two features of this result are particularly relevant to our work:

1. Since weight one cohomology occurs in more than one degree, the Hecke algebra acting on $\mathrm{H}^0(X, \omega)$ alone does not capture all the relevant information. One approach is to study Hecke operators as endomorphisms of $\mathrm{R}\Gamma(X, \omega)$ in the derived category [KT17].

2. Unramifiedness at p is a local–global compatibility statement. It yields a map $R \to \mathbb{T}_{\mathfrak{m}}$ from a Galois deformation ring with the corresponding local condition. Establishing an $R = \mathbb{T}_{\mathfrak{m}}$ theorem in this setting requires going beyond the numerical conditions of the original Taylor–Wiles method, as in the refined patching method of Calegari–Geraghty [CG18].

Compared with the spherical case, integral structures for weight one modular forms with level at p are less well understood. In his work on companion forms, Gross [Gro90] studied weight two forms as differentials on the Deligne–Rapoport integral model of the modular curve of level $\Gamma_1(p)$ [DR73]. Integral lattices for critical modular forms were also studied in [CWE25]. For weight one forms of level $\Gamma_1(p)$, however, the choice of an appropriate integral structure remains a central question. One can define a lattice by requiring q-expansion coefficients to be p-adic integers, but this description does not directly provide a geometric construction that extends to broader settings. We therefore seek an integral structure arising from the cohomology of a line bundle on an integral model. Once the generic-fiber line bundle is fixed, any two extensions, when they exist, differ by a Cartier divisor supported on the special fiber.

An arbitrary extension need not have the desired properties. Twisting by vertical divisors can introduce additional torsion into integral cohomology and hence additional classes after reduction modulo p, with no reason for these classes to satisfy the expected local–global compatibility. Hida theory provides a way to study the ordinary parts of these cohomology groups through the action of U_p and its variation in p-adic families. Boxer–Pilloni [BP22, BP26] developed a geometric approach, known as higher Hida theory, that treats all cohomological degrees. Their approach uses cohomological correspondences and the geometry of integral models of Shimura varieties to analyze the action of

U_p. The Cousin complex then relates ordinary cohomology to cohomology supported on suitable strata, which can be interpreted in terms of higher p-adic automorphic forms. This provides a framework for studying ordinary integral structures in all degrees.

The goal of this paper is to construct an optimal integral structure for weight one modular forms of level $\Gamma_1(p)$ and establish the corresponding local–global compatibility at p. We first develop higher Hida theory for the integral weight one line bundle we construct, and then use it to prove optimality. As applications, we establish local–global compatibility both for the Galois representations attached to mod p eigenforms and for the corresponding Hecke-algebra-valued representations. We also establish classicality theorems and vanishing results for ordinary cohomology in weights $k \neq 1$.

1.2 Main results

Let $\mathcal{X}_0(p)$ be the compactified integral modular curve of level $\Gamma_1(N) \cap \Gamma_0(p)$ over $\mathbb{Z}_p$. Write

$$\mathcal{X} = \mathcal{X}_0(p), \qquad X = \mathcal{X} \otimes_{\mathbb{Z}_p} \mathbb{F}_p, \qquad \mathfrak{X} = \widehat{\mathcal{X}},$$

where the completion is along the special fibre. Let $\mathcal{X}_0(p^2)$ be the Katz-Mazur integral model for modular curve of level $\Gamma_1(N) \cap \Gamma_0(p^2)$ [KM85], which gives a correspondence

$$\mathcal{X} \xleftarrow{p_1} \mathcal{X}_0(p^2) \xrightarrow{p_2} \mathcal{X}.$$

Let $[\cdot] : \mathbb{F}_p^\times \to \mathbb{Z}_p^\times$ be the Teichmüller character, and write $\chi = [\cdot]^a$, with $0 \leq a \leq p-2$. Here χ is a Nebentypus at p; the Nebentypus away from p is allowed to vary. I index the eigensheaves so that their generic-fibre forms have p-part of the classical Nebentypus equal to χ^{-1}. An informal statement of the main theorem follows.

Theorem 1.2. *There is a unique integral line bundle ω_χ on $\mathcal{X}$ satisfying the following properties:*

1. *The space $\mathrm{H}^0(\mathcal{X}, \omega_\chi)$ is naturally identified with the space of classical weight one modular forms of level $\Gamma_1(Np)$ and Nebentypus χ^{-1}.*

2. *A cohomological correspondence induces an action of U_p on $\mathrm{R}\Gamma(\mathcal{X}, \omega_\chi)$; see Theorem 1.3.*

3. *The complex $\mathrm{R}\Gamma(\mathcal{X}, \omega_\chi)$ is computed by a Cousin complex whose terms are spaces of (higher) ordinary p-adic modular forms; see Theorem 1.4.*

4. *The associated Galois representations satisfy local–global compatibility; see Theorem 1.7 and Theorem 1.9.*

1.2.1 Integral structures and a Cousin complex

Theorem 1.3 (5.6)**.** *There is an integral cohomological correspondence*

$$U_p : p_2^*\omega_\chi \longrightarrow p_1^!\omega_\chi$$

whose reduction to the special fibre is nonzero, and induces an action U_p on $\mathrm{R}\Gamma(\mathcal{X}, \omega_\chi)$.

The following Cousin complex makes the difference of the different integral structures explicit.

Write $X^{F,\mathrm{ord}}$ and $X^{V,\mathrm{ord}}$ for the ordinary Frobenius and Verschiebung loci, and use the same superscripts for their formal lifts. Denote the normalized Frobenius operator on the Verschiebung term by F, and the ordinary projectors by $e(U_p)$ and $e(F)$; H^1_c denotes the compactly supported coherent cohomology used in higher Hida theory.

Theorem 1.4 (5.26)**.** *U_p acts invertibly on* $\mathrm{R}\Gamma(X, \omega_\chi)$ *and* $\mathrm{R}\Gamma(\mathcal{X}, \omega_\chi)$*. If $\chi \neq [\cdot]^{p-2}$, there are U_p-equivariant quasi-isomorphisms*

$$\mathrm{R}\Gamma(X, \omega_\chi) \simeq \left[e(U_p)\mathrm{H}^0(X^{F,\mathrm{ord}}, \omega_\chi) \xrightarrow{\delta} e(F)\mathrm{H}^1_c(X^{V,\mathrm{ord}}, \omega_\chi)\right],$$
$$\mathrm{R}\Gamma(\mathcal{X}, \omega_\chi) \simeq \left[e(U_p)\mathrm{H}^0(\mathfrak{X}^{F,\mathrm{ord}}, \omega_\chi) \xrightarrow{\delta} e(F)\mathrm{H}^1_c(\mathfrak{X}^{V,\mathrm{ord}}, \omega_\chi)\right].$$

The two-term complexes are placed in degrees zero and one. The exceptional character $\chi = [\cdot]^{p-2}$ requires a modified complex.

Remark 1.5. Classically, local representation theory implies that weight one forms of level $\Gamma_1(p)$ are ordinary. This remains true for our optimal integral structure, allowing us to study the entire space of integral weight one forms using higher Hida theory.

We can now explain why the integral structure we pick is optimal. With the normalization of U_p fixed, twisting by $\omega_0 \otimes \omega_1^{-1}$ destroys the required nonzero integral correspondence on the special fibre. Twisting in the opposite direction by $\omega_0^{-1} \otimes \omega_1$ replaces δ by $p\delta$, under the natural identifications of the terms. After reduction, the connection map therefore vanishes, potentially increasing the dimensions of H^0 and H^1. These two competing effects explain the optimality and uniqueness of ω_χ within this family of twists.

In the case of trivial Nebentypus, i.e. $a = 0$, we obtain the following doubling result.

Theorem 1.6 (3.19)**.**

1. *There is a quasi-isomorphism of complexes of $\mathbb{Z}_p$-modules*
$$\mathrm{R}\Gamma(X, \omega)^{\oplus 2} \simeq \mathrm{R}\Gamma(X_0(p), \omega_1),$$
which is T_l-equivariant for every prime $l \neq p$.

2. *Under this identification, the action of U_p on* $\mathrm{R}\Gamma(X_0(p), \omega_1)$ *is given by*
$$\begin{pmatrix} 0 & -\langle p \rangle \\ 1 & T_p^t \end{pmatrix},$$
where T_p^t denotes the transpose of T_p.

Consequently, the Galois representations arising from weight one forms of level $\Gamma_0(p)$ with our integral structure are precisely those arising at spherical level.

1.2.2 Local-global compatibility and modularity

The operators T_ℓ for $\ell \nmid Np$ and $\langle d\rangle$ for $d \in (\mathbb{Z}/N\mathbb{Z})^\times$ act on these complexes. Write $G_p = G_{\mathbb{Q}_p}$, let $\bar{\varepsilon}_p : G_{\mathbb{Q}} \to \mathbb{F}_p^\times$ be the mod-p cyclotomic character, and put $\widetilde{\omega}_p = [\bar{\varepsilon}_p]$. For a unit u in the coefficient ring, let $\lambda(u)$ denote the unramified character of G_p sending arithmetic Frobenius to u.

Theorem 1.7 (Local-global compatibility6.4)**.** *Let f be a cuspidal eigenform in $\mathrm{H}^0(X_{\overline{\mathbb{F}}_p}, \omega_\chi)$ with non-Eisenstein eigensystem. Suppose*

$$T_\ell f = a_\ell f, \qquad U_p f = a_p f, \qquad \langle d\rangle f = \epsilon(d) f,$$

where $\epsilon : (\mathbb{Z}/N\mathbb{Z})^\times \to \overline{\mathbb{F}}_p^\times$. Ordinarity is automatic for this integral structure, so $a_p \neq 0$. There is a continuous irreducible representation $\overline{\rho}_f : G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_p)$, unramified outside Np, such that, for $\ell \nmid Np$,

$$\operatorname{tr} \overline{\rho}_f(\mathrm{Frob}_\ell) = a_\ell, \qquad \det \overline{\rho}_f(\mathrm{Frob}_\ell) = \epsilon(\ell)\ell^{-a}.$$

Moreover,

$$\overline{\rho}_f|_{G_p} \simeq \begin{cases} \bar{\varepsilon}_p^{-a}\lambda\big(\epsilon(p)/a_p\big) \oplus \lambda(a_p), & \text{if } \chi \neq 1, \\ \text{an unramified representation}, & \text{if } \chi = 1. \end{cases}$$

By constructing a pair of companion forms on two components of $X_{\mathbb{F}_p}$ and verifying the gluing condition using the multiplicity-one result for Serre's mod-p quaternionic modular forms inspired by [Bao26], I also obtain a converse under the corresponding local hypothesis.

Theorem 1.8 (6.7)**.** *Let $\overline{\rho} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_p)$ be a continuous, irreducible, odd representation. Suppose that $\overline{\rho}|_{G_p}$ is a direct sum of two characters, exactly one of which is unramified. Then $\overline{\rho} \simeq \overline{\rho}_f$ for a cuspidal eigenform f as in Theorem 1.7, for some level N prime to p and some non-trivial character χ.*

The local splitting result also holds over the Hecke algebra. Let χ be a non-trivial character, and

$$\mathbb{T} \subset \mathrm{End}_{D(\mathbb{Z}_p)}\big(\mathrm{R}\Gamma(\mathcal{X}, \omega_\chi)\big)$$

be the $\mathbb{Z}_p$-algebra generated by T_ℓ for $\ell \nmid Np$, U_p, and the tame diamond operators. Let $\mathfrak{m}$ be a non-Eisenstein maximal ideal, and write $\mathbb{T}_\mathfrak{m}$ for the localization. Automatic ordinarity ensures that U_p is a unit in $\mathbb{T}_\mathfrak{m}$.

Theorem 1.9 (6.16)**.** *There is a continuous representation $\rho_\mathfrak{m} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{T}_\mathfrak{m})$, unramified outside Np, satisfying*

$$\operatorname{tr} \rho_\mathfrak{m}(\mathrm{Frob}_\ell) = T_\ell, \qquad \det \rho_\mathfrak{m}(\mathrm{Frob}_\ell) = \langle \ell\rangle [\bar{\ell}]^{-a} \quad (\ell \nmid Np).$$

Its restriction to G_p is

$$\rho_{\mathfrak{m}}|_{G_p} \simeq \widetilde{\omega}_p^{-a}\lambda\big(\langle p\rangle/U_p\big) \oplus \lambda(U_p).$$

Remark 1.10. By applying the Calegari–Geraghty patching method [CG18], we establish a minimal $R = \mathbb{T}$ theorem for Galois representations that are split at p. The details will appear in forthcoming work.

Acknowledgments

I would like to thank especially my advisor George Boxer for suggesting this problem, generously sharing his knowledge, and constant help during this project. I am also grateful to my other advisor Fred Diamond for his interest in my work and helpful discussions. I also thank Chengyang Bao, Martin Ortiz, Calle Sönne for discussions related to this work.

2 The Geometry of the modular curve $X_0(p^n)$

2.1 The Katz-Mazur integral model

We first recall the the construction of the Katz-Mazur integral modular curves of level $\Gamma_0(p^n)$ following[KM85], which are defined as moduli space parametrizing elliptic curves with certain Drinfeld level structure.

Definition 2.1. Let E/S be an elliptic curve over S, $G \subset E$ be a closed subgroup scheme which is finte locally free of rank N. G is called cyclic if there exists some fppf morphism $T \to S$ and a point $P \in E(T)$, such that P is a generator of G_T/T. That is to say:

$$G = \sum_{i=1}^{N} [iP]$$

as effective Cartier divisors on E.

Definition 2.2. A $\Gamma_0(N)$-structure on E/S is an isogeny $f : E \to E'$ of degree N such that $\ker f$ is cyclic.

Let $\mathcal{E}ll$ be the moduli stack of elliptic curves, i.e. for an arbitrary scheme S, $\mathcal{E}ll(S)$ is a groupoid whose objects are elliptic curves $p : E \to S$ and morphisms are isomorphisms of elliptic curves over S. Let $\mathcal{Y}_0(N)$ be the moduli stack of elliptic curves with $\Gamma_0(N)$-structure, i.e. $\mathcal{Y}_(N)(S)$ is a groupoid whose objects are cyclic N-isogenies $(E \xrightarrow{\pi} E')$, where E and E' are elliptic curves over S, and a morphism between $(E_1 \xrightarrow{\pi_1} E_1')$ and $(E_2 \xrightarrow{\pi_2} E_2')$ is a pair of isomorphisms of elliptic curves $a : E_1 \to E_2$ and $a' : E_1' \to E_2'$ such that $\pi_2 \circ a = a' \circ \pi_1$. We obtain the following functor:

$$s : \mathcal{Y}_0(N) \to \mathcal{E}ll$$
$$(E/S \xrightarrow{\pi} E'/S) \mapsto E/S$$

Theorem 2.3. *The moduli stacks $\mathcal{Y}_0(N)$ and $\mathcal{E}ll$ are 2-dimensional Deligne-Mumford stacks. The functor $s : \mathcal{Y}_0(N) \to \mathcal{E}ll$ is relatively representable, finite flat of degree $\psi(N) = N \cdot \prod_{l|N}(1 + l^{-1})$, and regular. Moreover, s is etale over $\mathbb{Z}[1/N]$.*

Proof. See [KM85] Theorem 5.1.1. □

To work with schemes instead of stacks, we also introduce the moduli problem with $\Gamma_1(N)$-structure.

Definition 2.4. A $\Gamma_1(N)$-structure on E/S is a morphism

$$\phi : \mathbb{Z}/N\mathbb{Z} \to E[N](S)$$

such that the effective Cartier divisor $\sum_{a=1}^{N}[\phi(a)]$ is an S-subgroup scheme of E/S.

Similarly as above, we denote $\mathcal{Y}_1(N)$ as the moduli stack of elliptic curves of $\Gamma_1(N)$-structure. The corresponding functor $\mathcal{Y}_1(N) \to \mathcal{E}ll$ is also relatively representable, finite flat and regular of dimension 2, by [KM85]Theorem 5.1.1.

Lemma 2.5. *If $N \geq 4$, then the $\Gamma_1(N)$-structure is rigid, i.e. if ϕ is an automorphism of an elliptic curve E/S over a connected scheme S, and ϕ induces identity on a cyclic subgroup $G \subset E$ of rank N, then $\phi = id$.*

Proof. G is contained in the kernel of $\phi - 1$, therefore $N|\deg(1-\phi) = 1 - \mathrm{tr}(\phi) + \deg(\phi) = 2 - \mathrm{tr}(\phi)$. On the other hand $(\mathrm{tr}(\phi)^2 \leq 4\deg(\phi)$. If $N \geq 5$, the only option is $\mathrm{tr}(\phi) = 2$ and $\phi = id$. If $N = 4$, and $\mathrm{tr}(\phi) = -2$, then $\deg(\phi) + 1 = 2 + \mathrm{tr}(\phi) = 0$, hence $\phi = -id$, and $G \subset \ker(1 - \phi) = E[2]$, which is a contradiction since G is cyclic. □

Corollary 2.6.
1. *If $N \geq 4$, the moduli stack $\mathcal{Y}_1(N)$ is represented by an affine regular curve $Y_1(N)$ over $\mathbb{Z}$, and $Y_1(N) \otimes \mathbb{Z}[1/N]$ is smooth over $\mathbb{Z}[1/N]$.*
2. *If $N \geq 4$, the moduli stack of elliptic curves with $\Gamma_1(N) \cap \Gamma_0(M)$-structure is represented by an affine regular scheme over $\mathbb{Z}$.*

Proof. Representablity follows from rigidity of $\Gamma_1(N)$-structure and relative representability of $s : \mathcal{Y}_0(N) \to \mathcal{E}ll$. Affiness follows from the finiteness of s. See [KM85] 4.7 for more details. □

By an abuse of notation, we will now write $Y_0(p^n)$ for the regular curve over $\mathbb{Z}_p$ representing the moduli problem of elliptic curves with $\Gamma_1(N) \cap \Gamma_0(p^n)$-structure for some N satisfying $N \geq 4$ and $p \nmid N$.

The j-invariant induces a finite flat morphism of $\mathbb{Z}_p$-schemes

$$j : Y_0(p^n) \to \mathbb{A}^1_{\mathbb{Z}_p}.$$

The compactified integral modular curve of level $\Gamma_1(N) \cap \Gamma_0(p^n)$, denoted by $X_0(p^n)$, is the normalization of $\mathbb{P}^1_{\mathbb{Z}_p}$ in $Y_0(p^n)$ via j. The curve $X_0(p^n)$ is a projective regular scheme over $\mathbb{Z}_p$. A moduli construction of the stack $\mathcal{X}_0(N)$ in terms of level structures on generalized elliptic curves can be found in [vCc17].

2.2 Components of the special fiber

We first develop the structure theory of cyclic subgroup schemes and cyclic isogenies of elliptic curves.

Theorem 2.7. *Let E/S be an elliptic curve, $G \subset E/S$ a cyclic subgroup scheme of rank N. Then*

1. *For any $d \mid N$, there exists a unique "standard" cyclic subgroup scheme $G_d \subset G$ of rank d, such that fppf-locally, G_d is generated by $(N/d)P$, where P is any generator of G.*
2. *The quotient $G' = G/G_d$ is a cyclic subgroup scheme of $E' = E/G_d$ with rank N/d. And if P is a generator of G, the its image P' is a generator of E'.*

Proof. This is proved in [KM85] 6.7. □

Therefore, given a cyclic N-isogeny $\phi : E \to E''$, we obtain a standard factorization of ϕ:

$$E \xrightarrow{\phi_d} E' \xrightarrow{\phi'_d} E''$$

where ϕ_d is a cyclic d-isogeny, and ϕ'_d is a cyclic N/d-isogeny. We call such a pair of composable isogenies (ϕ_d, ϕ'_d) cyclic in standard order. Generally, a sequence of composable cyclic isogenies

$$E_0 \xrightarrow[\deg=d_1]{\phi_1} E_1 \xrightarrow[\deg=d_2]{\phi_2} \cdots \xrightarrow[\deg=d_n]{\phi_n} E_n$$

is called cyclic in standard order if $\phi = \phi_n \circ \cdots \circ \phi_1$ is cyclic, and $\ker(\phi_i \circ \cdots \circ \phi_1) = G_{d1 \cdots d_i}$ for any $1 \le i \le n$, where $G = \ker \phi$.

Example 2.8. Let E/S be an elliptic curve over an $\mathbb{F}_p$-scheme S, then the morphism $[p]$ is cyclic and its standard factorization is

$$E \xrightarrow{F_{E/S}} E^{(p)} \xrightarrow{V_{E/S}} E,$$

If E/S is moreover ordinary, then the factorization

$$E^{(p)} \xrightarrow{V_{E/S}} E \xrightarrow{F_{E/S}} E^{(p)}$$

is non-standard. Where $F_{E/S}$ is the relative Frobenius and $V_{E/S}$ is the relative Verschiebung.

Theorem 2.9. *Let $(\phi_1, \ldots, \phi_n)$ be a sequence of composable isogenies*

$$E_0 \xrightarrow[\deg=d_1]{\phi_1} E_1 \xrightarrow[\deg=d_2]{\phi_2} \cdots \xrightarrow[\deg=d_n]{\phi_n} E_n$$

such that $d_1, \ldots, d_n$ have the same prime factors. Then $(\phi_1, \ldots, \phi_n)$ is cyclic in standard order if and only if (ϕ_i, ϕ_{i+1}) is cyclic in standard order for any $1 \le i \le n-1$.

Proof. This is proved in [KM85]Theorem 6.7.15. □

Proposition 2.10. *Consider a pair of composable cyclic isogenies (ϕ_1, ϕ_2) of elliptic curves over S:*

$$E \xrightarrow{\phi_1} E' \xrightarrow{\phi_2} E''.$$

Then

1. *(ϕ_1, ϕ_2) is cyclic in standard order if and only if the dual isogenies $(\phi_2^\vee, \phi_1^\vee)$ is cyclic in standard order.*
2. *If ϕ_2 is etale, (ϕ_1, ϕ_2) is cyclic in standard order if and only if $\phi = \phi_2 \circ \phi_1$ is a cyclic isogeny.*

Proof. See [KM85] Proposition 6.7.8 and Theorem 6.7.9. □

We will further investigate the characteristic p theory for cyclic isogenies. Let E/S be an ordinary elliptic curve over an $\mathbb{F}_p$-scheme S. Then for any integer $n \geq 1$, $\ker(V^n)$ is a finite etale S-group, etale locally isomorphic to $\mathbb{Z}/p^n\mathbb{Z}$, and $\ker(F^n)$ is its Cartier dual, which is etale locally isomorphic to μ_{p^n}.

Theorem 2.11. *Let $\phi : E \to E'$ be a p^n-isogeny of ordinary elliptic curves over an $\mathbb{F}_p$-scheme S. Then*

1. *There exists a unique pair (a, b) of non-negative integers, such that we have the factorization of ϕ:*

$$E \xrightarrow{F^a} E^{(p^a)} \underset{\sim}{\xrightarrow{\epsilon}} E'^{(p^b)} \xrightarrow{V^b} E'$$

   where ϵ is an isomorphism.

2. *If ϕ is also a cyclic isogeny, then the above factorization is standard with respect to the divisor p^a of p^n. Moreover, the refined factorization*

$$E \xrightarrow{F} \cdots \xrightarrow{F} E^{(p^a)} \underset{\sim}{\xrightarrow{\epsilon}} E'^{(p^b)} \xrightarrow{V} \cdots \xrightarrow{V} E'$$

   is also standard.

Proof. 1follows from [KM85]Theorem 13.3.3.
For 2, since $\ker(V^b)$ is etale, by [KM85]Propostion 6.7.8, (F^a, V^b) is in standard order. For the second statement, since $ker(V^n)$ and $\ker(F^n)$ are both cyclic for any $n \geq 1$, by 2.9, we only need to show that the composition of the middle isogenies $E^{(p^{a-1})} \xrightarrow{F} E^{(p^a)} \underset{\sim}{\xrightarrow{\epsilon}} E^{(p^b)} \xrightarrow{V} E^{(p^{b-1})}$ is cyclic. This again follows from [KM85]Propostion 6.7.8. □

Write $Y_0(p^n)_1$ and $X_0(p^n)_1$ for the special fibers of $Y_0(p^n)$ and $X_0(p^n)$, and $Y_0(p^n)_1^{\mathrm{ord}} \subset Y_0(p^n)_1$ and $X_0(p^n)_1^{\mathrm{ord}} \subset X_0(p^n)_1$ for the corresponding ordinary locus. For any (a, b) pair of non-negative integers with $a + b = n$, we

denote $Y_0(a,b)^{\mathrm{ord}}$ for the closed subscheme of $Y_0(p^n)_1^{\mathrm{ord}}$, whose points correspond to cyclic isogenies $\phi : E \to E'$ satisfying the conditions in 2.11. Then $Y_0(p^n)_1^{\mathrm{ord}} = \coprod_{a+b=n} Y_0(a,b)^{\mathrm{ord}}$ as a disjoint union. We can further extend this construction to the supersingular locus and cusps by imposing an (a,b)-cyclic condition[KM85] 13.4.3, which is the same as the conditions in 2.11 when restricted to the ordinary locus. And one obtained closed subschemes $X_0(a,b)$ of $X_0(p^n)_1$.

Theorem 2.12. *$X_0(p^n)_1 = \cup_{(a,b)} X_0(a,b)$, and $X_0(a,b)$ are the irreducible components of $X_0(p^n)_1$, with crossings at the supersingular points.*

Proof. See [KM85] Theorem 13.4.6 for the proof of the case $Y_0(p^n)$. The main tool is to use a general theorem in algebraic geometry on "crossings"[KM85] Theorem 13.1.3. For treatments of cusps see [GZ86] III.1. □

2.3 Deformation rings at ordinary points

If $x \in X_0(p^n)(\overline{\mathbb{F}}_p))$ is a geometric point of the special fiber, we use the Serre-Tate parameter to describe the deformation ring $\widehat{\mathcal{O}}_{X_0(p^n),x}$.

Let $k = \overline{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$, and $\mathcal{C}_k$ be the category of local Artinian rings with residue field k, whose morphisms are local ring homomorphisms compatible with the reduction to k. Then any object of $\mathcal{C}_k$ is endowed with an algebra structure over W, where $W = W(k)$.

Let E_0/k be an ordinary elliptic curve, and R be an object in $\mathcal{C}_k$. We define the deformation functor $\mathrm{Ell}_{E_0} : \mathcal{C}_k \to \mathrm{Sets}$ to be:

$$R \mapsto \{(E,\iota) : E/R \text{ an elliptic curve}, \iota : E \otimes_R k \to E_0 \text{ an isomorphism}\}/\sim .$$

Theorem 2.13.

There is an isomorphism of functors on the category $\mathcal{C}_k$:

$$\mathrm{Ell}_{E_0}(-) \cong Hom_{\mathbb{Z}_p}(T_pE(k) \otimes_{\mathbb{Z}_p} T_pE(k), \hat{\mathbb{G}}_m(-)).$$

Proof. See [Del81]Theorem 2.1. □

Let Q be a generator of the Tate module $T_pE_0(k)$, and notice that $\hat{\mathbb{G}}_m(R) = 1 + m_R$, where m_R is the maximal ideal of R. The isomorphism sends a deformation E/R of E_0 to a bilinear form:

$$q(E/R;\cdot,\cdot) : T_pE_0(k) \times T_pE_0(k) \to \hat{\mathbb{G}}_m(R),$$

which is equivalent to determine $q(E/R,Q,Q) \in 1 + m_R$. Therefore, $\mathrm{Ell}_{E_0} = \mathrm{Spf}(W[\![X]\!])$, where $X = q(E_{univ}/\mathrm{Ell}_{E_0}; Q, Q) - 1$.

Theorem 2.14. *Let E_0 and E_0' be ordinary elliptic curves over k, let $\pi_0 : E_0 \to E_0'$ be an isogeny, and $\pi_0^t : E_0' \to E_0$ be its dual. Let E and E' be liftings of E_0 and E_0' over R respectively. Then π_0 lifts to an isogeny $\pi : E \to E'$ if and only if:*

$$q(E/R;\alpha,\pi_0^t(\beta)) = q(E';\pi_0(\alpha),\beta)$$

for every $\alpha \in T_pE_0(k)$ and $\beta \in T_pE_0'(k)$.

Proof. See [Kat73]Theorem 2.1. □

If $E \to E'$ is a lift of the n-th Frobenius $F^n : E_0 \to E_0^{(p^n)}$ over R, let P be a generator of the Tate module $T_pE_0(k)$, and since $\ker(F^n)$ is connected, $Q = F^n(P)$ is a generator of $T_pE_0^{(p^n)}(k)$. By the above theorem,

$$q(E'; Q, Q) = q(E'; F^n(P), Q) = q(E; P, V^n(Q)) = q(E; P, P)^{p^n}.$$

Lemma 2.15. *Let W be a discrete valuation ring with residue characteristic p and uniformizer π. Let S be an integral noetherian scheme, quasi-separated and flat over W. Let G be a finite locally free group scheme over S of order p^n, which is a closed subgroup scheme of an elliptic curve E/S. If the isogeny $\pi_G : E \to E/G$ doesn't factor through the multiplication by p morphism of E, then G/S is a cyclic group scheme.*

Proof. We give a sketch of the proof, for more details see [Zhu25]4.1. The condition that π_G factors through $[p]_E$ is a closed condition on S. Therefore, over the generic point ξ of S, G_ξ doesn't factor through $[p]_{E_\xi}$. The finite group scheme G_ξ is etale since $\kappa(\xi)$, the residue field of ξ, is of characteristic 0. Hence $G_\xi \simeq \prod_{i=1}^k \mathbb{Z}/p^{a_i}\mathbb{Z}$, where $\sum_{i=1}^k a_i = n$. G_ξ doesn't contain $\ker([p]_{E_\xi})$ implies $k = 1$ and $G_\xi \simeq \mathbb{Z}/p^n\mathbb{Z}$. By [KM85]Theorem 6.4.1 there exists a closed subscheme $S_0 \subset S$ such that for any $T \to S$, G_T/T is cyclic if and only if $T \to S$ factors through S_0. Then $\xi \in S_0$, and $S_0 = S$. □

Theorem 2.16. *Let $x \in X_0(p^n)(k)$ be an ordinary point on the special fiber, then $x \in X_0(a, b)$ for some unique pair of non-negative integers (a, b) satisfying $a + b = n$ by 2.12.*

1. *When $a, b \geq 1$, denote by $\widehat{\mathcal{O}}_{X_0(p^n),x}$ the complete local ring of $X_0(p^n)$ at x, then there is an $W(k)$-linear isomorphism:*

$$\widehat{\mathcal{O}}_{X_0(p^n),x} \cong W[\![X, Y]\!] / \left(\frac{(1+X)^{p^a} - (1+Y)^{p^b}}{(1+X)^{p^{a-1}} - (1+Y)^{p^{b-1}}} \right).$$

2. *When $a = 0$ or $b = 0$, there is an $W(k)$-linear isomorphism:*

$$\widehat{\mathcal{O}}_{X_0(p^n),x} \cong W[\![X]\!].$$

Proof. We only prove the case when $a, b \geq 1$, the case when $a = 0$ or $b = 0$ is by a similar argument. x is given by a cyclic isogeny $\phi_0 : E_0 \to E_0'$, whose standard factorization is of the form

$$E_0 \xrightarrow{F} \cdots \xrightarrow{F} E_0^{(p^a)} \xrightarrow[\sim]{\epsilon} E_0'^{(p^b)} \xrightarrow{V} \cdots \xrightarrow{V} E_0'.$$

Let P, Q be generators of $T_pE_0(k)$ and $T_pE_0'(k)$ respectively, such that $\epsilon \circ F^a(P) = V^b(Q)$. Let $\mathcal{C}$ be the category of complete local noetherian W-algebras whose residue field is $\overline{\mathbb{F}}_p$. Then the functor $\mathcal{F} : \mathcal{C} \to \text{Sets}$ which assigns to R

the set of isomorphism classes of cyclic p^n-isogenies $\phi : E \to E'$ that lift ϕ_0, is representable by $\widehat{\mathcal{O}}_{X_0(p^n),x}$. By 2.9, $\mathcal{F}$ is isomorphic to the moduli problem of lifting the standard factorization of ϕ_0 to

$$E \xrightarrow{\phi_1} E_1 \xrightarrow{\phi_2} \cdots \xrightarrow{\phi_{n-1}} E_{n-1} \xrightarrow{\phi_n} E'$$

over R such that (ϕ_i, ϕ_{i+1}) is cyclic in standard order for any $1 \leq i \leq n-1$. Let

$$E_{univ} \xrightarrow{\phi_1} E_1 \xrightarrow{\phi_2} \cdots \xrightarrow{\phi_{n-1}} E_{n-1} \xrightarrow{\phi_n} E'_{univ}$$

be universal lift of the standard factorization of ϕ_0 over $\widehat{\mathcal{O}}_{X_0(p^n),x}$. For $1 \leq i \leq a-1$ and $a+1 \leq i \leq n$, (ϕ_i, ϕ_{i+1}) is a lift of either (F, F) or (V, V) over $\widehat{\mathcal{O}}_{X_0(p^n),x}$, which doesn't factor through the isogeny of multiplication by p when E_0 is ordinary, therefore by 2.15 $\phi_{i+1} \circ \phi_i$ is cyclic. Since for ordinary elliptic curves $\ker V$ is etale and it deform uniquely over $\widehat{\mathcal{O}}_{X_0(p^n),x}$, by 2.10 (ϕ_i, ϕ_{i+1}) is cyclic in standard order. For (ϕ_a, ϕ_{a+1}), the condition that $\phi_{a+1} \circ \phi_a$ is cyclic is equivalent to E_{a-1} is not isomorphic to E_{a+1} by 2.15. And since ϕ_{a+1} is etale, by2.10 (ϕ_a, ϕ_{a+1}) is cyclic in standard order if and only if $\phi_{a+1} \circ \phi_a$ is cyclic. For an object $R \in \mathcal{C}$, we have $R = \text{colim} R/m_R^n$, and $R/m_R \in C_k$ therefore 2.132.14 also works for $\mathcal{C}$. We have

$$q(E_{univ}; P, P)^{p^a} = q(E_a; F^a(P), F^a(P)) = q(E'_{univ}; Q, Q)^{p^b},$$

and the condition E_{a-1} is not isomorphic to E_{a+1} implies

$$q(E_{univ}; P, P)^{p^{a-1}} = q(E_{a-1}; F^{a-1}(p), F^{a-1}(P))) \neq q(E_{a+1}; V^{b-1}(Q), V^{b-1}(Q)) = q(E'_{univ}; Q, Q)^{p^{b-1}}.$$

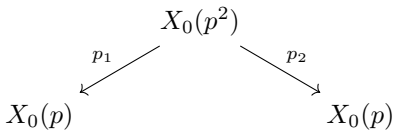
Let $X = q(E_{univ}; P, P) - 1$ and $Y = q(E'_{univ}; Q, Q) - 1$, then we have $\widehat{\mathcal{O}}_{X_0(p^n),x} \cong W[\![X, Y]\!] / \left(\frac{(1+X)^{p^a} - (1+Y)^{p^b}}{(1+X)^{p^{a-1}} - (1+Y)^{p^{b-1}}} \right)$. □

3 Modular forms on $X_0(p)$

$X_0(p)$ is the simplest non spherical level case where we can work on. We'll develop higher Hida theory and then focus on weight one modular forms on $X_0(p)$.

3.1 Cohomological Correspondences

Let $X_0(p)$ and $X_0(p^2)$ be the compactified modular curve over $\text{Spec}\mathbb{Z}_p$ of level $\Gamma_1(N) \cap \Gamma_0(p)$ and $\Gamma_1(N) \cap \Gamma_0(p^2)$ respectively. $X_0(p^2)$ parametrizes degree p^2 cyclic isogenies $E_0 \xrightarrow{\phi} E_2$, and we write its standard factorization as $E_0 \xrightarrow{\phi_1} E_1 \xrightarrow{\phi_2} E_2$. We have the following correspondence:



where p_1 and p_2 are given by:

$$p_1 : (E_0 \xrightarrow{\phi_1} E_1 \xrightarrow{\phi_2} E_2, P) \mapsto (E_1 \xrightarrow{\phi_2} E_2, \phi_1(P)),$$
$$p_2 : (E_0 \xrightarrow{\phi_1} E_1 \xrightarrow{\phi_2} E_2, P) \mapsto (E_0 \xrightarrow{\phi_1} E_1, P).$$

For both $i = 1, 2$, we have p_i is proper, quasi finite, hence a finite morphism. On the other hand, the modular curves are integral regular(which implies Cohen-Macaulay) schemes of same dimensional, by the "miracle flatness" criterion we obtain that p_i is flat. Since flat morphism between regular schemes is local complete intersection morphism, and therefore Gorenstein, we have $p_i^!\mathcal{O}_{X_0(p)}$ is an invertible sheaf on $X_0(p^2)$.

Let $E_0^{univ} \to E_1^{univ}$ be the universal isogeny between the universal semi-abelian schemes over $X_0(p)$, and for $i = 0, 1$, denote e_i to be the unit section of $E_i^{univ} \to X_0(p)$. Let $\omega_i = e_i^*\Omega^1_{E_i^{univ}/X}$, and $\omega(k, a) = \omega_1^{\otimes k} \otimes \omega_0^{\otimes(k-a)}$ for $(k, a) \in \mathbb{Z}^2$. $\omega(k, a)$ are the candidates for weight-k automorphic line bundles over the integral modular curve $X_0(p)$. Denote also $X \to \mathrm{Spec}\mathbb{Z}_p$ to be the modular curve of level $\Gamma_1(N)$, X_1 be its special fiber, and ω be the modular line bundle.

Lemma 3.1. *There is a rational cohomological correspondence*

$$U_p^{\mathrm{naive}} : p_2^*\omega(k, a) \dashrightarrow p_1^!\omega(k, a)$$

over $X_0(p^2)$.

Proof. By definition of the correspondence, the standard factorization of the universal isogeny over $X_0(p^2)$ is given by $p_2^*E_0 \xrightarrow{\phi_1} p_2^*E_1 \cong p_1^*E_0 \xrightarrow{\phi_2} p_1^*E_1$. Let $\phi_1^\vee : p_1^*E_0 \to p_2^*E_0$ and $\phi_2^\vee : p_1^*E_1 \to p_2^*E_1$ be the dual isogenies. The pull back on differentials induces $p_2^*\omega_0 \to p_1^*\omega_0$ and $p_2^*\omega_1 \to p_1^*\omega_1$. Therefore we get a rational map $p_2^*\omega(k, a) \dashrightarrow p_1^*\omega(k, a)$ for any (k, a). On the other hand we have the trace map $\mathrm{tr}_{p_1} : \mathcal{O}_{X_0(p^2)} \to p_1^!\mathcal{O}_{X_0(p)}$, and U_p^{naive} is given by the tensor product. □

Theorem 3.2. *Let* $U_p = p^{-\inf\{1,k,a\}}U_p^{\mathrm{naive}}$*, then* U_p *is a cohomological correspondence* $p_2^*\omega(k, a) \to p_1^!\omega(k, a)$.

Proof. U_p is a rational map between invertible sheaves over the regular scheme $X_0(p^2)$, so we only need to check it's defined outside codimension 2. Over $\mathbb{Q}_p$ the maps between differentials are all isomorphisms since p is invertible, and U_p is defined over the generic fiber of $X_0(p^2)$. It's enough to show that U_p is defined locally at every ordinary points on the special fiber. Let $x \in X_0(p^2)(\overline{\mathbb{F}}_p)$ be an ordinary geometric point.

1. If $x \in X_0(2, 0)$, then over $\widehat{\mathcal{O}}_{X_0(p^2),x}$, $\phi_1^\vee$ and $\phi_2^\vee$ are both etale, and induces isomorphism $(p_2^*\omega(k, a))_x \xrightarrow{\sim} (p_1^*\omega(k, a))_x$. By adjunction, tr_{p_1} :

$(\mathcal{O}_{X_0(p^2)})_x \to (p_1^!\mathcal{O}_{X_0(p)})_x$ is induced by $\mathrm{tr} : (p_{1*}\mathcal{O}_{X_0(p^2)})_{p_1(x)} \to (\mathcal{O}_{X_0(p)})_{p_1(x)}$. By 2.16 it is given by the trace of:

$$\begin{aligned} W[\![X]\!] &\to W[\![Y]\!] \\ X &\mapsto (1+Y)^p - 1 \end{aligned}$$

Therefore tr_{p_1} factors through $(\mathcal{O}_{X_0(p^2)})_x \xrightarrow{\sim} p(p_1^!\mathcal{O}_{X_0(p)})_x \subset (p_1^!\mathcal{O}_{X_0(p)})_x$.

2. If $x \in X_0(1,1)$, then $\phi_1^\vee$ is etale and $\phi_2^\vee$ is multiplicative. The differential map induces $p_2^*\omega_0 \xrightarrow{\sim} p_1^*\omega_0$ and $p_2^*\omega_1 \xrightarrow{\sim} p \cdot p_1^*\omega_1 \subset p_1^*\omega_1$ over $\widehat{\mathcal{O}}_{X_0(p^2),x}$. Hence the we have a factorization $(p_2^*\omega(k,a))_x \to p^a \cdot (p_1^*\omega(k,a))_x$. By 2.16 the trace map is induced by:

$$\begin{aligned} W[\![t]\!] &\to W[\![X,Y]\!] / \left(\frac{(1+X)^p - (1+Y)^p}{(1+X)-(1+Y)} \right) \\ t &\mapsto Y \end{aligned}$$

since the degree of the map is $p-1$, the trace map doesn't factor through multiplication by p.

3. If $x \in X_0(0,2)$, then $\phi_1^\vee$ and $\phi_2^\vee$ are both multiplicative and induces $(p_2^*\omega(k,a))_x \xrightarrow{\sim} p^k \cdot (p_1^*\omega(k,a))_x \subset (p_1^*\omega(k,a))_x$. On the other hand, $\mathrm{tr}_{p_1} : (\mathcal{O}_{X_0(p^2)})_x \to (p_1^!\mathcal{O}_{X_0(p)})_x$ is an isomorphism.

Therefore we have $p^{-\inf\{1,k,a\}}U_p^{\mathrm{naive}} : p_2^*\omega(k,a) \to p_1^!\omega(k,a)$ is a ccohomological correspondence. □

From now on, we only consider the case when $a \geq \inf\{1,k\}$, then $U_p = p^{-\inf\{1,k\}}U_p^{\mathrm{naive}}$. In particular when $k \geq 1$, over the generic fiber, U_p induces

$$U_p(f)(E,H,\omega) = p^{k-1} \cdot \sum_{C \subset E[p], C \neq H} f(E/C, E[p]/C, \omega_C)$$

on Katz modular forms by 2.15 and the construction of the correspondence. This coincides with the definition in [Kat73].

3.2 Higher Hida theory

In this section we develop the higher Hida theory for weights $k \geq 2$ and $k \leq 0$ forms. Let $X_0(p)_1^F := X_0(1,0)$ and $X_0(p)_1^V := X_0(0,1)$ be the Frobenius and Verschiebung component on the special fiber $X_0(p)_1$ respectively.

Proposition 3.3. *We have the following factorizations of cohomological correspondences over $X_0(p^2)$:*

1. *If $k \geq 2$,*

$$\begin{array}{ccc} p_2^*\omega(k,a) & \xrightarrow{U_p} & p_1^!\omega(k,a) \\ \downarrow & \nearrow_{U_p} & \downarrow \\ p_2^*\omega(k,a+1) & \xrightarrow{U_p} & p_1^!\omega(k,a+1) \end{array}$$

Where the left vertical arrow is given by $p_2^*\omega_0 \to p_2^*\omega_1$ induced from $\phi_1^\vee$, and the right vertical arrow is given by $p_1^*\omega_0 \to p_1^*\omega_1$ induced by $\phi_2^\vee$.

2. *If $k \leq 0$,*

$$\begin{array}{ccc} p_2^*\omega(k,a) & \xrightarrow{U_p} & p_1^!\omega(k,a) \\ \uparrow & \searrow & \uparrow \\ p_2^*\omega(k,a+1) & \xrightarrow{U_p} & p_1^!\omega(k,a+1) \end{array}$$

Where the left vertical arrow is given by $p_2^*\omega_1 \to p_2^*\omega_0$ induced from ϕ_1, and the right vertical arrow is given by $p_1^*\omega_1 \to p_1^*\omega_0$ induced from ϕ_2.

Proof. We only provide the proof when $k \geq 2$, the case $k \leq 0$ is similar. When $k \geq 2$, $U_p = p^{-1}U_p^{\text{naive}}$. Let $\pi : p_2^*\omega(k,a+1) \dashrightarrow p_1^*\omega(k,a)$ be the rational map induced by differentials of isogenies and $\mathrm{tr}_{p_1} : \mathcal{O}_{X_0(p^2)} \to p_1^!\mathcal{O}_{X_0(p)}$ be the trace map. It suffices to show that the rational correspondence

$$p^{-1} \cdot (\pi \otimes \mathrm{tr}_{p_1}) : p_2^*\omega(k,a+1) \dashrightarrow p_1^!\omega(k,a)$$

is well defined. As in 3.2, we only need to prove the factorization exists at every ordinary points on the special fiber. π is the tensor product of $p_2^*\omega_1^a \to p_1^*\omega_1^a$ induced by $\phi_2^\vee$, $p_2^*\omega_0^{k-a-1} \to p_1^*\omega_0^{k-a-1}$ induced by $\phi_1^\vee$, and the natural isomorphism $p_2^*\omega_1 \xrightarrow{\sim} p_1^*\omega_0$.

1. If $x \in X_0(2,0)$, then over $\widehat{\mathcal{O}}_{X_0(p^2),x}$, $\phi_1^\vee$ and $\phi_2^\vee$ are both etale and induces isomorphism $(p_2^*\omega(k,a+1))_x \xrightarrow{\sim} (p_1^*\omega(k,a))_x$. And tr_{p_1} factors through $(\mathcal{O}_{X_0(p^2)})_x \xrightarrow{\sim} p(p_1^!\mathcal{O}_{X_0(p)})_x \subset (p_1^!\mathcal{O}_{X_0(p)})_x$ by 3.2.
2. If $x \in X_0(1,1)$, then $\phi_1^\vee$ is etale and $\phi_2^\vee$ is multiplicative. Hence the we have a factorization $(p_2^*\omega(k,a+1))_x \to p^a \cdot (p_1^*\omega(k,a))_x$. Since we assume that $a \geq \inf\{1,k\} = 1$, $p^{-1} \cdot (\pi \otimes \mathrm{tr}_{p_1})$ is well defined over $\widehat{\mathcal{O}}_{X_0(p^2),x}$.
3. If $x \in X_0(0,2)$, then $\phi_1^\vee$ and $\phi_2^\vee$ are both multiplicative and induces $(p_2^*\omega(k,a+1))_x \xrightarrow{\sim} p^{k-1} \cdot (p_1^*\omega(k,a))_x \subset (p_1^*\omega(k,a))_x$. Since $k \geq 2$, $p^{-1} \cdot (\pi \otimes \mathrm{tr}_{p_1})$ is well defined over $\widehat{\mathcal{O}}_{X_0(p^2),x}$.

□

Corollary 3.4. *We have the following factorization of diagrams:*

1. *If $k \geq 2$,*

$$\begin{array}{ccc} \mathrm{R}\Gamma(X_0(p),\omega(k,a)) & \xrightarrow{U_p} & \mathrm{R}\Gamma(X_0(p),\omega(k,a)) \\ \downarrow & \nearrow & \downarrow \\ \mathrm{R}\Gamma(X_0(p),\omega(k,a+1)) & \xrightarrow{U_P} & \mathrm{R}\Gamma(X_0(p),\omega(k,a+1)) \end{array}$$

Where the vertical arrows are given by $\omega_0 \to \omega_1$ induced from the dual of the universal isogeny $E_0 \to E_1$.

2. If $k \leq 0$,

$$\begin{array}{ccc} \mathrm{R}\Gamma(X_0(p),\omega(k,a)) & \xrightarrow{U_p} & \mathrm{R}\Gamma(X_0(p),\omega(k,a)) \\ \uparrow & \searrow & \uparrow \\ \mathrm{R}\Gamma(X_0(p),\omega(k,a+1)) & \xrightarrow{U_P} & \mathrm{R}\Gamma(X_0(p),\omega(k,a+1)) \end{array}$$

Where the vertical arrows are given by $\omega_1 \to \omega_0$ induced from the universal isogeny $E_0 \to E_1$.

We first formulate the mod p higher Hida theory. Notice over the special fiber, the vanishing locus of $\omega_0 \to \omega_1$ induced by the dual of the universal isogeny $E_0 \to E_1$ is the etale component $X_0(p)_1^V$. Hence

$$\mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}},\omega(k,a)) = \operatorname*{colim}_n \mathrm{R}\Gamma(X_0(p)_1,\omega(k,a)\otimes(\omega_1\otimes\omega_0^{-1})^{\otimes n}),$$

where the colimit is induced by multiplication by the section corresponding to $\omega_0 \to \omega_1$. On the other hand, the vanishing locus of $\omega_1 \to \omega_0$ induced by the universal isogeny $E_0 \to E_1$ is the multiplicative component $X_0(p)_1^F$. We have for cohomology with compact support

$$\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}},\omega(k,a)) = \lim_n \mathrm{R}\Gamma(X_1,\omega(k,a)(-nSS)) = \lim_n \mathrm{R}\Gamma(X_0(p)_1,\omega(k,a)\otimes(\omega_1\otimes\omega_0^{-1})^{\otimes n}),$$

where we identify $X_0(p)^{V,\mathrm{ord}}$ with X_1^{ord} on the spherical level, and the limit is induced by $\omega_1 \to \omega_0$. Since $\omega(k,a)\otimes(\omega_1\otimes\omega_0^{-1}) \simeq \omega(k,a+1)$, we deduce the following classicality theorem.

Proposition 3.5. *1. If $k \geq 2$, $e(U_p)\mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}},\omega^k) = e(U_p)\mathrm{R}\Gamma(X_0(p)_1,\omega(k,a))$.*

2. If $k \leq 0$, $e(U_p)\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}},\omega^k) = e(U_p)\mathrm{R}\Gamma(X_0(p)_1,\omega(k,a))$.

In particular, the space $e(U_p)\mathrm{R}\Gamma(X_0(p)_1,\omega(k,a))$ and $e(U_p)\mathrm{R}\Gamma(X_0(p)_1,\omega(k,a))$ are independent from the choice of $a \geq \inf\{1,k\}$.

Proof. Notice that we can identify $X_0(p)_1^{F,\mathrm{ord}}$ and $X_0(p)_1^{V,\mathrm{ord}}$ with the ordinary locus of spherical level modular curve X_1, and there exists and isomorphism $\omega(k,a)|_{X_1} \simeq \omega^k$ induced by the Hasse invariant. U_p acts on $\mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}},\omega(k,a))$ and $\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}},\omega(k,a))$ via the terms in the colimit and limit above. The theorem follows from 3.4. □

3.3 Comparison with spherical level

We now compare the action of U_p and T_p on spherical level modular curve. In general, we can construct the $*$ and !-pullbacks of cohomological correspondence as follows. All the items below should be understood in the derived category of quasi coherent sheaves. Assume that we have a commutative diagram

$$\begin{array}{ccccc} X_1 & \xleftarrow{q_2} & W_1 & \xrightarrow{q_1} & Y_1 \\ \downarrow{\scriptstyle f_X} & & \downarrow{\scriptstyle f_W} & & \downarrow{\scriptstyle f_Y} \\ X_2 & \xleftarrow{p_2} & W_2 & \xrightarrow{p_1} & Y_2 \end{array}$$

such that p_1, p_2, q_1, q_2 are proper.

1. If the right square is Cartesian and f_Y is flat. Then we can define the $*$-pullback of a bottom cohomological correspondence $T_{W_2} : p_2^* \mathcal{F} \to p_1^! \mathcal{G}$ as the decomposition:

$$f_W^* T_{W_2} : q_2^* f_X^* \mathcal{F} = f_W^* p_2^* \mathcal{F} \xrightarrow{T_{W_2}} f_W^* p_1^! \mathcal{G} \to q_1^! f_Y^* \mathcal{G},$$

where the last map is given by the flat base change. The induced maps on cohomology are compatible in the sense that we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{R}\Gamma(X_2, \mathcal{F}) & \xrightarrow{T_{W_2}} & \mathrm{R}\Gamma(Y_2, \mathcal{G}) \\ \downarrow & & \downarrow \\ \mathrm{R}\Gamma(X_1, f_X^* \mathcal{F}) & \xrightarrow{f_W^* T_{W_2}} & \mathrm{R}\Gamma(Y_1, f_Y^* \mathcal{G}) \end{array}$$

2. If the left square is Cartesian and p_2 is flat. Then we can define the $!$-pullback of a bottom cohomological correspondence $T_{W_2} : p_2^* \mathcal{F} \to p_1^! \mathcal{G}$ as the decomposition:

$$f_W^! T_{W_2} : q_2^* f_X^! \mathcal{F} \to f_W^! p_2^* \mathcal{F} \xrightarrow{T_{W_2}} f_W^! p_1^! \mathcal{G} = q_1^! f_Y^! \mathcal{G},$$

where the first map is given by the flat base change. Regarding compatibility, if we assume all the vertical maps are proper, there is a commutative diagram:

$$\begin{array}{ccc} \mathrm{R}\Gamma(X_2, \mathcal{F}) & \xrightarrow{T_{W_2}} & \mathrm{R}\Gamma(Y_2, \mathcal{G}) \\ \uparrow & & \uparrow \\ \mathrm{R}\Gamma(X_1, f_X^! \mathcal{F}) & \xrightarrow{f_W^! T_{W_2}} & \mathrm{R}\Gamma(Y_1, f_Y^! \mathcal{G}) \end{array}$$

where the vertical maps are induced by the adjunction morphisms $(f_X)_* f_X^! \to id$ and $(f_Y)_* f_Y^! \to id$.

We denote $\mathfrak{X}_0(p)$ as the formal completion of $X_0(p)$ along its special fiber, and $\mathfrak{X}_0(p)^{\mathrm{ord}}$ be its ordinary part. Then $\mathfrak{X}_0(p)^{\mathrm{ord}}$ is a disjoint union of two types of irreducible components:

$$\mathfrak{X}_0(p)^{\mathrm{ord}} = \mathfrak{X}_0(p)^{\mathrm{ord},F} \amalg \mathfrak{X}_0(p)^{\mathrm{ord},V}$$

such that on the first component the universal isogeny is multiplicative and on the second it is etale. As in [BP22]4.2, we can define $F = p^{-k} F^{\mathrm{naive}}$ and $U_p^{\mathrm{sph}} = p^{-1} U_p^{\mathrm{sph,naive}}$ as cohomological correspondences for ω^k supported respectively on $\mathfrak{X}_0(p)^{\mathrm{ord},F}$ and $\mathfrak{X}_0(p)^{\mathrm{ord},V}$, using the same strategy as in 3.4.

Lemma 3.6. *1. If $k \geq 2$, U_p action on $\mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}}, \omega(k,a))$ and U_p^{sph} action on $\mathrm{R}\Gamma(X_1^{\mathrm{ord}}, \omega^k)$ are compatible.*

2. *If $k \leq 0$, U_p action on $\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}}, \omega(k,a))$ and F action on $\mathrm{R}\Gamma_c(X_1^{\mathrm{ord}}, \omega^k)$ are compatible.*

Proof. 1. Consider the following diagram

$$\begin{array}{ccccc} X_0(p)_1^{F,\mathrm{ord}} & \xleftarrow{q_2} & X_0(2,0)^{\mathrm{ord}} & \xrightarrow{q_1} & X_0(p)_1^{F,\mathrm{ord}} \\ \downarrow & & \downarrow{\scriptstyle j} & & \downarrow \\ X_0(p) & \xleftarrow{p_2} & X_0(p^2)_1 & \xrightarrow{p_1} & X_0(p) \end{array}$$

where the vertical maps are the open immersions. Apply i^*-pullback, we obtain a commutative diagram:

$$\begin{array}{ccc} \mathrm{R}\Gamma(X_0(p)_1, \omega(k,a)) & \xrightarrow{U_p} & \mathrm{R}\Gamma(X_0(p)_1, \omega(k,a)) \\ \downarrow & & \downarrow \\ \mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}}, \omega^k) & \xrightarrow{i^*U_p} & \mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}}, \omega^k) \end{array}$$

Identify the bottom row with spherical level objects, we get

$$i^*U_p = p^{-1}i^*U_p^{\mathrm{naive}} = p^{-1}U_p^{\mathrm{sph,naive}} = U_p^{\mathrm{sph}}.$$

2. Consider the following diagram

$$\begin{array}{ccccc} X_0(p)_1^{V} & \xleftarrow{q_2} & X_0(0,2) & \xrightarrow{q_1} & X_0(p)_1^{V} \\ \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i'} & & \downarrow{\scriptstyle i} \\ X_0(p)_1 & \xleftarrow{p_2} & X_0(p^2)_1 & \xrightarrow{p_1} & X_0(p)_1 \end{array}$$

where the vertical maps are the open immersions. Apply $(i')^!$-pullback, we get a commutative diagram:

$$\begin{array}{ccc} \mathrm{R}\Gamma(X_0(p)_1, \omega(k,a)) & \xrightarrow{U_p} & \mathrm{R}\Gamma(X_0(p)_1, \omega(k,a)) \\ \uparrow & & \uparrow \\ \mathrm{R}\Gamma(X_0(p)_1^{V}, i^!\omega(k,a)) & \xrightarrow{(i')^!U_p} & \mathrm{R}\Gamma(X_0(p)_1^{V}, i^!\omega(k,a)) \end{array}$$

Similarly, $(i')^!U_p = p^{-k}(i')^!U_p^{\mathrm{naive}} = p^{-k}F^{\mathrm{naive}} = F$. Since sections of a coherent sheaf which is scheme theoretically supported in a closed subscheme is set theoretically supported in the closed subset corresponding to the underlying space of the closed subscheme, the vertical maps factors through:

$$\mathrm{R}\Gamma(X_0(p)_1^{V}, i^!\omega(k,a)) \to \mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}}, \omega^k) \to \mathrm{R}\Gamma_{X_0(p)_1^{V}}(X_0(p)_1, \omega(k,a)) \to \mathrm{R}\Gamma(X_0(p)_1, \omega(k,a))$$

Moreover we have the diagram:

$$\begin{array}{ccc} \mathrm{R}\Gamma(X_0(p)_1,\omega(k,a)) & \xrightarrow{U_p} & \mathrm{R}\Gamma(X_0(p)_1,\omega(k,a)) \\ \uparrow & & \uparrow \\ \mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}},\omega^k) & \xrightarrow{F} & \mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}},\omega^k) \end{array}$$

□

Theorem 3.7. *1. If $k \geq 2$ and $a \geq 1$, $e(U_p^{\mathrm{sph}})\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^k) = e(U_p)\mathrm{H}^0(X_0(p),\omega(k,a))$.*

2. If $k \leq 0$ and $a \geq k$, $e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^k) = e(U_p)\mathrm{H}^1(X_0(p),\omega(k,a))$.

Proof. There are natural locally finite actions of U_p^{sph} and F, see [BP22]Theorem 4.15 for details. Since $\mathfrak{X}^{\mathrm{ord}}$ is affine, both $e(U_p^{\mathrm{sph}})\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^k)$ and $e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^k)$ are complete flat $\mathbb{Z}_p$-modules. By classicality results for spherical level(see [BP22]Corollary 4.5), modulo p they are finite, hence they are finite free $\mathbb{Z}_p$-modules. Consider the short exact sequence:

$$0 \to \omega(k,a) \xrightarrow{p} \omega(k,a) \to \omega(k,a)/p\omega(k,a) \to 0$$

The corresponding long exact sequence implies that $\mathrm{H}^0(X_0(p),\omega(k,a))$ is p-torsion free, and if

$$e(U_p)\mathrm{H}^0(X_0(p),\omega(k,a)/p\omega(k,a)) \simeq e(U_p)\mathrm{H}^0(X_0(p)_1,\omega(k,a)) = 0$$

then $e(U_p)\mathrm{H}^1(X_0(p),\omega(k,a))$ is p-torsion free. By 3.5, when $k \leq 0$, $e(U_p)\mathrm{H}^0(X_0(p)_1,\omega(k,a))$ vanishes, therefore $e(U_p)\mathrm{H}^1(X_0(p),\omega(k,a))$ is finite free $\mathbb{Z}_p$-module. Similarly, we also get from 3.5 that if $k \geq 2$, $e(U_p)\mathrm{H}^0(X_0(p),\omega(k,a))/p \simeq e(U_p)\mathrm{H}^0(X_0(p)_1,\omega(k,a))$, and $\mathrm{H}^1(X_0(p),\omega(k,a))/p \simeq \mathrm{H}^1(X_0(p)_1,\omega(k,a))$ since H^2 vanishes. Now in both cases we can construct natural morphism between finite free $\mathbb{Z}_p$-modules which is isomorphic modulo p by 3.5. □

3.4 Weight one forms

For general weights, even though we do not expect to have a classicality theorem, we show that there is a so-called Cousin complex as a substitute. Moreover, we prove that there is a unique optimal integral structure for weight-one forms in the sense that the corresponding torsions in cohomologies are less enough to provide integral analogues of classical results for weight one forms over characteristic 0 field. In particular, we obtain that under this integral structure, all weight one forms are ordinary, and the Galois representations associated to mod p weight one forms of level $\Gamma_0(p)$ are unramified.

3.4.1 The Cousin Complex

Denote $\mathfrak{X}_0(p)^V$ as the formal completion along $X_0(p)_1^V \subset X_0(p)$. For an abelian sheaf $\mathcal{F}$ over $\mathfrak{X}_0(p)$, let $\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p), \mathcal{F})$ be the subgroup of $\mathrm{H}^0(\mathfrak{X}_0(p), \mathcal{F})$ of sections set theoretically supported in $\mathfrak{X}_0(p)^V$, and $\mathrm{R}\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p), \mathcal{F})$ to be the cohomology with support in $\mathfrak{X}_0(p)^V$ given by the derived functor of $\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p), -)$. Let $\mathfrak{X}_0(p)^V \xrightarrow{i} \mathfrak{X}_0(p) \xleftarrow{j} \mathfrak{X}_0(p)^{F,\mathrm{ord}}$. Corresponding to the localization sequence $i_* i^! \to \mathrm{id} \to j_* j^*$, we have an exact triangle:

$$\mathrm{R}\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p), \mathcal{F}) \to \mathrm{R}\Gamma(\mathfrak{X}_0(p), \mathcal{F}) \to \mathrm{R}\Gamma(\mathfrak{X}_0(p)^{F,\mathrm{ord}}, \mathcal{F}) \xrightarrow{+1}$$

There is another exact triangle with respect to the cohomology with compact support. Let $\mathcal{I}$ be any sheaf of ideal on $X_0(p)_1$ whose vanishing locus is $X_0(p)_1^F$, and $\mathcal{F}$ be a coherent sheaf on $X_0(p)_1$. The inverse system of short exact sequence

$$0 \to \mathcal{I}^n \mathcal{F} \to \mathcal{F} \to \mathcal{F}/\mathcal{I}^n \mathcal{F} \to 0$$

satisfies the Mittag-Leffler condition and induces:

$$\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}}, \mathcal{F}) \to \mathrm{R}\Gamma(X_0(p), \mathcal{F}) \to \mathrm{R}\Gamma(\widehat{X_0(p)}_{X_0(p)_1^F}, \hat{\mathcal{F}}) \xrightarrow{+1}$$

where $\widehat{X_0(p)}_{X_0(p)_1^F}$ is the formal completion of $X_0(p)_1$ along $X_0(p)_1^F$. Similar constructions apply for the modulo p^n modular curve $X_0(p)_n \to \mathrm{Spec}\mathbb{Z}/p^n\mathbb{Z}$. After taking limits we obtain an exact triangle for coherent sheaf $\mathcal{F}$ over $\mathfrak{X}_0(p)$:

$$\mathrm{R}\Gamma_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}}, \mathcal{F}) \to \mathrm{R}\Gamma(\mathfrak{X}_0(p), \mathcal{F}) \to \mathrm{R}\Gamma(\widehat{\mathfrak{X}_0(p)}_{\mathfrak{X}_0(p)^F}, \hat{\mathcal{F}}) \xrightarrow{+1}$$

The goal of this section is to identify the ordinary parts of these two exact triangle for weight one forms.

Proposition 3.8. *There is a locally finite action of U_p on $\mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a))$. And for $k = 1$, the natural map $\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}}, \omega(1,a)) \to \mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(1,a))$ induces an isomorphism between finite dimensional $\mathbb{F}_p$ vector spaces:*

$$e(U_p)\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}}, \omega(1,a)) \simeq e(U_p)\mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(1,a)).$$

Proof. Since $p_2^{-1}(X_0(p)_1^V) = X_0(0,2) \subset p_1^{-1}(X_0(p)_1^V)$, there is a U_p action on $\mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a))$ given by:

$$\mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a)) \xrightarrow{p_2^*} \mathrm{R}\Gamma_{p_2^{-1}(X_0(p)_1^V)}(X_0(p^2)_1, p_2^*\omega(k,a)) \to$$
$$\mathrm{R}\Gamma_{p_1^{-1}(X_0(p)_1^V)}(X_0(p^2)_1, p_2^*\omega(k,a)) \to \mathrm{R}\Gamma_{p_1^{-1}(X_0(p)_1^V)}(X_0(p^2)_1, p_1^!\omega(k,a)) \xrightarrow{\mathrm{tr}} \mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a))$$

where the second arrow is given by change of support, the third arrow is the cohomological correspondence given by U_p, and the last map is the trace map (see[BP22]Lemma5.1). Let $i : X_0(p)_1^V \to X_0(p)_1$ be the closed immersion, and

$j : X_0(p)_1^{F,\mathrm{ord}} \to X_0(p)_1$ be its open complement. Consider the exact triangle given by localization sequence:

$$i_* i^! \omega(k,a) \to \omega(k,a) \xrightarrow{\alpha} j_* j^* \omega(k,a) \xrightarrow{+1}$$

We obtain a filtration on $\mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a))$ corresponding to the canonical filtration on $[\omega(k,a) \xrightarrow{\alpha} j_* j^* \omega(k,a)]$ in the derived category:

$$\mathrm{H}^0_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a)) = \mathrm{H}^0(X_0(p)_1, \ker \alpha)$$
$$0 \to \mathrm{H}^0(X_0(p)_1, \mathrm{coker}\alpha) \to \mathrm{H}^1_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a)) \to \mathrm{H}^1(X_0(p)_1, \ker \alpha) \to 0$$

The U_p-action on $\mathrm{R}\Gamma(X_0(p)_1, \ker \alpha)$ is locally finite since $\ker \alpha = i_*(i^*\omega(k,a)(-SS))$, where $SS \subset X_0(p)_1^V$ is the supersingular divisor, is coherent. Denote $\mathcal{L}$ to be line bundle $\omega(0,1) = \omega_1 \otimes \omega_0^{-1}$, there is a section $\mathcal{O}_{X_0(p)_1} \to \mathcal{L}$ induced by the differential of the dual isogeny $E_0^{univ} \to E_1^{univ}$ with vanishing locus $X_0(p)^V$. Therefore we obtain an U_p compatible colimit:

$$j_*\omega(k,a)|_{X_0(p)_1^{F,\mathrm{ord}}} \simeq \mathrm{colim}\ \omega(k,a) \otimes \mathcal{L}^n,$$

and in particular, the U_p action on $\mathrm{R}\Gamma(X_0(p)_1, \mathrm{coker}\alpha)$ is locally finite. As a conclusion, U_p action on $\mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p)_1, \omega(k,a))$ is locally finite.

Let $s : \mathcal{L} \to \mathcal{O}_{X_0(p)_1}$ be the morphism induced by the universal isogeny $E_0^{univ} \to E_1^{univ}$, therefore the vanishing locus of s is $X_0(p)_1^F$. Take $\mathcal{I} = \mathrm{Im}(s)$ to be the corresponding ideal sheaf. For $n \geq 1$, s induces

$$\omega(k,a) \otimes \mathcal{L}^n \to \mathcal{I}^n \omega(k,a) \to \omega(k,a) \otimes \mathcal{L}^{n-1}$$

which means the two inverse systems of $\omega(k,a) \otimes \mathcal{L}^n$ and $\mathcal{I}^n\omega(k,a)$ are cofinals of each other, and we obtain an U_p-equivaiant isomorphism:

$$\mathrm{R}\Gamma_c(X_0(p)_1^{V,\mathrm{ord}}, \omega(k,a)) = \lim\ \mathrm{R}\Gamma(X_0(p)_1, \mathcal{I}^n\omega(k,a)) \simeq \lim\ \mathrm{R}\Gamma(X_0(p)_1, \omega(k,a) \otimes \mathcal{L}^n)$$

Now we restrict to the case when $k = 1$, $a \geq 1$ and suppose $p \geq 3$. Since $-ap + a + 1 \leq -1$ we have:

$$\mathrm{H}^0(X_0(p)_1, \ker \alpha) \simeq \mathrm{H}^0(X_0(p)_1^V, i^*\omega(1,a)(-SS)) = \mathrm{H}^0(X_1, \omega^{-ap+a+1}) = 0$$

On the other hand $X_0(p)_1^{V,\mathrm{ord}}$ is affine, thus $\mathrm{H}^0_c(X_0(p)_1^{V,\mathrm{ord}}, \omega(1,a)) = 0$. By comparison with spherical level, the U_p action on $\mathrm{H}^1(X_0(p)_1, \ker \alpha)$ is identified with the F action on $\mathrm{H}^1(X_1, \omega^{-ap+a+1})$. Therefore, by [BP22]Corollary 4.5,

$$e(U_p)\mathrm{H}^1(X_0(p)_1, \ker \alpha) = e(F)\mathrm{H}^1(X_1, \omega^{-ap+a+1}) = e(F)\mathrm{H}^1_c(X_1^{\mathrm{ord}}, \omega^{-ap+a+1}),$$

is a finite dimensional $\mathbb{F}_p$-module. Also, the U_p action on $\mathrm{H}^1_c(X_0(p)_1^{V,\mathrm{ord}}, \omega(1,a))$ is identified with F action on

$$\mathrm{H}^1_c(X_1^{\mathrm{ord}}, \omega^{a+p(1-a)}) \simeq \mathrm{H}^1_c(X_1^{\mathrm{ord}}, \omega^{-ap+a+1}).$$

As a conclusion, we obtain $e(U_p)\mathrm{H}^1_c(X_0(p)_1^{V,\mathrm{ord}},\omega(1,a))=e(U_p)\mathrm{H}^1(X_0(p)_1,\ker\alpha)$. It suffices to show that $e(U_p)\mathrm{H}^0(X_0(p)_1,\mathrm{coker}\alpha)=0$. We have

$$\mathrm{coker}\alpha\simeq\mathrm{colim}\ \frac{\omega(1,a)\otimes\mathcal{L}^n}{\omega(1,a)}$$

is zero over $X_0(p)_1^{V,\mathrm{ord}}$. Denote $i':X_0(p)_1^F\to X_0(p)_1$ to be the closed immersion, then

$$\mathrm{coker}\alpha\simeq i'_*(i')^*\mathrm{coker}\alpha\simeq\mathrm{colim}\ \frac{\omega^{pa+(1-a)}(nSS)}{\omega^{pa+(1-a)}}.$$

The U_p action on $\mathrm{H}^0(X_0(p)_1,\mathrm{coker}\alpha)$ is identified with the U_p^{sph} action on $\mathrm{H}^0(X_1,\mathrm{colim}\ \frac{\omega^{pa+(1-a)}(nSS)}{\omega^{pa+(1-a)}})$. Since $pa+(1-a)\geq 3$, by [BP22]Corollary 4.5,

$$\begin{aligned}&e(U_p^{\mathrm{sph}})\mathrm{H}^0(X_1,\omega^{pa+(1-a)})=e(U_p^{\mathrm{sph}})\mathrm{H}^0(X_1^{\mathrm{ord}},\omega^{pa+(1-a)})\\&\simeq e(U_p^{\mathrm{sph}})\mathrm{H}^0(X_1^{\mathrm{ord}},\omega^{pa+(1-a)}(nSS))=e(U_p^{\mathrm{sph}})\mathrm{H}^0(X_1,\omega^{pa+(1-a)}(nSS)).\end{aligned}$$

Therefore, $e(U_p)\mathrm{H}^0(X_0(p)_1,\mathrm{coker}\alpha)=0$. □

Corollary 3.9. *There is a locally finite action of U_p on $\mathrm{R}\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p),\omega(k,a))$. And for $k=1$, the natural map $\mathrm{R}\Gamma_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega(1,a))\to\mathrm{R}\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p),\omega(1,a))$ induces an isomorphism between finite free $\mathbb{Z}_p$-modules:*

$$e(U_p)\mathrm{R}\Gamma_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega(1,a))\simeq e(U_p)\mathrm{R}\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p),\omega(1,a)).$$

Proof. The action of U_p is defined similarly for $X_0(p)_n\to\mathrm{Spec}\mathbb{Z}/p^n\mathbb{Z}$, and taking limit we get a locally finite U_p action on $\mathrm{R}\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p),\omega(k,a))$. The natural morphism $e(U_p)\mathrm{R}\Gamma(\widehat{\mathfrak{X}_0(p)}_{\mathfrak{X}_0(p)^F},\widehat{\omega(1,a)})\to e(U_p)\mathrm{R}\Gamma(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a))$ induced by the two exact triangles is isomorphic modulo p. Since $\mathfrak{X}_0(p)^{F,\mathrm{ord}}$ is affine, the cohomologies of

$$e(U_p)\mathrm{R}\Gamma(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a))$$

are complete flat $\mathbb{Z}_p$-module, which is finite dimensional modulo p, hence finite free $\mathbb{Z}_p$-modules. The morphism

$$e(U_p)\mathrm{R}\Gamma(\widehat{\mathfrak{X}_0(p)}_{\mathfrak{X}_0(p)^F},\widehat{\omega(1,a)})\to e(U_p)\mathrm{R}\Gamma(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a))$$

must map torsion into zero, but they are isomorphic modulo p, therefore

$$e(U_p)\mathrm{R}\Gamma(\widehat{\mathfrak{X}_0(p)}_{\mathfrak{X}_0(p)^F},\widehat{\omega(1,a)})$$

is torsion free, and moreover a finite free $\mathbb{Z}_p$-module. We obtain that

$$e(U_p)\mathrm{R}\Gamma(\widehat{\mathfrak{X}_0(p)}_{\mathfrak{X}_0(p)^F},\widehat{\omega(1,a)})\simeq e(U_p)\mathrm{R}\Gamma(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a)).$$

Chasing back the exact triangle, we get the isomorphism:

$$e(U_p)\mathrm{R}\Gamma_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega(1,a))\simeq e(U_p)\mathrm{R}\Gamma_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p),\omega(1,a)).$$

Since the left hand side is a finite free $\mathbb{Z}_p$-module, both of them are finite free $\mathbb{Z}_p$-modules. □

Theorem 3.10. *1. For mod p forms, we have:*

$$e(U_p)\mathrm{R}\Gamma(X_0(p)_1,\omega(1,a)) = [e(U_p)\mathrm{H}^0(X_0(p)_1^{F,\mathrm{ord}},\omega(1,a)) \xrightarrow{\delta(a)} e(U_p)\mathrm{H}^1_c(X_0(p)_1^{V,\mathrm{ord}},\omega(1,a))].$$

2. Integrally, we have:

$$e(U_p)\mathrm{R}\Gamma(X_0(p),\omega(1,a)) = [e(U_p)\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a)) \xrightarrow{\delta(a)} e(U_p)\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega(1,a))].$$

Proof. We have

$$\mathrm{R}\Gamma(\mathfrak{X}_0(p),\omega(1,a)) = \lim \mathrm{R}\Gamma(X_0(p)_n,\omega(1,a)) \simeq \lim \mathrm{R}\Gamma(X_0(p),\omega(1,a))/p^n = \mathrm{R}\Gamma(X_0(p),\omega(1,a)),$$

where the isomorphism is given by theorem on formal functions. The first statement follows from the exact triangle and 3.9. □

Remark 3.11. 1. Since there exists non-liftable mod p weight one forms, and we will show that all weight one forms are ordinary, hence $\mathrm{H}^0(X_0(p),\omega(1,a))/p \subsetneq \mathrm{H}^0(X_0(p)_1,\omega(1,a))$. This does not contradict to our theorem 3.10.

2. We have the following diagram:

$$\begin{array}{ccc} e(U_p)\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a)) & \xrightarrow{\delta(a)} & e(U_p)\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega(1,a)) \\ \downarrow & & \downarrow \\ e(U_p)\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a+1)) & \xrightarrow{\delta(a+1)} & e(U_p)\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega(1,a+1)) \end{array}$$

where the vertical arrows are induced by $\omega_0 \to \omega_1$ corresponding to the dual of the universal isogeny $E_0^{univ} \to E_1^{univ}$. $\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega(1,a))$ and $\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega(1,a))$ are independent of a, hence by identifying source and target of the vertical arrows, the left vertical is an isomorphism and the right vertical is multiplication by p. Therefore $\delta(a+1) = p\cdot\delta(a)$.

3. For $a \geq 2$, the mod p Cousin map

$$\delta(a) : e(U_p)\mathrm{H}^0(X_0(p)_1^{F,\mathrm{ord}},\omega(1,a)) \xrightarrow{\delta(a)} e(U_p)\mathrm{H}^1_c(X_0(p)_1^{V,\mathrm{ord}},\omega(1,a))$$

is zero. Therefore, $\omega(1,1) = \omega_1$ is the optimal integral structure for weight one modular forms on $X_0(p)$.

Let $i : X_0(p)_1^V \to X_0(p)_1$ and $i' : X_0(p)_1^F \to X_0(p)_1$ be the closed immersions. Then there are two short exact sequences associated to ω_1:

1. $0 \to i_*(i^*\omega_1(-SS)) \to \omega_1 \to i'_*(i')^*\omega_1 \to 0$. Identify $X_0(p)_1^V$ and $X_0(p)_1^F$ with X_1, we obtain:

$$0 \to i_*\omega^{2-p} \to \omega_1 \to i'_*\omega^p \to 0.$$

This induces long exact sequence:

$$0 \to \mathrm{H}^0(X_0(p)_1,\omega_1) \to \mathrm{H}^0(X_1,\omega^p) \to \mathrm{H}^1(X_1,\omega^{2-p}) \to \mathrm{H}^1(X_0(p)_1,\omega_1) \to 0 \tag{1}$$

2. Similarly, we have $0 \to i'_*((i')^*\omega_1(-SS)) \to \omega_1 \to i_*i^*\omega_1 \to 0$, which is identified with:
$$0 \to i'_*\omega \to \omega_1 \to i_*\omega \to 0.$$
And induces long exact sequence:
$$0 \to \mathrm{H}^0(X_1,\omega) \to \mathrm{H}^0(X_0(p)_1,\omega_1) \to \mathrm{H}^0(X_1,\omega) \to \mathrm{H}^1(X_1,\omega) \to \mathrm{H}^1(X_0(p)_1,\omega_1) \to \mathrm{H}^1(X_1,\omega) \to 0. \tag{2}$$

Lemma 3.12. *The mod p Cousin map* $\delta(1) : e(U_p)\mathrm{H}^0(X_0(p)_1^{F,\mathrm{ord}},\omega_1) \to e(U_p)\mathrm{H}^1_c(X_0(p)^{V,\mathrm{ord}},\omega_1)$ *is identified with the map:*
$$e(U_p)\mathrm{H}^0(X_1,\omega^p) \to e(U_p)\mathrm{H}^1(X_1,\omega^{2-p})$$
induced by 1.

Proof. Consider the following diagram between exact triangles:
$$\begin{array}{ccccccc}
\mathrm{R}\Gamma_{X_0(p)_1^V}(X_0(p),\omega_1) & \longrightarrow & \mathrm{R}\Gamma(X_0(p)_1,\omega_1) & \longrightarrow & \mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}},\omega_1) & \xrightarrow{+1} & \\
\uparrow{\scriptstyle f} & & \uparrow{\scriptstyle id} & & \uparrow{\scriptstyle res} & & \\
\mathrm{R}\Gamma(X_1,\omega^{2-p}) & \longrightarrow & \mathrm{R}\Gamma(X_0(p)_1,\omega_1) & \longrightarrow & \mathrm{R}\Gamma(X_1,\omega^p) & \xrightarrow{+1} &
\end{array}$$
where f is the morphism induced by id and restriction. From [BP22]Corollary 4.5 and comparison with spherical level, $e(U_p)\mathrm{R}\Gamma(X_1,\omega^p) = e(U_p)\mathrm{R}\Gamma(X_1^{\mathrm{ord}},\omega^p) \simeq e(U_p)\mathrm{R}\Gamma(X_0(p)_1^{F,\mathrm{ord}},\omega_1)$. Therefore after taking Hida ordinary projector $e(U_p)$, those exact triangles are isomorphic. □

3.4.2 Integral and mod p weight one forms

Let $\pi_1,\pi_2 : X_0(p) \to X$ be the correspondence for X given by :
$$\pi_1 : (E_0 \xrightarrow{\phi} E_1, P) \mapsto (E_0, P),$$
$$\pi_2 : (E_0 \xrightarrow{\phi} E_1, P) \mapsto (E_1, \phi(P)).$$
Follows from [BP22], we can define the cohomological correspondence
$$T_p : \pi_2^*\omega^k \to \pi_1^!\omega^k$$
as $p^{-\inf\{1,k\}T_p^{\mathrm{naive}}}$ for some naturally defined T_p^{naive} as we have done in 3.1. Let $\langle p\rangle$ be the diamond operator, then $T_p^t = \langle p\rangle^{-1}T_p$.

Denote $\pi_i^* : \mathrm{R}\Gamma(X,\omega^k) \to \mathrm{R}\Gamma(X_0(p),\pi_i^*\omega^k)$ to be the pull back maps, and $\phi_{21} : \mathrm{R}\Gamma(X_0(p),\omega_0) \to \mathrm{R}\Gamma(X_0(p),\omega_1)$ to be the map induced by $\omega_0 \to \omega_1$ corresponding to the differential map of the dual of the universal isogeny $E_0^{univ} \to E_1^{univ}$. Similarly, we have $\phi_{12} : \mathrm{R}\Gamma(X_0(p),\omega_1) \to \mathrm{R}\Gamma(X_0(p),\omega_0)$. Also, denote $\mathrm{tr}\,\pi_i : \mathrm{R}\Gamma(X_0(p),\pi_i^*\omega) \to \mathrm{R}\Gamma(X,\omega)$ to be the trace maps. Recall that there is a so called Atkin-Lehner involution $w : X_0(p) \to X_0(p)$ given by:
$$w : (\phi : E_0 \to E_1, P) \mapsto (\phi^\vee : E_1 \to E_0, \phi(P)).$$

Denote $w^* : \mathrm{R}\Gamma(X_0(p), \omega(k,a)) \to \mathrm{R}\Gamma(X_0(p), \omega^*(\omega(k,a)))$ to be the pull back map. We have $\pi_1 \circ w = \pi_2$, $\pi_2 \circ w = \langle p \rangle \pi_1$, $w^2 = \langle p \rangle$. Let $W : \mathrm{R}\Gamma(X_0(p), \omega_1) \xrightarrow{w^*} \mathrm{R}\Gamma(X_0(p), \omega_0) \xrightarrow{\phi_{21}} \mathrm{R}\Gamma(X_0(p), \omega_1)$. Then

$$W^2 = \phi_{21} \circ w^* \circ \phi_{21} \circ w^* = \phi_{21} \circ \phi_{12} \circ w^* \circ w^* = p\langle p \rangle.$$

Theorem 3.13. *1. Integrally, there is an isomorphism between $\mathbb{Z}_p$-modules:* $\mathrm{R}\Gamma(X, \omega)^2 \simeq \mathrm{R}\Gamma(X_0(p), \omega_1)$.

2. There is also an isomorphism between $\mathbb{F}_p$-vector spaces: $\mathrm{R}\Gamma(X_1, \omega)^2 \simeq \mathrm{R}\Gamma(X_0(p)_1, \omega_1)$.

Proof. We give the proof for the integral statement and the mod p statement is easier and follows along similar lines. Define $\psi : \mathrm{R}\Gamma(X, \omega)^2 \to \mathrm{R}\Gamma(X_0(p), \omega_1)$ to be $(\pi_2^*, \phi_{21} \circ \pi_1^*)$. We claim that $\mathrm{tr}_{\pi_2} \circ \phi_{21} : \mathrm{R}\Gamma(X_0(p), \omega_0) \to \mathrm{R}\Gamma(X, \omega)$ can be divided by p. By construction of the Hecke operator, we can identify this map with

$$\mathrm{R}\Gamma(X_0(p), \pi_1^* \omega) \xrightarrow{T_p^{t,\mathrm{naive}}} \mathrm{R}\Gamma(X_0(p), \pi_2^! \omega) \to \mathrm{R}\Gamma(X, \omega).$$

Since $T_p^{t,\mathrm{naive}} = p \cdot T_p^t$, the map can be divided by p. Therefore we can construct a map $\psi^\vee : \mathrm{R}\Gamma(X_0(p), \omega_1) \to \mathrm{R}\Gamma(X, \omega)^2$ by $\begin{pmatrix} \mathrm{tr}\, \pi_2 \\ \frac{1}{p} \mathrm{tr}\, \pi_2 \circ W \end{pmatrix}$. The composition $\psi^\vee \circ \psi$ is given by:

$$\begin{pmatrix} \mathrm{tr}\, \pi_2 \circ \pi_2^* & \mathrm{tr}\, \pi_2 \circ \phi_{21} \circ \pi_1^* \\ \frac{1}{p} \mathrm{tr}\, \pi_2 \circ W \circ \pi_2^* & \frac{1}{p} \mathrm{tr}\, \pi_2 \circ W \circ \phi_{21} \circ \pi_1^* \end{pmatrix} = \begin{pmatrix} p+1 & p\langle p \rangle^{-1} T_p \\ T_p & p+1 \end{pmatrix}.$$

The computation of each entry is given by:

1. $\mathrm{tr}\, \pi_2 \circ \pi_2^* = \deg \pi_2 = p + 1$
2. $\mathrm{tr}\, \pi_2 \circ \phi_{21} \circ \pi_1^* = T_p^{t,\mathrm{naive}} = p \cdot T_p^t = p\langle p \rangle^{-1} T_p$
3. $\frac{1}{p} \mathrm{tr}\, \pi_2 \circ W \circ \pi_2^* = \frac{1}{p} \mathrm{tr}\, \pi_2 \circ \phi_{21} \circ w^* \circ \pi_2^* = \frac{1}{p} \mathrm{tr}\, \pi_2 \circ \phi_{21} \circ \langle p \rangle \pi_1^* = T_p$
4. $\frac{1}{p} \mathrm{tr}\, \pi_2 \circ W \circ \phi_{21} \circ \pi_1^* = \frac{1}{p} \mathrm{tr}\, \pi_2 \circ W \circ \phi_{21} \circ \langle p \rangle^{-1} w^* \circ \pi_2^* = (p\langle p \rangle)^{-1} \mathrm{tr}\, \pi_2 \circ W^2 \circ \pi_2^* = \mathrm{tr}\, \pi_2 \circ \pi_2^* = \deg \pi_2 = p + 1$

We have $\det(\psi^\vee \circ \psi) = (p+1)^2 - p\langle p \rangle^{-1} T_p^2 \equiv 1 (\mathrm{mod}\ p)$, which means that $\psi^\vee \circ \psi$ is an isomorphism modulo p. Following from the lemma below, we obtain that $\psi^\vee \circ \psi$ is an isomorphism. We deduce that $\mathrm{R}\Gamma(X, \omega)^2$ is a direct summand of $\mathrm{R}\Gamma(X_0(p), \omega_1)$, and write $\mathrm{R}\Gamma(X_0(p), \omega_1) \simeq \mathrm{R}\Gamma(X, \omega)^2 \oplus V^\bullet$. The theorem follows from 3.15. □

Lemma 3.14. *Let M be a finitely generated $\mathbb{Z}_p$-module, $\phi \in \mathrm{End}_{\mathbb{Z}_p} M$, if ϕ is an isomorphism modulo p, then ϕ is an isomorphism.*

Proof. Since M is finitely generated, $M = M_1 \oplus M_2$, where M_1 is a free $\mathbb{Z}_p$-module, and $M_2 = M[p^\infty]$ is finite.

$$\phi = \begin{pmatrix} \phi_1 & 0 \\ \phi_3 & \phi_2 \end{pmatrix} \in \mathrm{End}_{\mathbb{Z}_p} M = \begin{pmatrix} \mathrm{End}_{\mathbb{Z}_p}(M_1) & \mathrm{Hom}(M_2, M_1) = 0 \\ \mathrm{Hom}(M_1, M_2) & \mathrm{End}(M_2) \end{pmatrix}.$$

ϕ is an isomorphism mod p implies ϕ_1 and ϕ_2 are isomorphism mod p. Therefore ϕ_1 is an isomorphism since M_1 is finite free. On the other hand $\mathrm{coker}\phi_2/p = 0$, we obtain that $\mathrm{coker}\phi_2 = 0$, therefore ϕ_2 is an isomorphism(M_2 is finite). □

Lemma 3.15. *The complement $V^\bullet = 0$.*

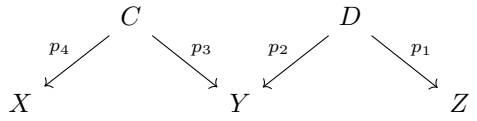
Proof. By the mod p short exact sequence we have:

$$\begin{aligned}
&\dim_{\mathbb{F}_p} \mathrm{H}^0(X_0(p)_1, \omega_1) = \dim_{\mathbb{F}_p} \mathrm{H}^0(X_0(p), \omega_1)/p + \dim_{\mathbb{F}_p} \mathrm{H}^1(X_0(p), \omega_1)[p], \\
&\dim_{\mathbb{F}_p} \mathrm{H}^0(X_1, \omega) = \dim_{\mathbb{F}_p} \mathrm{H}^0(X, \omega)/p + \dim_{\mathbb{F}_p} \mathrm{H}^1(X, \omega)[p], \\
&\dim_{\mathbb{F}_p} \mathrm{H}^1(X_0(p)_1, \omega_1) = \dim_{\mathbb{F}_p} \mathrm{H}^1(X_0(p), \omega)/p, \\
&\dim_{\mathbb{F}_p} \mathrm{H}^1(X_1, \omega) = \dim_{\mathbb{F}_p} \mathrm{H}^1(X, \omega)/p.
\end{aligned}$$

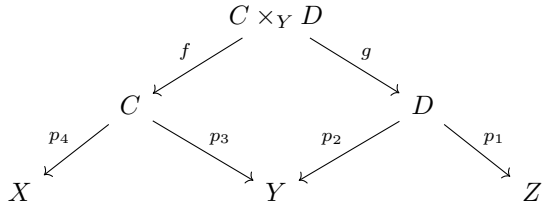
From 2 we have $\dim_{\mathbb{F}_p} \mathrm{H}^i(X_0(p)_1, \omega_1) \leq 2\dim_{\mathbb{F}_p} \mathrm{H}^i(X_1, \omega)$. Therefore $\dim_{\mathbb{F}_p} \mathrm{H}^1(X_0(p), \omega)/p = 2\dim_{\mathbb{F}_p} \mathrm{H}^1(X, \omega)/p$, and $\mathrm{H}^1(V^\bullet) = 0$. Moreover, $\dim_{\mathbb{F}_p} \mathrm{H}^1(X_0(p), \omega_1)[p] = 2\dim_{\mathbb{F}_p} \mathrm{H}^1(X, \omega)[p]$, and $\dim_{\mathbb{F}_p} \mathrm{H}^0(X_0(p), \omega_1)/p = 2\dim_{\mathbb{F}_p} \mathrm{H}^0(X, \omega)/p$, which implies $\mathrm{H}^0(V^\bullet) = 0$. □

Now we consider the U_p-action on $\mathrm{R}\Gamma(X_0(p), \omega_1)$. Before proving the properties, we recall two constructions of cohomological correspondence.

Lemma 3.16 (Definition)**.** *Let C, D be two algebraic correspondences given as follows:*



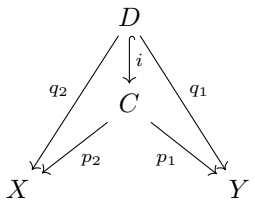
with cohomological correspondences $T_C : p_4^* \mathcal{F} \to p_3^! \mathcal{G}$ and $T_D : p_2^* \mathcal{G} \to p_1^! \mathcal{H}$. Assume that p_2 is flat, we can form the composition $T_{C \times_Y D}$ as a cohomological correspondence over $C \times_Y D$ given by:



$$T_D \circ T_C : f^* p_4^* \mathcal{F} \xrightarrow{T_C} f^* p_3^! \mathcal{G} \xrightarrow{\sim} g^! p_2^* \mathcal{G} \xrightarrow{T_D} g^! p_1^! \mathcal{H},$$

where the middle map is given by base change. Then on cohomology $T_{C \times_Y D} = T_D \circ T_C : \mathrm{R}\Gamma(X, \mathcal{F}) \to \mathrm{R}\Gamma(Z, \mathcal{H})$.

Lemma 3.17. *(restriction) Suppose $i : D \hookrightarrow C$ is a closed immersion, and we have algebraic correspondences:*



with cohomological correspondence $T_D : q_2^*\mathcal{F} \to q_1^!\mathcal{G}$, then we can form i_*T_D as:

$$i_*T_D : p_2^*\mathcal{F} \to i_*i^*p_2^*\mathcal{F} = i_*q_2^*\mathcal{F} \xrightarrow{T_D} i_*q_1^!\mathcal{G} = i_*i^!p_1^!\mathcal{G} \to p_1^!\mathcal{G},$$

where the first and third maps are given by adjunction. We say a cohomological correspondence $T_C : p_2^*\mathcal{F} \to p_1^!\mathcal{G}$ can be restricted to D if there exists a cohomological correspondence $T_D : q_2^*\mathcal{F} \to q_1^!\mathcal{G}$ such that $T_C = i_*T_D$. Then T_C can be restricted to D if and only if the morphism T_C is scheme theoretically supported on D.

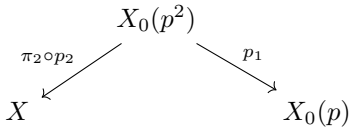
Proof. See [BP22]Proposition 2.3. □

Proposition 3.18. *Consider the U_p operator acting on $\mathrm{R}\Gamma(X_0(p), \omega_1)$ and $\mathrm{R}\Gamma(X_0(p)_1, \omega_1)$, we have:*

1. $U_p \circ \pi_2^* = \phi_{21} \circ \pi_1^*$,
2. $U_p \circ \phi_{21} \circ \pi_1^* = \phi_{21} \circ \pi_1^* \circ T_p^t - \langle p \rangle \pi_2^*$

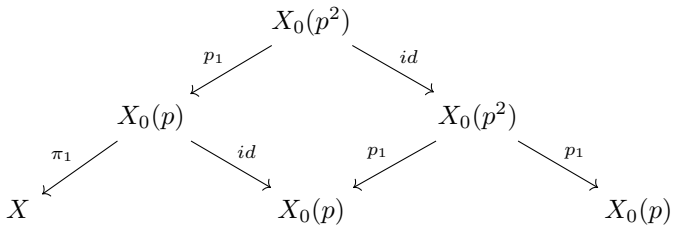
Proof. Our strategy is to show that both sides can be interpreted cohomological correspondences on an algebraic correspondence and with same source and target. Since all the correspondence we construct are naturally induced by maps on differentials and traces, it's easy to check that they are equal.

1. $U_p \circ \pi_2^*$ can be realized as a cohomological correspondence $T : p_2^*\pi_2^*\omega \xrightarrow{U_p} p_1^!\pi_2^*\omega$ over:



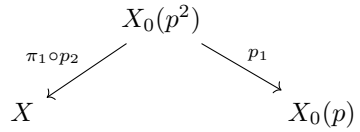
On the other hand, consider the composition $T' : p_1^*\pi_1^*\omega \to p_1^!\omega_1$ of

$\pi_1^*\omega \xrightarrow{\phi_{21}} id^*\pi_2^*\omega$ and $p_1^*\omega_1 \xrightarrow{\mathrm{tr}_{p_1}} p_1^!\omega_1$,

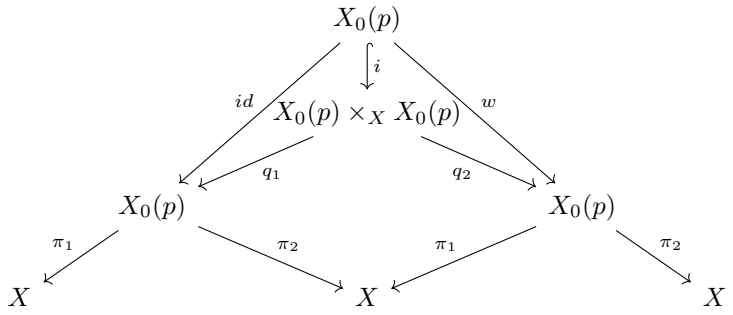


Using the same argument as in3.2, we obtain that the cohomological correspondence $p^{-1}T'$ is well defined. Since on cohomology, $\mathrm{R}\Gamma(X_0(p),\omega_1) \to \mathrm{R}\Gamma(X_0(p^2),p_1^*\omega) \xrightarrow{\mathrm{tr}\,p_1} \mathrm{R}\Gamma(X_0(p^2),p_1^!\omega) \to \mathrm{R}\Gamma(X_0(p),\omega_1)$ is multiplication by $\deg p_1 = p$, the cohomological correspondence $p^{-1}T'$ induces $\phi_{21}\circ\pi_1^*$. Since $\pi_2\circ p_2 = \pi_1\circ p_1$, we realize $\phi_{21}\circ\pi_1^*$ as a cohomological correspondence $p^{-1}T'$ on the same algebraic correspondence and with the same source and target as T. Tracking the definition of T and T', we obtain that $T = p^{-1}T'$.

2. $U_p\circ\phi_{21}\circ\pi_1^*$ can be interpreted as a cohomological correspondence T : $p_2^*\pi_1^*\omega \xrightarrow{\phi_{21}} p_2^*\pi_2^*\omega \xrightarrow{U_p} p_1^!\omega_1$ over:

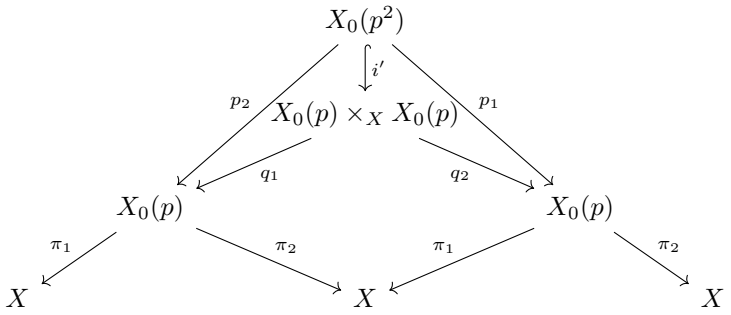


To construct a cohomological correspondence corresponding to $\phi_{21}\circ\pi_1^*\circ T_p^t - \langle p\rangle\pi_2^*$, consider the following diagram:



$\phi_{21}\circ\pi_1^*\circ T_p^t$ corresponds to the cohomological correspondence $T_1 : q_1^*\pi_1^*\omega \xrightarrow{T_p^t} q_1^*\pi_2^!\omega \xrightarrow{\sim} q_2^!\pi_1^*\omega \xrightarrow{\phi_{21}} q_2^!\omega_1$, and $\langle p\rangle\pi_2^*$ as a cohomological correspondence $T_2 : \pi_1^*\omega \xrightarrow{\langle p\rangle} \langle p\rangle\pi_1^*\omega = w^*\omega_1 = w^!\omega_1$. Let i_*T_2 be the pushforward of T_2, and denote $T' = T_1 - i_*T_2$. Consider the closed immersion

$i' : X_0(p^2) \hookrightarrow X_0(p) \times_X X_0(p)$ given by:



To interpret T' as a cohomological correspondence over $X_0(p^2)$, it suffices to show that T' is supported in $X_0(p^2)$ by4.15. Take x to be any geometric point on the complement of $X_0(p^2)$ in $X_0(p) \times_X X_0(p)$, then over $\widehat{\mathcal{O}}_{X_0(p)\times_X X_0(p),x}$ the corresponding isogeny $E_{0,x} \xrightarrow{\phi} E_{1,x} \xrightarrow{\phi^\vee} E_{0,x}$ is multiplication by p, and the trace map $\widehat{\mathcal{O}}_{X_0(p)\times_X X_0(p),x} \to q_2^! \widehat{\mathcal{O}}_{X_0(p),q_2(x)}$ is an isomorphism. Since $T_p^t = p^{-1} T_p^{t,\mathrm{naive}}$, $T_{1,x}$ is given by

$$(q_1^* \omega_0)_x \xrightarrow{\sim} (q_2^* \omega_1)_x \xrightarrow{\mathrm{tr}\, q_2} (q_2^! \omega_1)_x.$$

By the construction of T_2, it is always induced by isomorphism on differentials, therefore $T_{1,x} = (i_* T_2)_x$, and T' is supported in $X_0(p^2)$. By 4.15, there exists a cohomological correspondence $T'' : p_2^* \pi_1^* \omega \to p_1^! \omega_1$ such that $i'_* T'' = T'$. To check $T = T''$, it suffices to show that $i'_* T = T'$. Notice that $(X_0(p) \times_X X_0(p))_{\mathbb{Q}_p} = X_0(p^2)_{\mathbb{Q}_p} \sqcup X_0(p)_{\mathbb{Q}_p}$, hence over the generic fiber $i_* T_2$ is only supported on the complement of $X_0(p^2)$, and $T_1 = i'_* T$ over $X_0(p^2)_{\mathbb{Q}_p}$ by construction. As a conclusion $i'_* T = T'$ as they are equal generically.

□

Corollary 3.19. *After identifying* $\mathrm{R}\Gamma(X, \omega)^2 \xrightarrow[\sim]{\psi} \mathrm{R}\Gamma(X_0(p), \omega_1)$ *and* $\mathrm{R}\Gamma(X_1, \omega)^2 \xrightarrow[\sim]{\psi} \mathrm{R}\Gamma(X_0(p)_1, \omega_1)$*, the* U_p*-action on* $\mathrm{R}\Gamma(X_0(p), \omega_1)$ *and* $\mathrm{R}\Gamma(X_0(p)_1, \omega_1)$ *are given by the matrix* $\begin{pmatrix} 0 & -\langle p \rangle \\ 1 & T_p^t \end{pmatrix}$.

As a direct corollary, we also obtain that all weight one integral and mod p forms are ordinary.

Corollary 3.20. $e(U_p)\mathrm{R}\Gamma(X_0(p), \omega_1) = \mathrm{R}\Gamma(X_0(p), \omega_1)$ *and* $e(U_p)\mathrm{R}\Gamma(X_0(p)_1, \omega_1) = \mathrm{R}\Gamma(X_0(p)_1, \omega_1)$.

Proof. Since $\det \begin{pmatrix} 0 & -\langle p \rangle \\ 1 & T_p^t \end{pmatrix} = \langle p \rangle$, U_p is an isomorphism. □

Corollary 3.21. *Mod p Galois representations associated to* $\mathrm{R}\Gamma(X_0(p)_1, \omega_1)$ *are unramified at p.*

Proof. The isomorphism $\psi : \mathrm{R}\Gamma(X, \omega)^2 \xrightarrow{\sim} \mathrm{R}\Gamma(X_0(p), \omega_1)$ is T_l-equivariant by definition. Galois representations associated to weight one eigenform at spherical level are unramified at p are proved in [CG18], therefore $\rho_f \mid_{G_{\mathbb{Q}_p}}$ is unramified for any eigenform $f \in \mathrm{H}^0(X_0(p)_1, \omega_1)$. □

4 Modular forms of level $\Gamma_1(p)$

We develop the theory of modular forms of level $\Gamma_1(p)$ in characteristic p and integrally, equivalently, we study modular forms on $X_0(p)$ with Nebentypus. We construct such modular forms as global sections of certain line bundles on $X_0(p)$ by descending from some carefully picked integral model of $X_1(p)$. We will again mostly interested in weight one modular forms, since both H^0 and H^1 of the above line bundles are non-vanishing, and therefore there are more new arithmetic properties than those coming from spherical level. In particular, we prove the local Galois representations at p associated to such weight one forms with Nebentypus is tamely ramified, and is isomorphic to the direct sum of two characters, which can be viewed as a local global compatibility property. Moreover, we show that there exists a Galois representation valued in the Hecke algebra with the same local condition, and prove an $R = \mathbb{T}$ type theorem using the Calegari-Geraghty method as in [CG18].

4.1 Geometry of the modular curve of level $\Gamma_1(p^n)$

Instead of using the usual integral model for modular curve of level $\Gamma_1(p^n)$ over $\mathbb{Z}_p$, we introduce the balanced canonical model, which was constructed also in [KM85]. The benefit for this integral model is that the multiplicative and etale components on the special fiber are isomorphic, therefore provide us an ideal geometric object for the theory of ordinary modular forms, but the cost is that this integral model is only defined over $\mathbb{Z}_p[\zeta_{p^n}]$. From now on fix a system of primitive p^n-th root of unity $\{\zeta_{p^n}\}_{n\geq 1}$ such that $\zeta_{p^{n+1}}^p = \zeta_{p^n}$ for all $n \geq 1$.

Definition 4.1. Let S be a scheme over $\mathrm{Spec}\mathbb{Z}_p[\zeta_{p^n}]$, and E/S be an elliptic curve. A balanced canonical $\Gamma_1(p^n)$-structure on E/S is a tuple (ϕ, ϕ^t, P, Q), where $\phi : E \to E'$ is a cyclic p^n-isogeny, ϕ^t is the dual isogeny, P is a generator of $\ker \phi$, and Q is a generator of $\ker \phi^t$

$$E \underset{\phi^t}{\overset{\phi}{\rightleftarrows}} E'$$

such that $\langle P, Q\rangle_\phi = \zeta_{p^n}$, where $\langle \cdot, \cdot \rangle_\phi$ is the Weil paring of the isogeny ϕ.

Let $\mathcal{E}ll/\mathbb{Z}_p[\zeta_{p^n}]$ be the moduli stack of elliptic curves over $\mathbb{Z}_p[\zeta_{p^n}]$. To simplify the notation, we denote $\mathcal{Y}_1(p^n)$ to be the moduli stack of elliptic curves

with balanced canonical $\Gamma_1(p^n)$-structure. Since the usual $\Gamma_1(p^n)$-structure will not appear in this paper, there is no risk of confusion. Therefore we obtain the following diagram:

$$\begin{array}{ccc} \mathcal{Y}_1(p^n) & \xrightarrow{s} & \mathcal{E}ll/\mathbb{Z}_p[\zeta_{p^n}] \\ \pi\downarrow & \swarrow_{\pi'} & \\ \mathrm{Spec}\mathbb{Z}_p[\zeta_{p^n}] & & \end{array}$$

where for arbitrary scheme S over $\mathbb{Z}_p[\zeta_{p^n}]$, $s : (E/S, \phi, \phi^t, P, Q) \mapsto E/S$, $\pi : (E/S, \phi, \phi^t, P, Q) \mapsto \langle P, Q\rangle_\phi = \zeta_{p^n} \in \mu_{p^n}(S)$, and $\pi' : E/S \mapsto \zeta_{p^n}$.

Theorem 4.2. *The functor $s : \mathcal{Y}_1(p^n) \to \mathcal{E}ll/\mathbb{Z}_p[\zeta_{p^n}]$ is relatively representable*

Proof. See [KM85] Theorem 5.1.1 and Proposition 9.1.7. □

For simplification, we work with schemes rather than stacks, and consider the moduli problem of elliptic curves with a balanced canonical $\Gamma_1(p^n)$-structure and a $\Gamma_1(N)$-structure, for some fixed $N \geq 4$ and $p \nmid N$. By the above theorem, it is represented by a scheme over $\mathbb{Z}_p[\zeta_{p^n}]$, denoted by $\mathcal{Y}_1(p^n)$. We apologize for the duplicate notation, but from this point onward we work exclusively with schemes, so no confusion should arise. Using the j-invariant, [KM85]Proposition 9.3.1 shows that the structure morphism $\mathcal{Y}_1(p^n) \to \mathrm{Spec}\mathbb{Z}_p[\zeta_{p^n}]$ extends to a compactified modular curve $\mathcal{X}_1(p^n) \to \mathrm{Spec}\mathbb{Z}_p[\zeta_{p^n}]$. Denote $X_1(p^n) := \mathcal{X}_1(p^n) \times_{\mathbb{Z}_p[\zeta_{p^n}]} \mathbb{F}_p$ to be the special fiber.

Proposition 4.3. *Suppose $p \geq 5$. Then $\mathcal{X}_1(p^n)$ is a regular 2-dimensional scheme, which is smooth and geometrically connected over $\mathbb{Q}_p[\zeta_{p^n}]$. Moreover, $X_1(p^n)$ is smooth outside supersingular points.*

Proof. See [KM85] Corollary 10.9.2. □

To study the geometry of $X_1(p^n)$, we introduce the Igusa curves.

Definition 4.4. Let E/S be an elliptic curve over an $\mathbb{F}_p$-scheme S, $n \geq 1$ be an integer. An Igusa level p^n-structure on E/S is a generator $P \in E^{(p^n)}(S)$ of $\ker(V^n_{E/S})$. Denote $\mathrm{Ig}(p^n)$ the corresponding Igusa moduli stack over $\mathcal{E}ll/\mathbb{F}_p$.

Theorem 4.5 (Igusa)**.** *1. $\mathrm{Ig}(p^n)$ is relatively representable, finite and flat over $(\mathcal{E}ll/\mathbb{F}_p)$ of degree $p^{n-1}(p-1)$.*

2. It is regular one-dimensional.

Proof. See [KM85]Theorem 12.6.1. □

As before, we denote $\mathrm{Ig}(p^n)$ to be the scheme over $\mathbb{F}_p$ corresponding to the compactified moduli stack of Igusa p^n-structure and a $\Gamma_1(N)$ structure. The compactification of Igusa curve is discussed in [KM85]12.7. Denote $X/\mathbb{F}_p$ to be the compactified modular curve of level $\Gamma_1(N)$, we have the following theorem.

Theorem 4.6. $\mathrm{Ig}(p^n)$ *is smooth over* $\mathbb{F}_p$*. The natural map* $\mathrm{Ig}(p^n) \to X$ *is finite etale with Galois group* $(\mathbb{Z}/p^n\mathbb{Z})^\times$ *outside the supersingular points of* X*, and is fully ramified over each supersingular points of* X*.*

Proof. See [KM85]Corollary 12.7.2. □

We can now state the main theorem regarding the geometry of $X_1(p^n)$.

Theorem 4.7.

$$X_1(p^n) = \bigcup_{\substack{(a,b,u) \\ a+b=n, u\in(\mathbb{Z}/p^{\min(a,b)}\mathbb{Z})^\times}} X(a,b,u),$$

where $X(a,b,u)$ *are the irreducible components of* $X_1(p^n)$*, labeled by non-negative integers* (a,b) *such that* $a+b=n$*, and* $u \in (\mathbb{Z}/p^{\min(a,b)}\mathbb{Z})^\times$*. Moreover* $X(a,b,u)$ *is isomorphic to the compactified Igusa curve* $\mathrm{Ig}(p^{\max(a,b)})$*, with crossings at the supersingular points.*

Proof. See [KM85] Theorem 13.11.4. □

The following example is the main geometric object for our construction.

Example 4.8. $X_1(p) = \mathrm{Ig}(p)_1 \cup \mathrm{Ig}(p)_2$ with crossings at the supersingular points, where $\mathrm{Ig}(p)_i$ are isomorphic to the Igusa curve $\mathrm{Ig}(p)$. Ig_1 corresponds to the points $(E \xrightarrow{F} E^{(p)}, 0, Q)$ on $X_1(p)$, and Ig_2 corresponds to the points $(E^{(p)} \xrightarrow{V} E, P, 0)$ on $X_1(p)$.

4.2 Modular line bundles on $\mathcal{X}_0(p)$ with a fixed Nebentypus

In this section, we construct line bundles on $\mathcal{X}_0(p)$ whose coherent cohomologies correspond to modular forms of level $\Gamma_0(p)$ with a fixed weight and Nebentypus. Recall that in 3 we have studied modular forms of level $\Gamma_0(p)$ with trivial Nebentypus, now we develop the theory for arbitrary Nebentypus by descending from $\mathcal{X}_1(p)$.

Let $\pi : \mathrm{Ig}(p) \to X$ be the Igusa cover of the modular curve. Denote ω_X to be the modular line bundle on X, and let $\omega_{\mathrm{Ig}(p)} = \pi^*\omega_X$, we will simply use ω without label when there's no confusion. Recall that the Hasse invariant on X is weight $p-1$ modular form $A \in \mathrm{H}^0(X, \omega^{(p-1)})$, induced by the Verschiebung $V : E^{(p)} \to E$. The key property of the Hasse invariant is that it is nonvanishing at ordinary points, and has a simple zero at each supersingular points. In other words, the divisor of A is SS, the supersigular divisor, and $\omega^{p-1} \simeq \mathcal{O}(SS)$.

Theorem 4.9 (Serre)**.** *There exists a section* $a \in \mathrm{H}^0(\mathrm{Ig}(p), \omega)$ *such that* $a^{p-1} = A \in \mathrm{H}^0(\mathrm{Ig}(p), \omega^{p-1})$*, where* A *is the Hasse invariant on* $\mathrm{Ig}(p)$ *defined in the same way.*

Proof. See [Gro90]Proposition 5.2. □

Corollary 4.10. $\mathcal{O}_{\mathrm{Ig}(p)}(SS) \simeq \omega_{\mathrm{Ig}(p)}$

Proof. Since π is totally ramified at each supersingular point of degree $p-1$, $\omega_{\mathrm{Ig}(p)}^{p-1} = \pi^*\mathcal{O}_X(SS) \simeq \mathcal{O}_{\mathrm{Ig}(p)}((p-1)SS)$. Therefore by 4.9, $(p-1)\mathrm{div}(a) = \mathrm{div}(A) = (p-1)SS$, and we obtain $\mathrm{div}(a) = SS$.

□

There is a $\mathbb{F}_p^\times$ action on $\mathrm{Ig}(p)$ given by $d \cdot (E, P) = (E, d^{-1}P)$, where P is a generator of $\ker V$, for $V : E^{(p)} \to E$, and $d \in \mathbb{F}_p^\times$. Under this action, $\pi : \mathrm{Ig}(p) \to X$ is a cyclic cover of degree $p-1$. This gives an action of $\mathbb{F}_p^\times$ on $\pi_*\mathcal{O}_{\mathrm{Ig}(p)}$, and We can consider the decomposition

$$\pi_*\mathcal{O}_{\mathrm{Ig}(p)} = \bigoplus_{\chi:\mathbb{F}_p^\times \to \mathbb{F}_p^\times} \mathcal{O}_\chi,$$

where $\mathcal{O}_\chi$ is the subsheaf of $\pi_*\mathcal{O}_{\mathrm{Ig}(p)}$ with $\mathbb{F}_p^\times$ action given by χ, which is actually a line bundle since the order of the group is prime to p. To simplify the notation, denote by $[i] : \mathbb{F}_p^\times \to \mathbb{F}_p^\times$ the character that maps d to d^i, and $\mathcal{O}(i)$ to be the character line bundle $\mathcal{O}_{[i]}$.

Proposition 4.11. *1. $a \in \mathrm{H}^0(\mathrm{Ig}(p), \omega)(-1)$, i.e the $(p-1)$-th root of Hasse invariant lies in the eigenspace given by the character $[-1]$.*

2.

$$\pi_*\mathcal{O}_{\mathrm{Ig}(p)} = \bigoplus_{0 \le k \le p-2} \omega_X^{-k},$$

and moreover $\omega_X^{-k} = \mathcal{O}(-i)$.

Proof. See [KM85] Corollary 12.8.5 and [Gro90] Proposition 5.2. □

To define the Nebentypus line bundle, we also need some facts about the local ring of $\mathcal{X}_1(p)$ at supersingular points. Let $\varpi = \zeta_p - 1$ to be uniformizer of $\mathbb{Z}_p[\zeta_p]$.

Theorem 4.12. *$\mathcal{X}_1(p)$ is regular of dimension 2. Let $x \in \mathcal{X}_1(p)(\bar{\mathbb{F}}_p)^{\mathrm{ss}}$ be a supersingular point, then $\widehat{\mathcal{O}}_{\mathcal{X}_1(p),x} \simeq W(\bar{\mathbb{F}}_p)[\zeta_p][\![X, Y]\!]/(XY - \varpi)$.*

Proof. This is essentially in [DR73]. The case of $\mathcal{X}_0(p)$ is well known, and the proof is essentially the same. For completeness, we recall the proof here. □

Let $[\,] : \mathbb{F}_p^\times \to \mathbb{Z}_p^\times$ be the Teichmuller character. $d \in \mathbb{F}_p^\times$ acts on $\mathcal{X}_1(p)$ by

$$d \cdot (\ E \underset{\phi^t}{\overset{\phi}{\rightleftarrows}} E'\ , P, Q) = (\ E \underset{\phi^t}{\overset{\phi}{\rightleftarrows}} E'\ , dP, d^{-1}Q).$$

we have the following modification.

Corollary 4.13. *There is a choice of coordinate such that*

$$\widehat{\mathcal{O}}_{\mathcal{X}_1(p),x} \simeq W(\bar{\mathbb{F}}_p)[\zeta_p][\![X,Y]\!]/(XY-\varpi),$$

and the action of $d \in \mathbb{F}_p^\times$ *is given by*

$$X \mapsto [d^{-1}]X, Y \mapsto [d]Y.$$

Moreover Y *generates the ideal corresponding to* Ig_1*, and* X *generates the ideal corresponding to* Ig_2*.*

Proof. □

The irreducible components Ig_1 and Ig_2 of $X_1(p)$ can be viewed as Cartier divisors on $\mathcal{X}_1(p)$. Denote $\mathcal{O}(\mathrm{Ig}_i)$ to be corresponding line bundles on $\mathcal{X}_1(p)$ for $i = 1, 2$. Notice that $\mathrm{Ig}_1 + \mathrm{Ig}_2 = (\varpi)$ as divisor, $\mathcal{O}(\mathrm{Ig}_1) \simeq \mathcal{O}(-\mathrm{Ig}_2)$. As before, for a coherent sheaf $\mathcal{F}$ on $\mathcal{X}_0(p)$ with an $\mathbb{F}_p^\times$-action, we denote $\mathcal{F}(i)$ to be character subsheaf such tha $\mathbb{F}_p^\times$ acts via the i-th power of the Teichmuller character. We now prove the main theorem of this section.

Theorem 4.14. *Let* $\pi : \mathcal{X}_1(p) \to \mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}$ *be the natural forgetful cover. Then* $\pi_*\mathcal{O}(\mathrm{Ig}_1)(-1)$ *is a line bundle on* $\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}$*, and* $\pi_*\mathcal{O}(\mathrm{Ig}_1)(i)$ *for* $i \neq -1$ *are not line bundles.*

Proof. Let $\mathcal{X}_1(p)^{\mathrm{ord}}$ and $\mathcal{X}_0(p)^{\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$ be the open subscheme of the integral models excluding the supersingular points on the special fiber. Since $\pi : \mathcal{X}_1(p)^{\mathrm{ord}} \to \mathcal{X}_0(p)^{\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$ is a finite flat $\mathbb{F}_p^\times$-cover, $\pi_*\mathcal{O}(\mathrm{Ig}_1)$ decompose into direct sum of character line bundles. Therefore, we only need to check that for a supersingular point $x \in \mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}(\bar{\mathbb{F}}_p)^{\mathrm{ss}}$, the complete local ring $\pi_*\mathcal{O}(\mathrm{Ig}_1)(-1)\hat{}_x$ is a free $\widehat{\mathcal{O}}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]},x} = (\widehat{\mathcal{O}}_{\mathcal{X}_1(p),x})^{\mathbb{F}_p^\times} \simeq W(\bar{\mathbb{F}}_p)[\zeta_p][\![X^{p-1}, Y^{p-1}]\!]/(X^{p-1}Y^{p-1} - \varpi^{p-1})$ module of rank one. Locally at x, the function Y cuts out the divisor Ig_1, therefore by the divisor line bundle correspondence, $\mathcal{O}(\mathrm{Ig}_1)\hat{}_x$ is given by $Y^{-1}\cdot\widehat{\mathcal{O}}_{\mathcal{X}_1(p)}, x$ viewed as a submodule inside the fraction ring. Then it follows from a direct computation that

1. $Y^{-1}\cdot W(\bar{\mathbb{F}}_p)[\zeta_p][\![X,Y]\!]/(XY-\varpi)(-1) = Y^{-1}\cdot W(\bar{\mathbb{F}}_p)[\zeta_p][\![X^{p-1}, Y^{p-1}]\!]/(X^{p-1}Y^{p-1} - \varpi^{p-1})$
2. $Y^{-1}\cdot W(\bar{\mathbb{F}}_p)[\zeta_p][\![X,Y]\!]/(XY-\varpi)(i) = (Y^i, X^{p-2-i}Y^{-1})\cdot W(\bar{\mathbb{F}}_p)[\zeta_p][\![X^{p-1}, Y^{p-1}]\!]/(X^{p-1}Y^{p-1} - \varpi^{p-1})$ for $i = 0, 1, \cdots, p-3$.

Therefore, $\pi_*\mathcal{O}(\mathrm{Ig}_1)_x(i)$ is free of rank one if and only if $i = -1$. □

Denote $\mathcal{L}$ to be the line bundle $\pi_*\mathcal{O}(\mathrm{Ig}_1)(-1)$, and let L be it's restriction to the special fiber $X_0(p)$. Recall that $X_1 := X_0(p)^F$ and $X_2 := X_0(p)^V$ are the two irreducible components of $X_0(p)$ corresponding to the Frobenius and Verschiebung isogeny, intersecting at supersingular points.

Corollary 4.15. *Consider the restriction of L to X_1 and X_2, then $L\mid_{X_1}\simeq\omega^{-1}$, and $L\mid_{X_2}\simeq\omega$.*

Proof. Since $\mathcal{O}(\mathrm{Ig}_1)\simeq\mathcal{O}(-\mathrm{Ig}_2)$, $\mathcal{O}(\mathrm{Ig}_1)\mid_{\mathrm{Ig}_1}\simeq\mathcal{O}_{\mathrm{Ig}_1}(-SS)$. Consider the diagram:

$$\begin{array}{ccc}\mathrm{Ig}_1 & \hookrightarrow & X_1(p)\\ {\scriptstyle\pi_1}\downarrow & & \downarrow{\scriptstyle\pi}\\ X_1 & \hookrightarrow & X_0(p)\end{array}$$

By proper base change, $L\mid_{X_1}\simeq\pi_{1,*}\mathcal{O}_{\mathrm{Ig}_1}(-SS)(-1)$. Viewing ω^{-1} and $\mathcal{O}_{\mathrm{Ig}_1}(-SS)$ as subsheaf of the sheaf of rational functions, then $a:\omega^{-1}\xrightarrow{\sim}\mathcal{O}(-SS)$ induces the isomorphism. Since 4.11 tells us $\mathbb{F}_p^\times$ acts on a via the character $[\ ]^{-1}$, $\pi_{1,*}\mathcal{O}_{\mathrm{Ig}_1}(-SS)(-1)\simeq\pi_{1,*}\omega^{-1}(0)\simeq\omega^{-1}$. For the restriction to X_2, notice that the action of $\mathbb{F}_p^\times$ restricted to Ig_2 is the inverse of the action we defined on Igusa curves. The rest of the arguments are the same. □

Combining the proof of 4.14 and 4.15, we can also get the following corollary.

Corollary 4.16. *Let $i_1:X_1\to X_0(p)$ and $i_2:X_2\to X_0(p)$ be the closed embeddings. Then $\pi_*\mathcal{O}(\mathrm{Ig}_1)(-i)\simeq i_{1,*}\omega^{-i}\oplus i_{2,*}\omega^{i+1-p}$, for $i=2,3,\cdots,(p-1)$.*

we can now give the definition of a modular form with Nebentypus. For a fixed weight $k,a\in\mathbb{Z},i\in\{0,1,\cdots,p-2\}$, we call $\omega_{k,a,i}:=\omega(k,a)\otimes L^i$ the line bundle of weight k, with integral structure $\omega(k,a)$ and Nebentypus $[-i]$.

Definition 4.17. The space of modular forms of weight k, integral structure $\omega(k,a)$ and Nebentypus $[-i]$ is $\mathrm{R}\Gamma(\mathcal{X}_0(p),\omega_{k,a,i})$. Similarly, we can define the mod p^n modular forms with a tame Nebentypus.

5 The U_p-operator and Cousin Complexes

5.1 The U_p-operator

In this section, we construct the U_p operator acting on the space of modular forms with Nebentypus we just defined. First, we need to introduce the Hecke correspondence for the modular curve $\mathcal{X}_1(p^n)$.

Definition 5.1. Let S be a scheme over $\mathrm{Spec}\mathbb{Z}_p[\zeta_{p^n}]$, and E/S be an elliptic curve. A $\Gamma_1(p^{n+1};p^n)$-structure on E/S the following data:

1. $E_0:=E\to E_{n+1}$ a cyclic isogeny of degree p^{n+1} with standard factorization:
$$E_0\underset{\lambda_{1,0}}{\overset{\pi_{0,1}}{\rightleftarrows}}E_1\underset{\lambda_{2,1}}{\overset{\pi_{1,2}}{\rightleftarrows}}\cdots\underset{}{\overset{}{\rightleftarrows}}E_n\underset{\lambda_{n+1,n}}{\overset{\pi_{n,n+1}}{\rightleftarrows}}E_{n+1}$$
2. P: a generator of $\ker\pi_{1,n+1}$, and Q a generator of $\ker\lambda_{n,0}$.

3. $\zeta_{p^n} = \langle P, \pi_{n,n+1}(Q)\rangle_{\pi_{1,n+1}} = \langle \lambda_{1,0}(P), Q\rangle_{\pi_{0,n}}$, where $\pi_{a,b} := \pi_{a,a+1} \circ \cdots \circ \pi_{b-1,b}$, and $\lambda_{b,a} := \lambda_{b,b-1} \circ \cdots \lambda_{a+1,a}$ for $a < b$.

Remark 5.2. By properties of the Weil paring, $\langle P, \pi_{n,n+1}(Q)\rangle_{\pi_{1,n+1}} = \langle \lambda_{1,0}(P), Q\rangle_{\pi_{0,n}}$ holds automatically, the condition is that the value of this paring is a fixed primitive p^n-th root of unity ζ_{p^n}.

As before, we denote $s : \mathcal{Y}_1(p^{n+1}; p^n) \to \mathcal{E}ll/\mathbb{Z}_p[\zeta_{p^n}]$ to be the corresponding morphism of moduli stacks, which is relatively representable [KM85]9.4. And for simplicity, we denote $\mathcal{X}_1(p^{n+1}; p^n)$ to be the scheme over $\mathbb{Z}_p[\zeta_{p^n}]$ that represents the moduli problem of elliptic curves with a $\Gamma_1(p^{n+1}, p^n)$-structure and a $\Gamma_1(N)$-structure.

Theorem 5.3. *$\mathcal{X}_1(p^{n+1}; p^n)$ is a regular scheme of dimension 2, which is smooth outside the supersingular points on the special fiber, and geometrically connected over $\mathbb{Q}_p(\zeta_{p^n})$.*

Proof. See [KM85] Theorem 10.12.2. □

The integral model $\mathcal{X}_1(p^{n+1}; p^n)$ provides us the Hecke correspondence for $\mathcal{X}_1(p^n)$ as follows. Consider the map $q_1 : \mathcal{X}_1(p^{n+1}; p^n) \to \mathcal{X}_1(p^n)$ given by:

$$(E_0 \underset{\lambda_{1,0}}{\overset{\pi_{0,1}}{\rightleftarrows}} E_1 \underset{\lambda_{2,1}}{\overset{\pi_{1,2}}{\rightleftarrows}} \cdots \rightleftarrows E_n \underset{\lambda_{n+1,n}}{\overset{\pi_{n,n+1}}{\rightleftarrows}} E_{n+1}, P, Q) \mapsto (\pi_{1,n+1} : E_1 \to E_{n+1}, P, \pi_{n,n+1}(Q)).$$

This map is well defined since $\pi_{n,n+1}(Q)$ is a generator of $\ker(\lambda_{n+1,1})$ by the standard order criterion[KM85]6.7.13, and $\langle P, \pi_{n,n+1}(Q)\rangle_{\pi_{1,n+1}} = \zeta_{p^n}$ by our assumption. Similarly, define $q_2 : \mathcal{X}_1(p^{n+1}; p^n) \to \mathcal{X}_1(p^n)$ given by:

$$(E_0 \underset{\lambda_{1,0}}{\overset{\pi_{0,1}}{\rightleftarrows}} E_1 \underset{\lambda_{2,1}}{\overset{\pi_{1,2}}{\rightleftarrows}} \cdots \rightleftarrows E_n \underset{\lambda_{n+1,n}}{\overset{\pi_{n,n+1}}{\rightleftarrows}} E_{n+1}, P, Q) \mapsto (\pi_{0,n} : E_0 \to E_n, \lambda_{1,0}(P), Q).$$

Therefore, we obtain the following Hecke correspondence:

$$\begin{array}{ccccc} & & \mathcal{X}_1(p^{n+1}, p^n) & & \\ & \swarrow^{q_1} & & \searrow^{q_2} & \\ \mathcal{X}_1(p^n) & & & & \mathcal{X}_1(p^n) \end{array}$$

Let $X_1(p^{n+1}, p^n) := \mathcal{X}_1(p^{n+1}; p^n) \times_{\mathbb{Z}_p[\zeta_{p^n}]} \mathbb{F}_p$ be the special fiber. We have the following theorem.

Theorem 5.4. *We have the decomposition of $X_1(p^{n+1}, p^n)$ into irreducible components:*

$$X_1(p^{n+1}, p^n) = \mathrm{Ig}_1(p^n) \cup \bigcup_{\substack{(a,b,u) \\ a+b=n+1, a,b\geq 1, u\in(\mathbb{Z}/p^{\min(a,b)}\mathbb{Z})^\times}} X(a,b,u) \cup \mathrm{Ig}_2(p^n),$$

where $\mathrm{Ig}_1(p^n)$ *and* $\mathrm{Ig}_2(p^n)$ *are the irreducible components corresponding to cyclic isogeny of type* $(n+1,0)$ *and* $(0,n+1)$ *in 3.4, which are isomorphic to* $\mathrm{Ig}(p^n)$. $X(a,b,u)$ *is a component isomorphic to* $\mathrm{Ig}(p^{\max(a,b)})$, *corresponding to cyclic isogeny of type* (a,b), *for* $a+b=n+1, a,b\geq 1$, *and* u *parametrize the relation of* P *and* Q. *Moreover, each two component intersects at the supersingular points.*

Proof. We briefly explain where these components come from. For details see [KM85]Corollary 13.12.4. □

Example 5.5. The key example which we will use in this paper is $X_1(p^2,p)$. By the above theorem,

$$X_1(p^2,p)=\mathrm{Ig}_1(p)\cup\bigcup_{u\in(\mathbb{Z}/p\mathbb{Z})^\times}\mathrm{Ig}_u(p)\cup\mathrm{Ig}_2(p),$$

where all the components are isomorphic to the Igusa curve $\mathrm{Ig}(p)$. $\mathrm{Ig}_1(p)$ corresponds to the points $(E\xrightarrow{F}E^{(p)}\xrightarrow{F}E^{(p^2)},0,Q)$, and $\mathrm{Ig}_2(p)$ corresponds to the points $(E^{(p^2)}\xrightarrow{V}E^{(p)}\xrightarrow{V}E,P,0)$. $\mathrm{Ig}_u(p)$ corresponds to the points $(E\xrightarrow{F}E^{(p)}\xrightarrow{V}E,P,Q)$ such that $P=u\cdot Q$. These points satisfies the relation $\langle P,Q\rangle=\zeta_p$ since in characteristic p, $\zeta_p=1$.

Recall that we have the line bundle ω_1 on $\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}$ defined in 3.1, and the Nebentypus line bundle $\mathcal{L}=\pi_*\mathcal{O}(\mathrm{Ig}_1)(-1)$. The goal of this section to proof the following theorem:

Theorem 5.6. *There is a cohomological correspondence* $U_p: p_2^*(\omega_1\otimes\mathcal{L}^{\otimes a})\to p_1^!(\omega_1\otimes\mathcal{L}^{\otimes a})$ *for all* $a=0,1,\cdots,(p-2)$, *which is non-trivial on the special fiber.*

We first construct a cohomological correspondence on $\mathcal{X}_1(p^2,p)$.

Lemma 5.7. *There is a rational cohomological correspondence* $U_p: q_2^*(\mathcal{O}_{\mathcal{X}_1(p)}(a\mathrm{Ig}_1))\dashrightarrow q_1^!(\mathcal{O}_{\mathcal{X}_1(p)}(a\mathrm{Ig}_1))$ *on* $\mathcal{X}_1(p^2,p)$ *for* $0\leq a\leq p-2$..

Proof. To give a rational map $q_2^*(\mathcal{O}_{\mathcal{X}_1(p)}(a\mathrm{Ig}_1))\dashrightarrow q_1^*(\mathcal{O}_{\mathcal{X}_1(p)}(a\mathrm{Ig}_1))$, is equivalent to give a rational section of $\mathcal{O}_{\mathcal{X}_1(p^2,p)}(q_1^*(a\mathrm{Ig}_1)-q_2^*(a\mathrm{Ig}_1))$. As a divisor

$$q_1^*(a\mathrm{Ig}_1)-q_2^*(a\mathrm{Ig}_1)=-a\sum_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p).$$

Take $s:\mathcal{O}\dashrightarrow\mathcal{O}(-a\sum_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p))$ to be a rational section, and denote $\mathrm{tr}_{q_1}:\mathcal{O}_{\mathcal{X}_1(p^2,p)}\to q_1^!\mathcal{O}_{\mathcal{X}_1(p)}$ to be the trace map. We define U_p to be $s\otimes\mathrm{tr}_{q_1}$. □

In order to prove 5.6, we show that the rational morphism U_p is regular, and further more, it descents to a cohomological correspondence $U_p: p_2^*(\omega_1\otimes\mathcal{L}^{\otimes a})\to p_1^!(\omega_1\otimes\mathcal{L}^{\otimes a})$. Consider the $\mathbb{F}_p^\times$-action on $\mathcal{X}_1(p^2,p)$ given by

$$d\cdot(\ E_0\ \underset{\lambda_{1,0}}{\overset{\pi_{0,1}}{\rightleftarrows}}\ E_1\ \underset{\lambda_{2,1}}{\overset{\pi_{1,2}}{\rightleftarrows}}\ E_2\ ,P,Q)=(\ E_0\ \underset{\lambda_{1,0}}{\overset{\pi_{0,1}}{\rightleftarrows}}\ E_1\ \underset{\lambda_{2,1}}{\overset{\pi_{1,2}}{\rightleftarrows}}\ E_2\ ,dP,d^{-1}Q).$$

By the affineness of the open modular curve, we can denote $\mathcal{Z} := \mathcal{X}_1(p^2, p)$ to be the geometric quotient of $\mathcal{X}_1(p^2, p)$ by $\mathbb{F}_p^\times$ as a scheme. Since the morphisms q_1 and q_2 are $\mathbb{F}_p^\times$-equivariant, we have the following diagram:

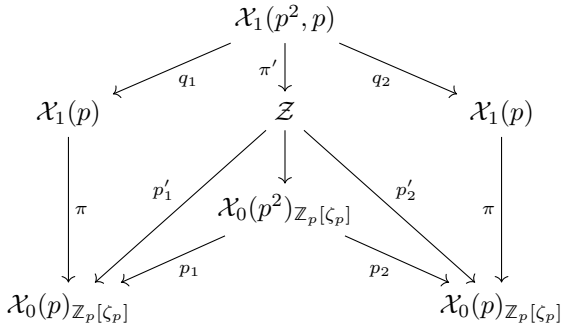


Figure 1: The full correspondences

Lemma 5.8. *1. Let X be a normal scheme, suppose a finite group G acts on X, and the geometric quotient $Y := X/G$ exists. Then Y is normal.*

2. A finite birational morphism $f : X \to Y$ of integral schemes with Y normal is an isomorphism,

Proof. 1. Let $U \subset Y$ be any open, $V := f^{-1}(U) \subset X$. Denote $A := \mathcal{O}(V)$, and $B := \mathcal{O}(U)$, then $B = A^G$, and $\mathrm{Frac}(B) = \mathrm{Frac}(A)^G$. If $f \in \mathrm{Frac}(B)$ is integral over B, then f is integral over A, hence $f \in A \cap \mathrm{Frac}(B) = B$.

2. Since finite morphism is integral, and X and Y have the same fraction fields, for any $U \subset Y$, and $V := f^{-1}(U) \subset X$, $f_* : \mathcal{O}(U) \to \mathcal{O}(V)$ is an isomorphism.

□

Corollary 5.9. *$\mathcal{Z}$ is isomorphic to $\widetilde{\mathcal{X}}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$, the normalization of $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$. Moreover, The upper left and right diagrams of 1 are Cartesian.*

Proof. By 5.8 $\mathcal{Z}$ is normal, and therefore the morphism $\mathcal{Z} \to \mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$ factors through the normalization. Since they are isomorphic over the generic fiber, by 5.8 we get an isomorphism $\mathcal{Z} \simeq \widetilde{\mathcal{X}}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$. Since the geometric quotient $\mathcal{Z}$ is isomorphic to the categorical quotient, the Cartesianess follows from the moduli description. □

Denote $\nu : \mathcal{Z} \to \mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$ to be the normalization map. Take $\alpha : \mathcal{O}_{\mathcal{X}_1(p)} \dashrightarrow \mathcal{O}_{\mathcal{X}_1(p)}(a\mathrm{Ig}_1)$ to be a rational section such that the $\mathbb{F}_p^\times$-action is given by the character $[\ ]^{-a}$(a^{-1}-th power of Teichmuller character). Since q_1, q_2 are $\mathbb{F}_p^\times$-equivariant, $\frac{q_1^*\alpha}{q_2^*\alpha} : \mathcal{O} \dashrightarrow \mathcal{O}(-a\sum_{u\in\mathbb{F}_p^\times} \mathrm{Ig}_u(p))$ is a rational section such that $\mathbb{F}_p^\times$ acts trivially, and therefore descent to a section $\frac{q_1^*\alpha}{q_2^*\alpha} : \mathcal{O}_{\mathcal{Z}} \dashrightarrow$

$\pi'_*(\mathcal{O}(-a\sum_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p)))(0)$. By the proper base change theorem, $(p'_1)^*\mathcal{L}^{\otimes a}\simeq \pi'_*q_1^*\mathcal{O}(\mathrm{Ig}_1)(-a)$, and $(p'_2)^*\mathcal{L}^{\otimes a}\simeq \pi'_*q_2^*\mathcal{O}(\mathrm{Ig}_1)(-a)$. Therefore $\frac{q_1^*\alpha}{q_2^*\alpha}$ can be viewed as a rational section $(p'_2)^*\mathcal{L}^{\otimes a}\dashrightarrow (p'_1)^*\mathcal{L}^{\otimes a}$. Let $\mathrm{tr}_{p_1}:\mathcal{O}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}}\to p_1^!\mathcal{O}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}}$ to be the trace map. Define $U_p:(p'_2)^*\mathcal{L}^{\otimes a}\dashrightarrow (p'_1)^*\mathcal{L}^{\otimes a}\otimes\nu^*p_1^!\mathcal{O}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}}$ to be $\frac{q_1^*\alpha}{q_2^*\alpha}\otimes\nu^*\mathrm{tr}_{p_1}$. We will prove that U_p is regular and descent to $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$.

Since the rational section $s:\mathcal{O}\dashrightarrow\mathcal{O}(-a\sum_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p))$ is regular outside the middle components $\cup_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p)$, we only need to check that for any $x\in\mathcal{Z}(\bar{\mathbb{F}}_p)$ that lies in the image $\pi'(\cup_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p))$, locally $U_{p,x}$ is a regular function. In other words, it suffices to show that $U_{p,x}$ is given by a function in $\widehat{\mathcal{O}}_{\mathcal{Z},x}$. By our local ring calculation in 2.16, $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$ is normal outside the component $X_0(1,1)$ on the special fiber. Therefore in order to show that the regular section U_p descent to $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$, we only need to show for any $x\in X_0(1,1)(\overline{\mathbb{F}}_p)$, $U_{p,\nu^{-1}(x)}\in\bigoplus_{y\in\nu^{-1}(x)}\widehat{\mathcal{O}}_{\mathcal{Z},y}$ lies in the image of $\widehat{\mathcal{O}}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]},x}$. Our propose is to prove 5.6, so we only need to show the above conditions hold for ordinary point x. This is because the algebraic Hartog's theorem works for S_2-schemes($\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$ is S_2 since $\mathcal{X}_0(p^2)$ is regular), and we only need to prove the section is regular outside codimension 2.

Regarding the trace map, we have the following lemma.

Lemma 5.10. *Let $x\in X_0(1,1)(\overline{\mathbb{F}}_p)^{\mathrm{ord}}$ be an ordinary point on the $(1,1)$-component on the special fiber. By 2.16 $\widehat{\mathcal{O}}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]},x}\simeq W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}$, then $\mathrm{tr}_{p_1,x}$ is divided by $(X-Y)^{(p-2)}$, in other words $\mathrm{tr}_{p_1,x}$ factors through*

$$\widehat{\mathcal{O}}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]},x}\to(X-Y)^{p-2}\cdot p_1^!\widehat{\mathcal{O}}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p],p_1(x)}}\subset p_1^!\widehat{\mathcal{O}}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p],p_1(x)}}.$$

Proof. We may assume the map $\widehat{\mathcal{O}}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p],p_1(x)}}\to p_{1,*}(\widehat{\mathcal{O}}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]},x})$ is given by $W[\zeta_p][Y]\to W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}$. By adjunction, we may consider $\mathrm{tr}_{p_1,x}\in\mathrm{Hom}_{W[\zeta_p][\![Y]\!]}(W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y},W[\zeta_p][\![Y]\!])$ to be the usual trace, and view $\mathrm{Hom}_{W[\zeta_p][\![Y]\!]}(W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y},W[\zeta_p][\![Y]\!])$ as an free $W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}$-module of rank 1. Let $T:=X-Y$, then

$$W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}=W[\zeta_p][\![T,Y]\!]/\Pi_{i=1}^{p-1}(T+(1-\zeta_p^i)(1+Y)).$$

An easy calculation gives

$$\Pi_{i=1}^{p-1}(T+(1-\zeta_p^i)(1+Y))=T^{p-1}+pT^{p-2}(1+Y)+\cdots+p(1+Y)^{p-1}.$$

Since $1,T,\cdots,T^{p-2}$ forms a basis of $W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}$ as a $W[\zeta_p][\![Y]\!]$-module, by the equation above, we get $\mathrm{tr}(1)=p-1$, and $\mathrm{tr}(T^i)$ is divided by p for all $i=1,2,\cdots,(p-2)$. Therefore $T\cdot\mathrm{tr}=p\cdot f$ for some $f\in\mathrm{Hom}_{W[\zeta_p][\![Y]\!]}(W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y},W[\zeta_p][\![Y]\!])$. The equation implies that $T^{p-1}=p\cdot u$ for some unit $u\in W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}$. Hence $T\cdot\mathrm{tr}=$

$T^{p-1}u^{-1}\cdot f$, and $T\cdot(\mathrm{tr}\,-T^{p-2}u^{-1}\cdot f)=0$. But T is not an annihilator, therefore $\mathrm{tr}\,-T^{p-2}u^{-1}\cdot f=0$, and tr is divided by $T^{p-2}=(X-Y)^{p-2}$. □

The morphism q_1 maps the middle components $\cup_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p)$ to Ig_2, and q_2 maps $\cup_{u\in\mathbb{F}_p^\times}\mathrm{Ig}_u(p)$ to Ig_1, so we take $x\in X_0(p)^{V,\mathrm{ord}}(\overline{\mathbb{F}}_p)$, and $y\in X_0(p)^{F,\mathrm{ord}}(\overline{\mathbb{F}}_p)$, such that $y\in p_2\circ p_1^{-1}(x)$. Assume that x corresponds to $V:E^{(p)}\to E$, and y corresponds to $F:E\to E^{(p)}$. Denote $\pi^{-1}(x)=\{x_1,\cdots,x_{p-1}\}$, where x_i corresponds to $(V:E^{(p)}\to E,iP,0)$, and P is a fixed generator of V. Similarly, write $\pi^{-1}(y)=\{y_1,\cdots,y_{p-1}\}$, where y_j corresponds to $(F:E\to E^{(p)},0,jQ)$, and Q is a fixed generator of V. The action of $d\in\mathbb{F}_p^\times$ on $\pi^{-1}(x)$ maps x_i to x_{id}, and maps y_j to $y_{jd^{-1}}$. Therefore we can view

$$([1^{-1}],[2^{-1}],\cdots,[(p-1)^{-1}])\in\bigoplus_{i=1}^{p-1}\widehat{\mathcal{O}}_{\mathcal{X}_1(p),x_i}$$

as a local section of $\mathcal{O}(\mathrm{Ig}_1)$ at $\pi^{-1}(x)$, with $\mathbb{F}_p^\times$ given by $[\,]^{-1}$. Similarly, we can view

$$(1-\zeta_p)\cdot([1],[2],\cdots,[(p-1)])\in\bigoplus_{j=1}^{p-1}\widehat{\mathcal{O}}_{\mathcal{X}_1(p),y_j}$$

as a local section of $\mathcal{O}(\mathrm{Ig}_1)$ at $\pi^{-1}(y)$, with $\mathbb{F}_p^\times$ given by $[\,]^{-1}$. As $\pi^{-1}(y)\subset\mathrm{Ig}_1$, we need the section of $\mathcal{O}(\mathrm{Ig}_1)$ vanishes on the points $\pi^{-1}(y)$, and this is where the factor $(1-\zeta_p)$ comes from.

Denote $z_{i,j}\in\mathcal{X}_1(p^2,p)(\overline{\mathbb{F}}_p)$ to be the point corresponding to $(E\xrightarrow{F}E^{(p)}\xrightarrow{V}E,iP,jQ)$, and $z_i:=\pi'(z_{1,i})\in\mathcal{Z}(\overline{\mathbb{F}}_p)$ to be the point on $\mathcal{Z}$ corresponding to the image of the $\mathbb{F}_p^\times$-orbit $(E\xrightarrow{F}E^{(p)}\xrightarrow{V}E,dP,d^{-1}iQ)$ on $\mathcal{Z}$. From the discussion above, we obtain a rational section $(p_2')^*\mathcal{L}^a\dashrightarrow(p_1')^*\mathcal{L}^a$ locally given by

$$(\frac{1}{(1-\zeta_p)[1]},\frac{1}{(1-\zeta_p)[2]},\cdots,\frac{1}{(1-\zeta_p)[p-1]})^a\in\bigoplus_{i=1}^{p-1}\mathrm{Frac}(\widehat{\mathcal{O}}_{\mathcal{Z},z_i}).$$

Denote $z:=p_1^{-1}(x)\cap p_2^{-1}(y)$ to be the point on $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$ corresponding to $(E\xrightarrow{F}E^{(p)}\xrightarrow{V}E)$.

Lemma 5.11. *Consider the map induced by the normalization: $\nu^\#:\widehat{\mathcal{O}}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]},z}\to\bigoplus_{i=1}^{p-1}\widehat{\mathcal{O}}_{\mathcal{Z},z_i}$, we can arrange the coordinates such that it is identified with*

$$W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}\to\prod_{i=1}^{p-1}W[\zeta_p][\![X,Y]\!]/((1+X)-\zeta_p^i(1+Y)),\tag{3}$$

where $W[\zeta_p][\![X,Y]\!]/((1+X)-\zeta_p^i(1+Y))$ is the local ring corresponding to the point $z_i\in\mathcal{Z}(\overline{\mathbb{F}}_p)$, for $i=1,2,\cdots,p-1$.

Proof. This follows from a careful analysis of the Serre–Tate coordinates. □

Lemma 5.12. *Consider the evaluation map*

$$\mathrm{ev} : \mathbb{Z}_p[\zeta_p][T] \to \prod_{i=1}^{p-1} \mathbb{Z}_p[\zeta_p], P(T) \mapsto (P(\zeta_p), P(\zeta_p^2), \cdots, P(\zeta_p^{p-1})).$$

The element $(\frac{1+\zeta_p+\cdots+\zeta_p^{p-1}}{[i]})_{i=1,2,\cdots,p-1} \in \prod_{i=1}^{p-1} \mathbb{Z}_p[\zeta_p]$ *lies in the image of* ev.

Proof. We first show that the inverse $(\frac{[i]}{1+\zeta_p+\cdots+\zeta_p^{p-1}})_{i=1,2,\cdots,p-1} \in \mathrm{Im}(\mathrm{ev})$. Consider the polynomial $P(T) := \sum_{i=1}^{p-1}((\frac{[i]}{1+\zeta_p+\cdots+\zeta_p^{p-1}}) \cdot \prod_{j\neq i}(\frac{T-\zeta_p^j}{\zeta_p^i-\zeta_p^j}))$, it suffices to show that the coefficients of P are integral. Let $\Phi_p(T) := \frac{T^p-1}{T-1}$ be the cyclotomic polynomial, then $\prod_{j\neq i}(\frac{T-\zeta_p^j}{\zeta_p^i-\zeta_p^j}) = \frac{\Phi_p(T)}{(T-\zeta_p^i)\Phi_p'(\zeta_p^i)} = \frac{\Phi_p(T)\zeta_p^i(\zeta_p^i-1)}{p(T-\zeta_p^i)}$.

$$P(T) = \sum_{i=1}^{p-1} \frac{\Phi_p(T)(\zeta_p-1)}{p} \cdot \frac{\zeta_p^i[i]}{T-\zeta_p^i} = \frac{\Phi_p(T)(\zeta_p-1)}{p} \cdot \sum_{i=1}^{p-1} \frac{[i]}{\zeta_p^{-i}T-1}.$$

Since $[i] \equiv i \pmod p$, it suffices to show that the valuation of the coefficients of $\sum_{i=1}^{p-1} \frac{i}{1-\zeta_p^{-i}T}$ are no less than $\mathrm{val}_\varpi(\frac{p}{\zeta_p-1}) = p-2$. We have $\sum_{i=1}^{p-1} \frac{i}{1-\zeta_p^{-i}T} = \sum_{n=0}^{\infty}(\sum_{i=1}^{p-1} i(\zeta_p^{-i})^n)T^n$. And

$$\sum_{i=1}^{p-1} i(\zeta_p^{-i})^n = \begin{cases} \frac{p}{\zeta_p^{-n}-1}, & \text{if } p \nmid n, \\ \frac{p(p-1)}{2} & \text{if } p \mid n. \end{cases}$$

Therefore $P(T) \in \mathbb{Z}_p[\zeta_p][T]$. The map ev factors through $\mathbb{Z}_p[\zeta_p][T]/(\Phi_p(T))$, whose maximal ideals are $(\zeta_p - 1, T - \zeta_p^i)$, for $i = 1, 2, \cdots, p-1$. Since $\frac{[i]}{1+\zeta_p+\cdots+\zeta_p^{p-1}} \equiv 1 \pmod \varpi$, $P(T)$ does not belong to any of the maximal ideals, therefore $P(T)$ is a unit in $\mathbb{Z}_p[\zeta_p][T]/(\Phi_p(T))$, and there exists $Q(T) = P(T)^{-1} \in \mathbb{Z}_p[\zeta_p][T]/(\Phi_p(T))$. The image of $Q(T)$ under ev is then $(\frac{1+\zeta_p+\cdots+\zeta_p^{p-1}}{[i]})_{i=1,2,\cdots,p-1} \in \prod_{i=1}^{p-1} \mathbb{Z}_p[\zeta_p]$. □

Now we can prove our main theorem.

Proof of 5.6. We first show that the rational section $U_p : (p_2')^*\mathcal{L}^{\otimes a} \dashrightarrow (p_1')^*\mathcal{L}^{\otimes a} \otimes \nu^* p_1^! \mathcal{O}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}}$ is regular and descent to $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$. Locally at the points z_i, this section is given by $(\frac{1}{(1-\zeta_p)[1]}, \frac{1}{(1-\zeta_p)[2]}, \cdots, \frac{1}{(1-\zeta_p)[p-1]})^a$ times the local trace map $((\nu^* \mathrm{tr}_{p_1})_{z_i})_{i=1,2,\cdots,p-1}$. By 5.10, the trace map has a local factor $(X-Y)^{p-2}$. Therefore the rational section is given by

$$(\frac{1}{(1-\zeta_p)[1]}, \frac{1}{(1-\zeta_p)[2]}, \cdots, \frac{1}{(1-\zeta_p)[p-1]})^a \cdot (X-Y)^{p-2} \in \mathrm{Frac}(\widehat{\mathcal{O}}_{\mathcal{Z},z_i}),$$

multiply with $\nu^*(\frac{\mathrm{tr}_{p_1}}{(X-Y)^{p-2}})$, which is a regular section that descent to $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$. It suffices to show that $(\frac{1}{(1-\zeta_p)[1]}, \frac{1}{(1-\zeta_p)[2]}, \cdots, \frac{1}{(1-\zeta_p)[p-1]})^a \cdot (X-Y)^{p-2} \in$

$\mathrm{Im}(\widehat{\mathcal{O}}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]},z} \to \bigoplus_{i=1}^{p-1} \widehat{\mathcal{O}}_{\mathcal{Z},z_i})$. By 5.11, $(\frac{1}{(1-\zeta_p)[1]}, \frac{1}{(1-\zeta_p)[2]}, \cdots, \frac{1}{(1-\zeta_p)[p-1]})^a \cdot (X-Y)^{p-2} = -(\frac{[i](1+Y)}{1+\zeta_p+\cdots+\zeta_p^{i-1}})_{i=1,2,\cdots,(p-1)} \cdot (X-Y)^{p-2-a} \in \prod_{i=1}^{p-1} W[\zeta_p][\![X,Y]\!]/((1+X)-\zeta_p^i(1+Y))$. Take $Q(T)$ in 5.12, then $-(1+Y)(X-Y)^{p-2-a}Q(\frac{1+X}{1+Y}) \in W[\zeta_p][\![X,Y]\!]/\frac{(1+X)^p-(1+Y)^p}{X-Y}$ maps to the element we need under $\nu^{\#}$.

The rest of the proof is similar to the proof of 3.2. Define a rational cohomological correspondence $U_p : p_2^*(\omega_1 \otimes \mathcal{L}^{\otimes a}) \dashrightarrow p_1^!(\omega_1 \otimes \mathcal{L}^{\otimes a})$ to be the composition of $\frac{1}{p}U_p^{\mathrm{naive}} : p_2^*\omega_1 \dashrightarrow p_1^*\omega_1$, where U_p^{naive} is the morphism induced by the universal isogeny, and $f : p_2^*\mathcal{L}^a \to p_1^!\mathcal{L}^a$, which we just defined. In order to prove U_p is regular, we only need to check on ordinary points on special fiber and the generic fiber. U_p is obviously regular over the generic fiber by our construction, and for points on the special fiber, we follow the proof in 3.2.

1. If $x \in X_0(2,0)$, then the trace map has a local factor p, $U_p = U_p^{\mathrm{naive}} \cdot (\frac{1}{p}f)$ is regular at x.
2. If $x \in X_0(1,1)$, then $\frac{1}{p}U_p^{\mathrm{naive}}$ is regular.
3. If $x \in X_0(0,2)$, then $\frac{1}{p}U_p^{\mathrm{naive}}$ is regular.

□

We can also consider the transpose correspondence by switching p_1, p_2, and construct a rational correspondence $U_p^t : p_1^*(\omega_0 \otimes \mathcal{L}^{-a}) \dashrightarrow p_2^!((\omega_0 \otimes \mathcal{L}^{-a})$ for $a = 0, 1, \cdots, (p-2)$ in the same way. Since $p \in \varpi^{p-1} \cdot \text{unit}$, the universal isogeny induces an isomorphism $\omega_0 \otimes \omega_1^{-1} = \mathcal{O}_{\mathcal{X}_1(p)}((p-1)\mathrm{Ig}_1)$. We obtain that $\mathcal{L}^{p-1} = \pi_*\mathcal{O}_{\mathcal{X}_1(p)}((p-1)\mathrm{Ig}_1)(p-1) \simeq \pi_*(\omega_0 \otimes \omega_1^{-1})(0) = \omega_0 \otimes \omega_1^{-1}$. Therefore $\omega_0 \otimes \mathcal{L}^{-a} \simeq \omega_1 \otimes \mathcal{L}^{(p-1-a)}$.

Corollary 5.13. *The transpose cohomological correspondence $U_p^t : p_1^*(\omega_0 \otimes \mathcal{L}^{-a}) \to p_2^!((\omega_0 \otimes \mathcal{L}^{-a})$ is a regular map for $a = 0, 1, \cdots, p-2$.*

Proof. The proof is the same as 5.6. □

Recall that we have an Atkin-Lehner involution $\omega : \mathcal{X}_0(p) \to \mathcal{X}_0(p)$, we can also define an Atkin-Lehner involution for the modular curve $\omega : \mathcal{X}_1(p) \to \mathcal{X}_1(p)$ given by:

$$(E \underset{\phi^\vee}{\overset{\phi}{\rightleftarrows}} E', P, Q) \longmapsto (E' \underset{\phi}{\overset{\phi^\vee}{\rightleftarrows}} E, Q, P) \ .$$

The action of $d \in \mathbb{F}_p^\times$ satisfies: $\omega(d \cdot (E \underset{\phi^\vee}{\overset{\phi}{\rightleftarrows}} E', P, Q)) = d^{-1} \cdot (E' \underset{\phi}{\overset{\phi^\vee}{\rightleftarrows}} E, Q, P)$.

We have a commutative diagram:

$$\begin{array}{ccc} \mathcal{X}_1(p) & \xrightarrow{\omega} & \mathcal{X}_1(p) \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ \mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]} & \xrightarrow{\omega} & \mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]} \end{array} \ .$$

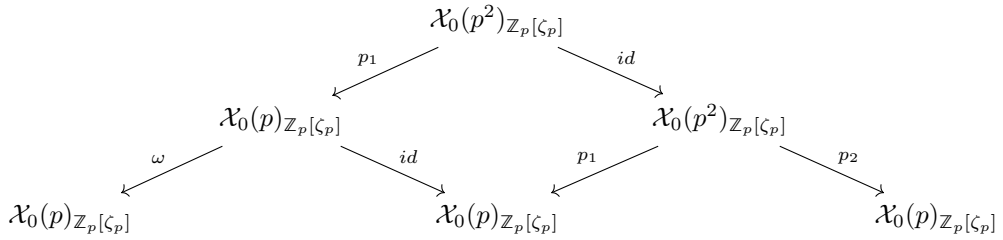
Since $\omega^*(\mathcal{O}(a\mathrm{Ig}_1)) = \mathcal{O}(a\mathrm{Ig}_2)$, by the proper base change theorem, we obtain that $\omega^*\mathcal{L} = \omega^*(\pi_*(\mathcal{O}(\mathrm{Ig}_1))(1)) = \pi_*(\omega^*\mathcal{O}(\mathrm{Ig}_1))(-1) = \pi_*(\mathcal{O}(\mathrm{Ig}_2))(-1) = \pi_*(\mathcal{O}(-\mathrm{Ig}_1))(-1) = \mathcal{L}^{-1}$. We also have $\omega^*(\omega_1 \otimes \mathcal{L}^a) = \omega_0 \otimes \mathcal{L}^{-a}$. In terms of the relation between U_p and U_p^t, we have the following propositions.

Proposition 5.14. *For $a = 0, 1, \cdots, p-1$, we have a commutative diagram:*

$$\begin{array}{ccc} \mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_1 \otimes \mathcal{L}^a) & \xrightarrow{\omega^*} & \mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_0 \otimes \mathcal{L}^{-a}) \\ \downarrow{\scriptstyle U_p} & & \downarrow{\scriptstyle U_p^t} \\ \mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_1 \otimes \mathcal{L}^a) & \xrightarrow[\omega^*]{} & \mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_0 \otimes \mathcal{L}^{-a}) \end{array}$$

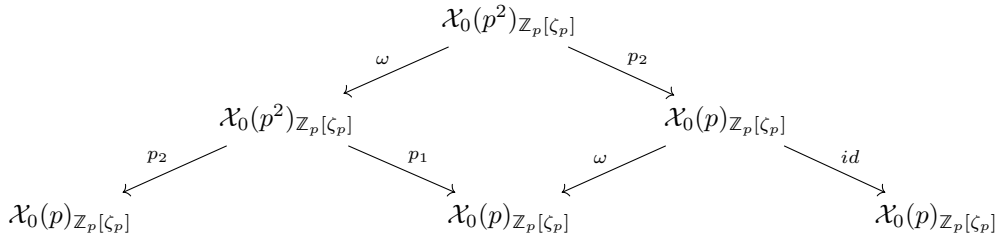
.

Proof. We can view $U_p^t \circ \omega^*$ as a cohomological correspondence, which is the composition of two cohomological correspondences:



,

where the first cohomological correspondence is the trivial map $\omega^*(\omega_1 \otimes \mathcal{L}^a) \xrightarrow{\sim} id^*(\omega_0 \otimes \mathcal{L}^{-a})$, which induces the pull back map ω^*. The second cohomological correspondence is $U_p^t : p_1^*(\omega_0 \otimes \mathcal{L}^{-a}) \to p_2^!(\omega_0 \otimes \mathcal{L}^{-a})$. Similarly, we can view $\omega^* \circ U_p$ as a composition:

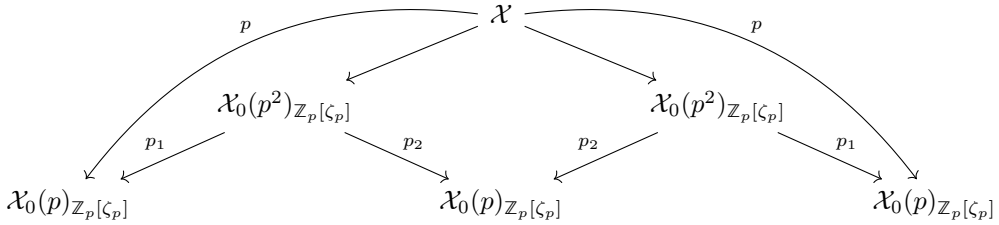


where the first cohomological correspondence is $U_p : p_2^*(\omega_1 \otimes \mathcal{L}^a) \to p_1^!(\omega_1 \otimes \mathcal{L}^a)$, and the second is $\omega^*(\omega_1 \otimes \mathcal{L}^a) \xrightarrow{\sim} id^*(\omega_0 \otimes \mathcal{L}^{-a})$. It's easy to check from the moduli description that the two composed cohomological correspondence diagrams are actually the same. □

Notice that $\omega_0 \otimes \mathcal{L}^{-a} \simeq \omega_1 \otimes \mathcal{L}^{p-1-a}$, therefore for $a = 1, 2, \cdots, p-2$, we have both U_p-action and U_p^t-action on $\mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_1 \otimes \mathcal{L}^a)$ induced by the cohomological correspondences.

Proposition 5.15. *$U_p \circ U_p^t = U_p^t \circ U_p = id$ as operators on $\mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_1 \otimes \mathcal{L}^a)$, for $a = 1, 2, \cdots, p-2$.*

Proof. We show that $U_p \circ U_p^t = id$, and the proof for the opposite one is similar. Consider $U_p \circ U_p^t$ as a new cohomological correspondence:



, where the upper diagram is Cartesian. Since $U_p = \frac{1}{p} U_p^{\mathrm{naive}}$, and $U_p^t = \frac{1}{p} U_p^{t,\mathrm{naive}}$, the composition $U_p \circ U_p^t$ is $\frac{1}{p^2} \operatorname{tr}_p : \mathcal{O}_{\mathcal{X}} \to p^! \mathcal{O}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}}$ times with $id : p^*(\omega_1 \otimes \mathcal{L}^a) \to p^*(\omega_1 \otimes \mathcal{L}^a)$. Therefore on cohomology, $U_p \circ U_p^t = \frac{1}{p^2} \deg(p) = \frac{1}{p^2} \deg(p_1) \cdot \deg(p_2) = 1$. As $\deg(p_1) = \deg(p_2) = p$. □

A direct corollary is the following:

Corollary 5.16. *$e(U_p) \mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_1 \otimes \mathcal{L}^a) = \mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_1 \otimes \mathcal{L}^a)$, which means all the weight one forms with non-trivial Nebentypus are ordinary.*

Remark 5.17. Combine with 3.20, we know that all the weight one modular forms with tame Nebentypus are ordinary. Here we consider the integral structure $\omega_1 \otimes \mathcal{L}^a$ for $a = 0, 1, \cdots, p-2$, and the U_p-operator we defined. This agrees with the classical theorem over $\mathbb{C}$, that all weight one modular forms of level $\Gamma_1(p)$ are ordinary at p.

5.2 Descent of $\mathcal{L}$ and U_p

We show that $\mathcal{L}$ and U_p both descent to $\mathbb{Z}_p$. For a scheme X over a DVR R, denote $\mathrm{VertDiv}(X)$ to be the group of vertical Cartier divisors supported on the special fiber of X.

Lemma 5.18. *Let $\pi : \mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]} \to \mathcal{X}_0(p)$ be the base change morphism, then $\mathrm{VertDiv}(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}) = \langle \pi^* \mathrm{VertDiv}(\mathcal{X}_0(p)), \operatorname{div}(\varpi) = [X_0(p)^F] + [X_0(p)^V] \rangle$. Therefore, we have a short exact sequence*

$$0 \to (\omega_1 \otimes \omega_0^{-1})^{\mathbb{Z}} \to \mathrm{Pic}(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}) \to \mathrm{Pic}(\mathcal{X}_0(p)_{\mathbb{Q}_p[\zeta_p]}).$$

Proof. This is by explicit calculation of the local ring at supersingular points. □

Corollary 5.19. *$\mathcal{L}$ descent to $\mathcal{X}_0(p)$.*

Proof. Since the modular curve $\mathcal{X}_1(p)_{\mathbb{Q}_p}$ already exists over $\mathbb{Q}_p$, it's clear that over the generic fiber $\mathcal{L}_{\mathbb{Q}_p[\zeta_p]}$ descents to $\mathcal{L}_{\mathbb{Q}_p}$. Since $\mathcal{X}_0(p)$ is regular, there is a short exact sequence obtained by taking closure of Cartier divisors on the generic fiber:

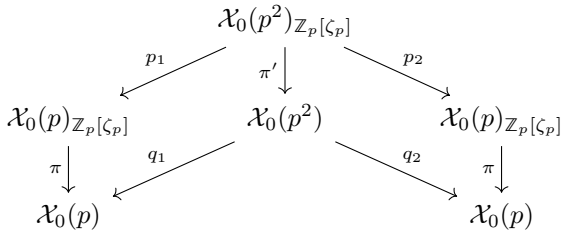
$$0 \to (\omega_1 \otimes \omega_0^{-1})^{\mathbb{Z}} \to \mathrm{Pic}(\mathcal{X}_0(p)) \to \mathrm{Pic}(\mathcal{X}_0(p)_{\mathbb{Q}_p}) \to 0.$$

Therefore, we have a line bundle $\mathcal{L}'$, such that $\mathcal{L}'_{\mathbb{Q}_p} \simeq \mathcal{L}_{\mathbb{Q}_p}$. By 5.18, $\pi^*\mathcal{L}' \otimes \mathcal{L}^{-1} \simeq (\omega_1 \otimes \omega_0^{-1})^n$ for some integer n. So the required descent is given by $\mathcal{L}' \otimes \omega_0^n \otimes \omega_1^{-n}$. □

From now on, we will denote $\mathcal{L}$ for the line bundle over $\mathcal{X}_0(p)$. For simplicity, we denote ω_a to be the line bundle $\omega_1 \otimes \mathcal{L}^a$ on $\mathcal{X}_0(p)$, and on $\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}$, when there is no ambiguity.

Remark 5.20. The same argument also tells us that $\mathcal{L}$ is only differed from the construction in [DS22] by some power of $\mathcal{O}(X_0(p)^F)$.

Consider the diagram:



By base change theorems, we have $p_2^*\omega_a = \pi^* p_2^* \omega_a = (\pi')^* q_2^* \omega_a$, and $p_1^! \omega_a = p_1^! \pi^* \omega_a = (\pi')^* q_1^! \omega_a$.

Lemma 5.21. *Let L_1, L_2 be line bundles over $\mathcal{X}_0(p^2)$, then $\phi \in \mathrm{Hom}((\pi')^*L_1, (\pi')^*L_2)$ descents to $\mathcal{X}_0(p^2)$ if and only if $g \cdot \phi = \phi$ for all $g \in \mathbb{F}_p^\times = \mathrm{Gal}(\mathbb{Z}_p[\zeta_p]/\mathbb{Z}_p)$.*

Proof. We have

$$\begin{aligned}
\mathrm{Hom}\big((\pi')^*L_1, (\pi')^*L_2\big) &\cong \mathrm{Hom}\Big(\mathcal{O}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}}, (\pi')^*(L_2 \otimes L_1^{-1})\Big) \\
&\overset{\text{projection formula}}{\cong} \mathrm{Hom}\Big(\mathcal{O}_{\mathcal{X}_0(p^2)}, L_2 \otimes L_1^{-1} \otimes \pi'_* \mathcal{O}_{\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}}\Big) \\
&\cong \mathrm{Hom}\Big(\mathcal{O}_{\mathcal{X}_0(p^2)}, (L_2 \otimes L_1^{-1}) \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\zeta_p]\Big) \\
&\cong \mathrm{Hom}\big(\mathcal{O}_{\mathcal{X}_0(p^2)}, L_2 \otimes L_1^{-1}\big) \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\zeta_p].
\end{aligned}$$

Since $p - 1 = |\mathrm{Gal}(\mathbb{Z}_p[\zeta_p]/\mathbb{Z}_p)|$ is a unit in $\mathbb{Z}_p$, averaging for the Galois action gives that $\mathrm{Hom}\big((\pi')^*L_1, (\pi')^*L_2\big)^{\mathbb{F}_p^\times} = \mathrm{Hom}(L_1, L_2)$. □

Theorem 5.22. *The U_p-operators and U_p^t-operators all descent to $\mathbb{Z}_p$.*

Proof. It suffices to show that $g \cdot U_p = U_p$, and $g \cdot U_p^t = U_p^t$ for all $g \in \mathbb{F}_p^\times$. By construction $p_2^*\mathcal{L}$ and $p_1^*\mathcal{L}$ agrees over the generic fiber, and it is clear that $p_2^*\omega_1 \to p_1^!\omega_1$ descents to $\mathbb{Z}_p$. Therefore, over the generic fiber U_p descents to $\mathbb{Q}_p$, and $g \cdot U_p - U_p = 0$ over the generic fiber. Since the scheme is integral, $g \cdot U_p = U_p$ over $\mathcal{X}_0(p^2)_{\mathbb{Z}_p[\zeta_p]}$, and U_p descents to $\mathbb{Z}_p$. The same arguments work for U_p^t. □

From now on, we will simply denote U_p and U_p^t as the cohomological correspondences over $\mathcal{X}_0(p^2)$.

5.3 Cousin complex for weight one modular forms with Nebentypus

We establish the Cousin complex for weight one modular forms with a fixed tame Nebentypus in this section. This will be crucial for our applications to Galois representations. The strategy and techniques are the same as in section3.4.1, so we omit the proofs which are essentially the same to avoid redundancy.

Proposition 5.23. *For each $a = 0, 1, \cdots, p-3$, there is a locally finite action of U_p on $\mathrm{R}\Gamma_{X_0(p)^V}(X_0(p), \omega_a)$ induced by the U_p action on $\mathrm{R}\Gamma(X_0(p), \omega_a)$. And the natural map $\mathrm{R}\Gamma_c(X_0(p)^{V,\mathrm{ord}}, \omega_a) \to \mathrm{R}\Gamma_{X_0(p)^V}(X_0(p), \omega_a)$ induces an isomorphism between finite dimensional $\mathbb{F}_p$-vector spaces:*

$$e(U_p)\mathrm{R}\Gamma_c(X_0(p)^{V,\mathrm{ord}}, \omega_a) \xrightarrow{\sim} e(U_p)\mathrm{R}\Gamma_{X_0(p)^V}(X_0(p), \omega_a).$$

Proof. The proof is identical to that of Proposition3.8. □

By Corollary4.15, for $a = 0, 1, \cdots, p-2$, $\omega_a \mid_{X_0(p)^{V,\mathrm{ord}}} \simeq \omega^{1+a}$, and

$$e(U_p)\mathrm{R}\Gamma_c(X_0(p)^{V,\mathrm{ord}}, \omega_a) \simeq e(F)\mathrm{R}\Gamma_c(X^{\mathrm{ord}}, \omega^{1+a}) = e(F)\mathrm{R}\Gamma_c(X^{\mathrm{ord}}, \omega^{2+a-p}).$$

Similarly, $e(U_p)\mathrm{R}\Gamma(X_0(p)^{F,\mathrm{ord}}, \omega_a) \simeq e(U_p)\mathrm{R}\Gamma(X^{\mathrm{ord}}, \omega^{p-a})$. For the case ω_{p-2}, the proof of Proposition3.8 fails, because the classicality theorem of spherical level fails for $\mathrm{H}^0(X^{\mathrm{ord}}, \omega^2)$ and $\mathrm{H}^1_c(X^{\mathrm{ord}}, \mathcal{O})$, see [BP22]Corollary 4.5. Therefore we simply use the exact triangle

$$\mathrm{R}\Gamma_{X_0(p)^V}(X_0(p), \omega_{p-2}) \to \mathrm{R}\Gamma(X_0(p), \omega_{p-2}) \to \mathrm{R}\Gamma(X_0(p)^{F,\mathrm{ord}}, \omega_{p-2}) \xrightarrow{+1},$$

or the exact triangle

$$\mathrm{R}\Gamma_c(X_0(p)^{V,\mathrm{ord}}, \omega_{p-2}) \to \mathrm{R}\Gamma(X_0(p), \omega_{p-2}) \to \mathrm{R}\Gamma(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2}) \xrightarrow{+1}.$$

In order to make $\mathrm{H}^0_{X_0(p)^V}(X_0(p), \omega_{p-2})$ and $\mathrm{H}^1(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2})$ vanish, we consider a cuspidal sheaf instead, to replace the line bundle ω_{p-2}. This will not affect our subsequent discussion, since for applications to Galois representations we will always localize at a non-Eisenstein maximal ideal of the Hecke algebra. Denote D to be the boundary divisor of $\mathcal{X}_0(p)$. It's special fiber $D_{\mathbb{F}_p} = D^F \sqcup D^V$,

is the disjoint union of the cusps of X^F and X^V. Then $D = D_1 \amalg D_2$, where D_1, D_2 are effective Cartier divisors on $\mathcal{X}_0(p)$ such that D_1 contains D^V, and D_2 contains D^F. Consider the sheaf $\omega_{p-2}(-D_1)$. By construction, $\omega_{p-2}(-D_1)\mid_{X_0(p)^V} \simeq \omega^{p-1}(-D^V)$, and $\omega_{p-2}(-D_1)\mid_{X_0(p)^F} \simeq \omega^2$. Following from the proof of 3.8, $\mathrm{H}^0_{X_0(p)^V}(X_0(p), \omega_{p-2}(-D_1)) = \mathrm{H}^0(X_0(p)^V, \omega_{p-2}(-D_1)\mid_{X_0(p)^V} (-SS)) \simeq \mathrm{H}^0(X, \mathcal{O}(-\mathrm{cusp})) = 0$. For $\mathrm{H}^1(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2}(-D_1))$, denote $\mathcal{I}$ to be the ideal sheaf of $X_0(p)^F$ on $X_0(p)$, and $\mathrm{H}^1(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2}) \simeq \lim_n \mathrm{H}^1(X_0(p), \omega_{p-2}(-D_1)/\mathcal{I}^n\omega_{p-2}(-D_1))$. We have

$$\mathrm{H}^1(X_0(p), \omega_{p-2}(-D_1)/\mathcal{I}\omega_{p-2}(-D_1)) = \mathrm{H}^1(X_0(p)^F, \omega_{p-2}(-D_1)) = \mathrm{H}^1(X, \omega^2) = 0,$$

the last equality follows from Serre duality. Therefore, by the short exact sequence

$$0 \to \mathcal{I}^n\omega_{p-2}(-D_1)/\mathcal{I}^{n+1}\omega_{p-2}(-D_1)) \to \omega_{p-2}(-D_1)/\mathcal{I}^{n+1}\omega_{p-2}(-D_1)) \to \omega_{p-2}(-D_1)/\mathcal{I}^n\omega_{p-2}(-D_1)) \to 0,$$

we have

$$\mathrm{H}^1(X_0(p), \omega_{p-2}(-D_1)/\mathcal{I}^n\omega_{p-2}(-D_1)) = 0$$

for all n, and

$$\mathrm{H}^1(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2}(-D_1)) = 0.$$

We will simply denote ω_{p-2} to be the cuspidal sheaf $\omega_{p-2}(-D_1)$ from now on. The following lemma tells us there doesn't exist a Cousin complex for ω_{p-2} with the same shape as before.

Lemma 5.24. *The natural maps* $\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2}) \to \mathrm{H}^0(X_0(p)^{F,\mathrm{ord}}, \omega_{p-2})$, *and* $\mathrm{H}^1_c(X_0(p)^{V,\mathrm{ord}}, \omega_{p-2}) \to \mathrm{H}^1_{X_0(p)^V}(X_0(p), \omega_{p-2})$ *does not induce an ismorphism after taking the ordinary projector* $e(U_p)$.

Proof. We give the proof for $\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2})$, and the other is similar. Let $\omega'_{p-2} := \omega_{p-2} \otimes \omega_1 \otimes \omega_0^{-1}$, be the sheaf with twisted integral structure. The same argument as in Proposition3.8 gives an isomorphism

$$e(U_p)\mathrm{R}\Gamma_c(X_0(p)^{V,\mathrm{ord}}, \omega'_a) \xrightarrow{\sim} e(U_p)\mathrm{R}\Gamma_{X_0(p)^V}(X_0(p), \omega'_a).$$

By comparing the two exact triangles as in 3.4.1, we get an isomorphism

$$e(U_p)\mathrm{R}\Gamma(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}'_{p-2}) \simeq e(U_p)\mathrm{R}\Gamma(X_0(p)^{F,\mathrm{ord}}, \omega'_{p-2}).$$

Consider the following commutative diagram:

$$\begin{array}{ccc} e(U_p)\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}'_{p-2}) & \xrightarrow{\simeq} & e(U_p)\mathrm{H}^0(X_0(p)^{F,\mathrm{ord}}, \omega'_{p-2}) \\ \uparrow & & \simeq\uparrow \\ e(U_p)\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2}) & \longrightarrow & e(U_p)\mathrm{H}^0(X_0(p)^{F,\mathrm{ord}}, \omega_{p-2}) \end{array}$$

The universal isogeny induces a map $\omega_0 \to \omega_1$, and the vertical maps are induced by $\alpha : \omega_{p-2} \to \omega'_{p-2}$. It's clear that the $\omega_{p-2} \to \omega'_{p-2}$ is an isomorphism over $X_0(p)^{F,\mathrm{ord}}$, and therefore the right vertical arrow is an isomorphism. Consider the closed embedding $i : X_0(p)^V \hookrightarrow X_0(p)$, then $\ker(\alpha) = i_* i^* \omega_{p-2}(-SS) \simeq i_* \mathcal{O}(-D^V)$. By [Sta26, Lemma 30.23.4, Tag 0881], we have an exact sequence $0 \to i_* \widehat{\mathcal{O}(-D^V)} \to \hat{\omega}_{p-2} \to \hat{\omega}'_{p-2}$, and the kernel of the left vertical arrow is therefore $e(U_p)\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, i_* \widehat{\mathcal{O}(-D^V)})$. Using the long exact sequence, we have:

$$0 \to \mathrm{H}^0_c(X_0(p)^{V,\mathrm{ord}}, i_*\mathcal{O}(-D^V)) \to \mathrm{H}^0(X_0(p), i_*\mathcal{O}(-D^V)) \to \mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, i_* \widehat{\mathcal{O}(-D^V)})$$
$$\to \mathrm{H}^1_c(X_0(p)^{V,\mathrm{ord}}, i_*\mathcal{O}(-D^V)) \to \mathrm{H}^1(X_0(p), i_*\mathcal{O}(-D^V)).$$

The first term is 0, as $X_0(p)^{V,\mathrm{ord}}$ is affine. $\mathrm{H}^0(X_0(p), i_*\mathcal{O}(-D^V)) = \mathrm{H}^0(X_0(p)^V, \mathcal{O}(-D^V)) = 0$. Therefore, $e(U_p)\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, i_* \widehat{\mathcal{O}(-D^V)}) = \ker(e(U_p)\mathrm{H}^1_c(X_0(p)^{V,\mathrm{ord}}, \mathcal{O}(-D^V)) \to e(U_p)\mathrm{H}^1(X_0(p)^V, \mathcal{O}(-D^V))) \neq 0$. The kernel is non zero because the classicality theorem fails for weight 0. □

Remark 5.25. Although the natural maps between them are not an isomorphism, we can show that $\dim_{\mathbb{F}_p} e(U_p)\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F}, \hat{\omega}_{p-2}) = \dim_{\mathbb{F}_p} e(U_p)\mathrm{H}^0(X_0(p)^{F,\mathrm{ord}}, \omega_{p-2})$, and $\dim_{\mathbb{F}_p} e(U_p)\mathrm{H}^1_c(X_0(p)^{V,\mathrm{ord}}, \omega_{p-2}) = \dim_{\mathbb{F}_p} e(U_p)\mathrm{H}^1_{X_0(p)^V}(X_0(p), \omega_{p-2})$. Denote χ to be the Euler characteristic, then it suffices to show that

$$\chi(\mathrm{R}\Gamma(X_0(p), \omega_{p-2})) = \chi(e(U_p)\mathrm{R}\Gamma(X_0(p), \omega_{p-2}))$$
$$= \dim_{\mathbb{F}_p} e(U_p)\mathrm{H}^0(X_0(p)^{F,\mathrm{ord}}, \omega_{p-2}) - \dim_{\mathbb{F}_p} e(U_p)\mathrm{H}^1_c(X_0(p)^{V,\mathrm{ord}}, \omega_{p-2}),$$

where the first equality follows from the ordinarity of weight one forms. The line bundle ω_{p-2} admits a short exact sequence $0 \to \mathcal{O}_{X_0(p)^V}(-D^V) \to \omega_{p-2} \to \omega^2_{X_0(p)^F} \to 0$. Therefore, $\chi(\mathrm{R}\Gamma(X_0(p), \omega_{p-2}) = 0$ by Serre duality. On the other hand, the right hand side is also 0, by Serre duality of p-adic modular forms, see [BP22]Theorem 4.18.

Recall that we denote $\mathfrak{X}_0(p)$ to be the formal completion of $\mathcal{X}_0(p)$, and it's ordinary locus $\mathfrak{X}_0(p)^{\mathrm{ord}} = \mathfrak{X}_0(p)^{F,\mathrm{ord}} \sqcup \mathfrak{X}_0(p)^{V,\mathrm{ord}}$. As a Corollary we have the following Cousin complexes:

Corollary 5.26. *1. For $a = 0, 1, \cdots, p-3$, we have an U_p-equivariant quasi-isomorphism:*

$$e(U_p)\mathrm{R}\Gamma(\mathcal{X}_0(p), \omega_a) = [e(U_p)\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}}, \omega_a) \to e(F)\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}}, \omega_a)].$$

2. For $a = 0, 1, \cdots, p-3$, the above complex mod p gives an U_p-equivariant quasi-isomorphism:

$$e(U_p)\mathrm{R}\Gamma(X_0(p), \omega_a) = [e(U_p)\mathrm{H}^0(X^{\mathrm{ord}}, \omega^{p-a}) \to e(F)\mathrm{H}^1_c(X^{\mathrm{ord}}, \omega^{1+a})].$$

3. *For ω_{p-2}, $e(U_p)\mathrm{R}\Gamma(\mathcal{X}_0(p),\omega_{p-2})$ is quasi-isomorphic to the following two U_p-equivariant complexes:*

$$\begin{array}{ccc} e(U_p)\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega_{p-2}) & \longrightarrow & e(U_p)\mathrm{H}^1_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p),\omega_{p-2}) \\ \uparrow & & \uparrow \\ e(U_p)\mathrm{H}^0(\widehat{\mathfrak{X}_0(p)}_{\mathfrak{X}_0(p)^F},\hat{\omega}_{p-2}) & \longrightarrow & e(F)\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega_{p-2}) \end{array}$$

Moreover, the terms of the bottom are identified with a full rank lattice of the top terms.

4. *$e(U_p)\mathrm{R}\Gamma(X_0(p),\omega_{p-2})$ is quasi-isomorphic to the above complexes mod p:*

$$\begin{array}{ccc} e(U_p)\mathrm{H}^0(X^{\mathrm{ord}},\omega^2) & \longrightarrow & e(U_p)\mathrm{H}^1_{X_0(p)^V}(X_0(p),\omega_{p-2}) \\ \uparrow & & \uparrow \\ e(U_p)\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^F},\hat{\omega}_{p-2}) & \longrightarrow & e(F)\mathrm{H}^1_c(X^{\mathrm{ord}},\mathcal{O}(-cusp)) \end{array}$$

But the vertical arrows are no longer injective.

Proof. We only prove 3, and the rest of the proof are the same as 3.9. Since the two horizontal complexes are quasi isomorphic, we only need to prove the left arrow is injective by diagram chasing. Over $\mathcal{X}_0(p)$ we have an injective map $0\to\omega_{p-2}\to\omega_{p-2}\otimes\omega_1\otimes\omega_0^{-1}$ by the divisor short exact sequence. Same as the proof of 5.24, we obtain that the left vertical arrow is injective. It is actually a lattice because both terms are finite free $\mathbb{Z}_p$-modules, and they have the same dimension mod p by 5.24. □

For $a=1,2,\cdots p-2$, we can also consider the Cousin complex associated to the transpose operator U_p^t, the U_p^t-action over $\mathfrak{X}_0(p)^{V,\mathrm{ord}}$ is identified with $\langle p\rangle^{-1}U_p$, and the U_p^t-action over $\mathfrak{X}_0(p)^{F,\mathrm{ord}}$ is identified with $\langle p\rangle^{-1}F$ see[BP22, 3.2]. For ω_1 consider the sheaf $\omega_1(-D_2)$ instead. The same arguments as above give the following.

Corollary 5.27. *1. For $2\le a\le p-2$, we have an U_p^t-equivariant quasi-isomorphism:*

$$e(U_p^t)\mathrm{R}\Gamma(\mathcal{X}_0(p),\omega_a)=[e(\langle p\rangle^{-1}U_p)\mathrm{H}^0(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega_a)\to e(\langle p\rangle^{-1}F)\mathrm{H}^1_c(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega_a)].$$

2. *For $2\le a\le p-2$, the above complex mod p gives an U_p^t-equivariant quasi-isomorphism:*

$$e(U_p^t)\mathrm{R}\Gamma(X_0(p),\omega_a)=[e(\langle p\rangle^{-1}U_p)\mathrm{H}^0(X^{\mathrm{ord}},\omega^{1+a})\to e(\langle p\rangle^{-1}F)\mathrm{H}^1_c(X^{\mathrm{ord}},\omega^{p-a})].$$

3. *or $a = 1$, $e(U_p)\mathrm{R}\Gamma(\mathcal{X}_0(p), \omega_1)$ is quasi-isomorphic to the following two U_p^t-equivariant complexes:*

$$\begin{array}{ccc} e(\langle p\rangle^{-1}U_p)\mathrm{H}^0(\mathfrak{X}_0(p)^{V,\mathrm{ord}}, \omega_1) & \longrightarrow & e(U_p^t)\mathrm{H}^1_{\mathfrak{X}_0(p)^F}(\mathfrak{X}_0(p), \omega_1) \\ \uparrow & & \uparrow \\ e(U_p^t)\mathrm{H}^0(\widehat{\mathfrak{X}_0(p)}_{\mathfrak{X}_0(p)^V}, \hat{\omega}_1) & \longrightarrow & e(\langle p\rangle^{-1}F)\mathrm{H}^1_c(\mathfrak{X}_0(p)^{F,\mathrm{ord}}, \omega_1) \end{array}$$

Moreover, the terms of the bottom are identified with a full rank lattice of the top terms.

4. *$e(U_p^t)\mathrm{R}\Gamma(X_0(p), \omega_1)$ is quasi-isomorphic to the above complexes mod p:*

$$\begin{array}{ccc} e(\langle p\rangle^{-1}U_p)\mathrm{H}^0(X^{\mathrm{ord}}, \omega^2) & \longrightarrow & e(U_p^t)\mathrm{H}^1_{X_0(p)^V}(X_0(p), \omega_1) \\ \uparrow & & \uparrow \\ e(U_p^t)\mathrm{H}^0(\widehat{X_0(p)}_{X_0(p)^V}, \hat{\omega}_1) & \longrightarrow & e(\langle p\rangle^{-1}F)\mathrm{H}^1_c(X^{\mathrm{ord}}, \mathcal{O}(-cusp)) \end{array}$$

But the vertical arrows are no longer injective.

We can relate the terms of the Cousin complex with the spaces of p-adic modular forms of spherical level constructed in [BP22]. Let $\mathfrak{IG} \to \mathfrak{X}^{\mathrm{ord}}$ be the Igusa tower over the ordinary formal scheme of the modular curve of spherical level. This is a profinite etale, $\mathbb{Z}_p^\times$-cover. Let $\Lambda = \mathbb{Z}_p[\![\mathbb{Z}_p^\times]\!]$ be the Iwasawa algebra, and let $\kappa^{\mathrm{un}} : \mathbb{Z}_p^\times \to \Lambda^\times$ be the universal character. Let $\omega^{\mathrm{un}} = (\mathcal{O}_{\mathfrak{IG}}\hat{\otimes}\Lambda)^{\mathbb{Z}_p^\times}$ be the line bundle on $\mathfrak{X}^{\mathrm{ord}}_\Lambda$. Consider a p-adic weight $\kappa : \Lambda \to \mathbb{Z}_p$, and let $\omega^\kappa = \omega^{\mathrm{un}} \otimes_{\Lambda,\kappa} \mathbb{Z}_p$ to be the line bundle on $\mathfrak{X}^{\mathrm{ord}}$. For $a \in \mathbb{Z}/(p-1)\mathbb{Z}$, let $\kappa_a : \mathbb{Z}_p^\times \to \mathbb{Z}_p^\times$ be the character given by $x \mapsto x \cdot [\bar{x}]^a$, where $\bar{x}$ is the image of x in $\mathbb{F}_p^\times$, and $[\bar{x}]$ is the Teichmuller lift. Let $\kappa_a : \Lambda \to \mathbb{Z}_p$ be the corresponding algebra morphism.

Lemma 5.28. *1. For $a = 0, 1, \cdots, p-3$, the restriction of ω_a to $\mathfrak{X}_0(p)^{F,\mathrm{ord}}$ is isomorphic to $\omega^{\kappa_{-a}}$, and the restriction of ω_a to $\mathfrak{X}_0(p)^{V,\mathrm{ord}}$ is isomorphic to ω^{κ_a}.*

2. The restriction of ω_{p-2} to $\mathfrak{X}_0(p)^{F,\mathrm{ord}}$ is isomorphic to ω^{κ_1}, and the restriction of ω_a to $\mathfrak{X}_0(p)^{V,\mathrm{ord}}$ is isomorphic to $\omega^{\kappa_{-1}}(-D)$, where D is the cusp on $\mathfrak{X}^{\mathrm{ord}}$.

Proof. We proof the statement for ω_a restricted to $\mathfrak{X}_0(p)^{F,\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$, and the other cases are similar. $\omega_a = \omega_1 \otimes \mathcal{L}^a = \omega_0 \otimes \mathcal{L}^{a-(p-1)}$. $\mathfrak{X}_1(p)^{\mathrm{ord}} = \mathfrak{IG}^F \sqcup \mathfrak{IG}^V$, where $\mathfrak{IG}^F$ and $\mathfrak{IG}^V$ are isomorphic to $\mathfrak{IG}_{\mathbb{Z}_p[\zeta_p]}/(1+p\mathbb{Z}_p)$, and is an $\mathbb{F}_p^\times$-cover over $\mathfrak{X}^{F,\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$ and $\mathfrak{X}^{V,\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$, respectively. Let $\chi : \mathbb{Z}_p^\times \to \mathbb{F}_p^\times \xrightarrow{[\cdot]} \mathbb{Z}_p^\times$. Since $\mathcal{L} = \pi_*\mathcal{O}_{\mathcal{X}_1(p)}(\mathrm{Ig}_1)(-1) = \pi_*\mathcal{O}_{\mathcal{X}_1(p)}(-\mathrm{Ig}_2)(-1)$, we obtain that the restriction of $\mathcal{L}$ to $\mathfrak{X}^{F,\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$ is $\pi_*\mathcal{O}_{\mathfrak{IG}^F}(-1) \simeq \omega^{\chi^{-1}}$. Therefore the restriction of ω_a to $\mathfrak{X}_0(p)^{F,\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$ is isomorphic to $\omega^{\kappa_{-a}}$. □

6 Galois representations and local global compatibility

In this section, we construct Galois representations associated with weight-one modular forms with Nebentypus in characteristic p, as well as Galois representations valued in Hecke algebras. We also prove a local-global compatibility result showing that these Galois representations are split at p. This can be viewed as a generalization of the result of [Gro90].

6.1 Local global compatibility for weight one modular forms of level $\Gamma_1(p)$ in characteristic p

First, we need to construct the Hecke operators T_l for $l \neq p$, and also the diamond operators $\langle d \rangle$ for $d \in (\mathbb{Z}/N\mathbb{Z})^\times$ acting on $\mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}, \omega_a)$ and $\mathrm{R}\Gamma(X_0(p), \omega_a)$ for $a = 0, 1, \cdots, p-2$. For prime $l \neq p$, denote $\mathcal{X}_0(l)$ to be the modular curve of level $\Gamma_0(l) \cap \Gamma_1(N)$ over $\mathbb{Z}_p$. $\mathcal{X}_0(l)$ is smooth over $\mathcal{X}_1(N)$. Define $\mathcal{X}_0(pl) := \mathcal{X}_0(p) \times_{\mathcal{X}_1(N)} \mathcal{X}_0(l)$ to be the modular curve of level $\Gamma_0(pl) \cap \Gamma_1(N)$ over $\mathbb{Z}_p$. By definition, $\mathcal{X}_0(pl)$ represents the following moduli problem:

$$\left(\begin{array}{ccc} E_0 & \longrightarrow & E_1 \\ \downarrow & & \downarrow \\ E_0' & \longrightarrow & E_1' \end{array} , \alpha \right),$$

where E_0, E_1, E_0', E_1' are elliptic curves over a $\mathbb{Z}_p$-scheme S, the horizontal maps are p-isogenies, the vertical maps are l-isogenies, the diagram is commutative, and α is a $\Gamma_1(N)$-structure on E_0. Denote α' to be the induced $\Gamma_1(N)$-structure on E_0'.

As usual, we have a Hecke correspondence:

$$\begin{array}{ccccc} & & \mathcal{X}_0(pl) & & \\ & \overset{p_1}{\swarrow} & & \overset{p_2}{\searrow} & \\ \mathcal{X}_0(p) & & & & \mathcal{X}_0(p), \end{array}$$

where p_1 sends the above datum to $(E_0 \to E_1, \alpha)$, and p_2 sends the above datum to $(E_0' \to E_1', \alpha')$. Let $p_2^*\omega_1 \to p_1^*\omega_1$ be the map induced by the universal isogeny $E_1^{univ} \to E_1^{\prime, univ}$.

Lemma 6.1. *Consider the above Hecke correspondence after base change to $\mathbb{Z}_p[\zeta_p]$, then $p_2^*\mathcal{L} \simeq p_1^*\mathcal{L}$.*

Proof. Let $\mathcal{X}_1(p,l) = \mathcal{X}_1(p) \times_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}} \mathcal{X}_0(pl)_{\mathbb{Z}_p[\zeta_p]}$, parameterizing

$$\left(\begin{array}{ccc} E_0 & \overset{\psi}{\longrightarrow} & E_1 \\ {\scriptstyle \phi_0}\downarrow & & \downarrow{\scriptstyle \phi_1} \\ E_0' & \underset{\psi'}{\longrightarrow} & E_1' \end{array} , P, Q \right),$$

where the horizontal maps are degree p isogenies, and the vertical are degree l isogenies, P generator of $\ker\psi$, Q generator of $\ker\psi^\vee$, satisfying $e_\psi(P,Q)=\zeta_p$. Then $e_{\psi'}(\phi_0(P),l^{-1}\phi_1(Q))=e_\psi(P,\phi_1^\vee(l^{-1}\phi_1(Q)))=e_\psi(P,Q)=\zeta_p$. We can consider the maps

$$q_1:\mathcal{X}_1(p,l)\to\mathcal{X}_1(p),\left(\begin{array}{ccc}E_0&\xrightarrow{\psi}&E_1\\ {\scriptstyle\phi_0}\downarrow&&\downarrow{\scriptstyle\phi_1}\\ E_0'&\xrightarrow{\psi'}&E_1'\end{array},P,Q\right)\mapsto(E_0\to E_1,P,Q),$$

$$q_2:\mathcal{X}_1(p,l)\to\mathcal{X}_1(p),\left(\begin{array}{ccc}E_0&\xrightarrow{\psi}&E_1\\ {\scriptstyle\phi_0}\downarrow&&\downarrow{\scriptstyle\phi_1}\\ E_0'&\xrightarrow{\psi'}&E_1'\end{array},P,Q\right)\mapsto(E_0'\to E_1',\phi_0(P),l^{-1}\phi_1(Q)).$$

We have the following diagram:

$$\begin{array}{ccccc}&&\mathcal{X}_1(p,l)&&\\ &\swarrow^{q_1}&\downarrow{\scriptstyle\pi'}&\searrow^{q_2}&\\ \mathcal{X}_1(p)&&\mathcal{X}_0(pl)_{\mathbb{Z}_p[\zeta_p]}&&\mathcal{X}_1(p)\\ \downarrow{\scriptstyle\pi}&\swarrow^{p_1}&&\searrow^{p_2}&\downarrow{\scriptstyle\pi}\\ \mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}&&&&\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]},\end{array}$$

where the two arrows in the upper correspondence are $(\mathbb{Z}/p\mathbb{Z})^\times$-equivariant. $q_2^*(\mathrm{Ig}_1)=q_1^*(\mathrm{Ig}_1)$ as Cartier divisors on $\mathcal{X}_1(p)\times_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}}\mathcal{X}_0(pl)_{\mathbb{Z}_p[\zeta_p]}$, since both of them are identified with the divisor supported on the special fiber such that $E_0\to E_1$ and $E_0'\to E_1'$ are both Frobenius maps. Therefore $p_2^*\mathcal{L}=p_2^*(\pi_*\mathcal{O}(\mathrm{Ig}_1)(-1))=\pi_*'(q_2^*\mathcal{O}(\mathrm{Ig}_1))(-1)=\pi_*'(q_1^*\mathcal{O}(\mathrm{Ig}_1))(-1)=p_1^*(\pi_*\mathcal{O}(\mathrm{Ig}_1)(-1))=p_1^*\mathcal{L}$. □

Composing the morphisms $p_2^*\omega_1\to p_1^*\omega_1$, $p_2^*\mathcal{L}\simeq p_1^*\mathcal{L}$, and the trace map $\mathrm{tr}_{p_1}:\mathcal{O}_{\mathcal{X}_0(pl)_{\mathbb{Z}_p[\zeta_p]}}\to p_1^!\mathcal{O}_{\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}}$, we obtain $T_l^{\mathrm{naive}}:p_2^*\omega_a\to p_1^!\omega_a$. Let $T_l:=l^{-1}\cdot T_l^{\mathrm{naive}}$, then T_l induces an endomorphism of $\mathrm{R}\Gamma(\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]},\omega_a)$ and $\mathrm{R}\Gamma(X_0(p),\omega_a)$. The action of $(\mathbb{Z}/N\mathbb{Z})^\times$ on $\mathcal{X}_0(p)$ induces the diamond operators.

Let $X_0(p)_n$ denote the reduction of $\mathcal{X}_0(p)_{\mathbb{Z}_p[\zeta_p]}$ modulo ϖ^n, viewed as a scheme over $\mathrm{Spec}(\mathbb{Z}_p[\zeta_p]/(\varpi^n))$. We have $X_0(p)_1=X_0(p)=X^F\cup X^V$. Let $\mathcal{I}_F$ and $\mathcal{I}_V$ denote the ideal sheaves of X^F and X^V, respectively. Therefore, $X_0(p)_n=nX^F\cup nX^V$, where $nX^F=V(\mathcal{I}_F^n)$ and $nX^V=V(\mathcal{I}_V^n)$. In particular, the ordinary locus $X_0(p)_n^{\mathrm{ord}}=X_n^{F,\mathrm{ord}}\cup X_n^{V,\mathrm{ord}}$, where $X_n^{F,\mathrm{ord}}$ and $X_n^{V,\mathrm{ord}}$ are isomorphic to the reduction of the spherical level modular curve $\mathcal{X}^{\mathrm{ord}}_{\mathbb{Z}_p[\zeta_p]}$ modulo ϖ^n. Let $i_{F,n}:nX^F\to X_0(p)_n$, $i_{V,n}:nX^V\to X_0(p)_n$ be the closed embeddings, and $j_{F,n}:X_n^{F,\mathrm{ord}}\to X_0(p)_n$, $j_{V,n}:X_n^{V,\mathrm{ord}}\to X_0(p)_n$ be the open

embeddings. The Teichmüller character $[\cdot]:\mathbb{F}_p^\times\to\mathbb{Z}_p^\times$ modulo p^n is given by $\mathbb{F}_p^\times\to(\mathbb{Z}/p^n\mathbb{Z})^\times, a\mapsto a^{p^{n-1}}$. We can identify ω^k on $\mathfrak{X}^{\mathrm{ord}}$ with the line bundle descending from the Igusa cover $\mathfrak{IG}$ by the character $\mathbb{Z}_p^\times\to\mathbb{Z}_p^\times, a\mapsto a^k$, see for example [BP22][Lemma 4.8]. Therefore the restriction of $\mathcal{L}$ to $X_n^{F,\mathrm{ord}}$ is isomorphic to $\omega^{-p^{n-1}}$, and the restriction of $\mathcal{L}$ to $X_n^{V,\mathrm{ord}}$ is isomorphic to $\omega^{p^{n-1}}$.

Lemma 6.2. *For each $l\neq p$ and $n\geq 1$, we have the following commutative diagrams:*

1.
$$\begin{array}{ccccc} X_n^{F,\mathrm{ord}} & \xleftarrow{p_1'} & X_0(l)_n^{\mathrm{ord}} & \xrightarrow{p_2'} & X_n^{F,\mathrm{ord}} \\ \downarrow{\scriptstyle i_{F,n}} & & \downarrow{\scriptstyle i'_{F,n}} & & \downarrow{\scriptstyle i_{F,n}} \\ X_0(p)_n & \xleftarrow{p_1} & X_0(pl)_n & \xrightarrow{p_2} & X_0(p)_n \end{array}$$

induces

$$\begin{array}{ccc} p_2^*\mathcal{L} & \xrightarrow{\sim} & p_1^*\mathcal{L} \\ \downarrow & & \downarrow \\ (i'_{F,n})_*(p_2')^*\omega^{-p^{n-1}} & \xrightarrow{(i'_{F,n})_*T_l} & (i'_{F,n})_*(p_1')^*\omega^{-p^{n-1}} \end{array},$$

where we identify $i_{F,n}^*\mathcal{L}$ with $\omega^{-p^{n-1}}$, and the bottom T_l is the usual map induced by the universal isogeny $E\to E_l$ on $X_0(l)_n$

2.
$$\begin{array}{ccccc} X_n^{V,\mathrm{ord}} & \xleftarrow{p_1'} & X_0(l)_n^{\mathrm{ord}} & \xrightarrow{p_2'} & X_n^{V,\mathrm{ord}} \\ \downarrow{\scriptstyle i_{V,n}} & & \downarrow{\scriptstyle i'_{V,n}} & & \downarrow{\scriptstyle i_{V,n}} \\ X_0(p)_n & \xleftarrow{p_1} & X_0(pl)_n & \xrightarrow{p_2} & X_0(p)_n \end{array}$$

induces

$$\begin{array}{ccc} p_2^*\mathcal{L} & \xrightarrow{\sim} & p_1^*\mathcal{L} \\ \downarrow & & \downarrow \\ (i'_{V,n})_*(p_2')^*\omega^{p^{n-1}} & \xrightarrow{(i'_{V,n})_*l^{-p^{n-1}}\cdot T_l} & (i'_{V,n})_*(p_1')^*\omega^{p^{n-1}} \end{array},$$

where we identify $i_{V,n}^*\mathcal{L}$ with $\omega^{p^{n-1}}$, and the bottom T_l is the usual map induced by the universal isogeny $E\to E_l$ on $X_0(l)_n$.

Proof. We give the proof of 2, and the proof of 1 is similar. Let $\mathrm{Ig}_n\to X_n^{\mathrm{ord}}$ be the level n Igusa tower, parameterizing $(E/S,\varphi:(\mathbb{Z}/p^n\mathbb{Z})_S\xrightarrow{\sim}(H_n^{\mathrm{can}})^D)$, where S is a scheme over $\mathrm{Spec}(\mathbb{Z}/p^n\mathbb{Z})$, E/S ordinary elliptic curve, and H_n^{can} is the canonical subgroup of degree p^n. For simplicity denote E to be the universal elliptic curve over Ig_n, H_n^{can} to be the degree p^n canonical subgroup of $E[p^n]$,

and $\varphi : (\mathbb{Z}/p^n\mathbb{Z})_{\mathrm{Ig}_n} \xrightarrow{\sim} (H_n^{\mathrm{can}})^D)$ to be the universal level structure. There is a Hodge-Tate isomorphism $\mathrm{HT} : (H_n^{\mathrm{can}})^D \otimes_{(\mathbb{Z}/p^n\mathbb{Z})_{\mathrm{Ig}_n}} \mathcal{O}_{\mathrm{Ig}_n} \xrightarrow{\sim} \omega_{\mathrm{Ig}_n}$, which sends a local section $s : H_n^{\mathrm{can}} \to \mu_{p^n}$ to $s^*(\frac{dx}{x}) \in \omega_{H_n^{\mathrm{can}}} \simeq \omega_{\mathrm{Ig}_n}$, where $\frac{dx}{x}$ is the canonical generator of the co-Lie algebra of $\mu_{p^n} = \mathrm{Spec}\mathbb{Z}[x]/(x^{p^n} - 1)$. Therefore, $\varphi^\vee : H_n^{\mathrm{can}} \xrightarrow{\sim} \mu_{p^n}$ induces an isomorphism $\mathcal{O}_{\mathrm{Ig}_n^{\mathrm{ord}}} \xrightarrow{\sim} \omega_E$, which sends 1 to $(\varphi^\vee)^*(\frac{dx}{x})$. Let $\mathrm{Ig}_n(l) = X_0(l)_n^{\mathrm{ord}} \times_{X_n^{\mathrm{ord}}} \mathrm{Ig}_n$, there is an extended diagram:

$$\begin{array}{ccccc}
\mathrm{Ig}_n & \xleftarrow{q_1} & \mathrm{Ig}_n(l) & \xrightarrow{q_2} & \mathrm{Ig}_n \\
\downarrow{\scriptstyle \pi} & \lrcorner & \downarrow{\scriptstyle \pi'} & \llcorner & \downarrow{\scriptstyle \pi} \\
X_n^{V,\mathrm{ord}} & \xleftarrow{p_1'} & X_0(l)_n^{\mathrm{ord}} & \xrightarrow{p_2'} & X_n^{V,\mathrm{ord}} \\
\downarrow{\scriptstyle i_{V,n}} & \lrcorner & \downarrow{\scriptstyle i'_{V,n}} & \llcorner & \downarrow{\scriptstyle i_{V,n}} \\
X_0(p)_n & \xleftarrow{p_1} & X_0(pl)_n & \xrightarrow{p_2} & X_0(p)_n
\end{array}$$

The pull back $(i_{V,n} \circ \pi)^*$ of the isomorphism $p_2^*\mathcal{L} \simeq p_1^*\mathcal{L}$ in6.1 is given by the trivialization $\pi^*\omega^{p^{n-1}} \simeq \mathcal{O}_{\mathrm{Ig}_n}$ and the identification $q_2^*\mathcal{O}_{\mathrm{Ig}_n} = q_1^*\mathcal{O}_{\mathrm{Ig}_n}$. Let $\psi : E \to E_l$ be the degree l universal isogeny over $\mathrm{Ig}_n(l)$. Therefore, it suffices to show the following diagram is commutative:

$$\begin{array}{ccc}
p_2^*\mathcal{O}_{\mathrm{Ig}_n} & \xrightarrow{\sim} & p_2^*\omega^{p^{n-1}} \\
\| & & \downarrow{\scriptstyle l^{-p^{n-1}}(\psi^*)^{\otimes p^{n-1}}} \\
p_1^*\mathcal{O}_{\mathrm{Ig}_n} & \longrightarrow & p_1^*\omega^{p^{n-1}}
\end{array}$$

Let $\varphi_E^\vee : H_E^{\mathrm{can}} \xrightarrow{\sim} \mu_{p^n}$ and $\varphi_{E_l}^\vee : H_{E_l}^{\mathrm{can}} \xrightarrow{\sim} \mu_{p^n}$ be the level structure of E and E_l respectively. Then $\varphi_{E_l}^\vee = \varphi_E \circ \hat{\psi}$, the upper horizontal arrow maps 1 to $((\varphi_{E_l}^\vee)^*(\frac{dx}{x}))^{\otimes p^{n-1}} = (\hat{\psi}^* \circ (\varphi_E^\vee)^*(\frac{dx}{x}))^{\otimes p^{n-1}}$, and the lower horizontal arrow maps 1 to $((\varphi_E^\vee)^*(\frac{dx}{x}))^{\otimes p^{n-1}}$. Therefore the right vertical arrow is $((\hat{\psi}^*)^{-1})^{\otimes p^{n-1}}$. Since $(\hat{\psi}^*)^{-1} = l^{-1} \circ \psi^*$, we get the desire result. □

A direct corollary of the lemma is the following.

Corollary 6.3. *For all $l \neq p$, and $a = 0, 1, \cdots, p-2$, we have:*

1. *The restriction map $\mathrm{R}\Gamma(X_0(p), \omega_a) \to \mathrm{R}\Gamma(X_0(p)^F, \omega^{p-a})$ is T_l-equivariant.*
2. *The restriction map $\mathrm{R}\Gamma(X_0(p), \omega_a) \to \mathrm{R}\Gamma(X_0(p)^V, \omega^{1+a})$ is equivariant for T_l action on the left, and $l^{-a}T_l$ action on the right.*
3. *The restriction map $\mathrm{R}\Gamma(\mathcal{X}_0(p), \omega_a) \to \mathrm{R}\Gamma(\mathfrak{X}_0(p)^{F,\mathrm{ord}}, \omega^{\kappa_{-a}})$ is T_l-equivariant.*
4. *The restriction map $\mathrm{R}\Gamma(\mathcal{X}_0(p), \omega_a) \to \mathrm{R}\Gamma(\mathfrak{X}_0(p)^{V,\mathrm{ord}}, \omega^{\kappa_a})$ is equivariant for T_l action on the left, and $[l]^{-a}T_l$ action on the right, where $[l] \in \mathbb{Z}_p^\times$ is the Teichmüller lift of $l \in \mathbb{F}_p^\times$.*

Proof. We give the proof of 1, 4, and the proof of 2, 3 is similar. For 1, let $i'_{F,1} : X_0(l) \to X_0(pl)$ be the Frobenius closed embedding that extends $i'_{F,1}$. The restriction of the isomorphism $(i'_{F,1})^* p_2^* \mathcal{L} \simeq (i'_{F,1})^* p_1^* \mathcal{L}$ to the open subscheme $X_0(l)^{\mathrm{ord}}$ is identified with T_l by the above lemma. Since for any line bundle L over $X_0(l)$, the restriction map $\mathrm{H}^0(X_0(l), L) \to \mathrm{H}^0(X_0(l)^{\mathrm{ord}}, L)$ is injective, we obtain that $(i'_{F,1})^* p_2^* \mathcal{L} \simeq (i'_{F,1})^* p_1^* \mathcal{L}$ is also identified with T_l. On the other hand, both of the isomorphisms $\omega_1 \mid_{X_0(p)^F} \simeq \omega^p$ and $\omega_1 \mid_{X_0(p)^V} \simeq \omega$ are T_l-equivariant, therefore $\omega_1 \otimes \mathcal{L}^a \mid_{X_0(p)^F} \simeq \omega^{p-a}$ is T_l-equivariant.
For 4, notice that $[l] \equiv l^{p^{n-1}}$ modulo p^n, therefore by the above lemma, $\omega_1 \otimes \mathcal{L}^a \mid_{\mathfrak{X}_0(p)^{V,\mathrm{ord}}} \simeq \omega^{\kappa_a}$ is $(T_l, [l]^{-a} T_l)$-equivariant. □

Write $G_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$, let $\overline{\varepsilon}_p : G_{\mathbb{Q}} \to \mathbb{F}_p^\times$ be the mod-p cyclotomic character. For a unit u in the coefficient ring, let $\lambda(u)$ denote the unramified character of G_p sending arithmetic Frobenius to u. We have the following local global compatibility theorem.

Theorem 6.4 (Local Global Compatibility)**.** *Take any $a \in \{0, 1, \cdots, p-2\}$. Let f be a cuspidal non-Eisenstein eigenform in $\mathrm{H}^0(X_0(p), \omega_a)$ with respect to the Hecke operators T_l for $l \neq p$, U_p, and $\langle d \rangle$ for $d \in (\mathbb{Z}/N\mathbb{Z})^\times$. Suppose*

$$T_\ell f = a_\ell f, \qquad U_p f = a_p f, \qquad \langle d \rangle f = \epsilon(d) f,$$

where $\epsilon : (\mathbb{Z}/N\mathbb{Z})^\times \to \overline{\mathbb{F}}_p^\times$ is the Nebentypus character. There is a continuous irreducible representation $\overline{\rho}_f : G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_p)$, unramified outside Np, such that, for $\ell \nmid Np$,

$$\mathrm{tr}\, \overline{\rho}_f(\mathrm{Frob}_\ell) = a_\ell, \qquad \det \overline{\rho}_f(\mathrm{Frob}_\ell) = \epsilon(\ell) \ell^{-a}.$$

Moreover,

$$\overline{\rho}_f|_{G_p} \simeq \begin{cases} \overline{\varepsilon}_p^{-a} \lambda\big(\epsilon(p)/a_p\big) \oplus \lambda(a_p), & \text{if } a \neq 0, \\ \text{an unramified representation}, & \text{if } a = 0. \end{cases}$$

Proof. We know that U_p acts invertiblely on $\mathrm{R}\Gamma(X_0(p), \omega_a)$(3.20, 5.16), therefore $a_p \neq 0$. For the case $a = 0$, the statement follows from 3.21. Now fix $a \neq 0$, and a cuspidal non-Eisenstein eigenform $f \in \mathrm{H}^0(X_0(p), \omega_a)$. By 5.26, the restiction gives an U_p-equivariant embedding $i_1 : \mathrm{H}^0(X_0(p), \omega_a) \hookrightarrow e(U_p)\mathrm{H}^0(X^{\mathrm{ord}}, \omega^{p-a})$, which is also T_l-equivariant by 6.3, and $\langle d \rangle$-equivariant. We can also consider the U_p^t-operator, and the same arguments give an U_p^t-equivariant embedding $i_2 : \mathrm{H}^0(X_0(p), \omega_a) \hookrightarrow e(U_p^t)\mathrm{H}^0(X^{\mathrm{ord}}, \omega^{1+a})$, which is $(T_l, l^{-a} T_l)$-equivariant by 6.3, and $\langle d \rangle$-equivariant. Combining the first embedding with Deligne's theorem on Galois representations associated to ordinary cuspidal eigenforms of weight $2 \leq k \leq p+1$ in characteristic p, we obtain that there is an irreducible Galois representation $\overline{\rho}_f : G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_p)$ with the required eigensystem, such that

$$\overline{\rho}_f|_{G_p} \simeq \begin{pmatrix} \overline{\varepsilon}_p^{p-a-1} \lambda(\epsilon(p)/a_p) & * \\ 0 & \lambda(a_p) \end{pmatrix}$$

On the other hand, the restriction of U_p^t to $X_0(p)^{V,\mathrm{ord}}$ is identified with $\langle p\rangle^{-1}U_p$ action. Therefore under the second embedding, $U_p(i_2(f)) = i_2(\langle p\rangle U_p^t(f)) = i_2(\epsilon(p)/a_p \cdot f) = \epsilon(p)/a_p \cdot i_2(f)$, since $U_pU_p^t = id$. Therefore we obtain that

$$\overline{\rho}_f|_{G_p} \simeq \begin{pmatrix} \overline{\varepsilon}_p^a\lambda(a_p) & * \\ 0 & \lambda(\epsilon(p)/a_p) \end{pmatrix} \otimes \overline{\varepsilon}_p^{-a} \simeq \begin{pmatrix} \lambda(a_p) & * \\ 0 & \overline{\varepsilon}_p^{p-1-a}\lambda(\epsilon(p)/a_p) \end{pmatrix}.$$

From this we see that $\overline{\rho}_f|_{G_p} \simeq \overline{\varepsilon}_p^{-a}\lambda\big(\epsilon(p)/a_p\big) \oplus \lambda(a_p)$. □

6.2 A modularity result

We first establish a multiplicity-one result for Serre's mod p quaternionic modular forms, which is proved for larger range of weights but with extra conditions on the Galois representations in [Bao26], and we use it to prove a modularity result for Galois representations of $G_{\mathbb{Q}}$ that are split at p.

We first recall some of the properties in [Ser96]. Let SS be the supersingular locus of the modular curve $X/\mathbb{F}_p$ for some level $\Gamma_1(N), p \nmid N$, and the space of quaternionic modular forms of weight k is identified with $\mathrm{H}^0(SS, \omega^k)$. Let $\mathbb{T} \subset \mathrm{End}(\mathrm{H}^0(X, \omega^k))$ be the Hecke algebra generated by T_l for $l \nmid Np$, and $\langle d\rangle$ for $d \in (\mathbb{Z}/N\mathbb{Z})^\times$. Also denote $\mathbb{T}_D$ as the Hecke algebra acting on $\mathrm{H}^0(SS, \omega^k)$, generated by the same Hecke operators T_l, and $\langle d\rangle$.

Theorem 6.5. *Let $k \in \mathbb{Z}$ be any integer.*

1. *There is function $B \in \mathrm{H}^0(SS, \omega^{p+1})$, such that it induces an isomorphism $B : \mathrm{H}^0(SS, \omega^k) \xrightarrow{\sim} \mathrm{H}^0(SS, \omega^{k+p+1})$, satisfying $T_l(Bf) = lBT_l(f)$, for all $l \neq p$.*
2. *The map $V_p : \mathrm{H}^0(SS, \omega^k) \to \mathrm{H}^0(SS, \omega^{pk})$, given by raising to the p-th power is a Hecke equivariant isomorphism.*

Proof. For 1 see [Edi92](Proposition 7.2). and for 2 see [Ser96]. □

Let $\overline{\rho} : \mathrm{G}_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_p)$ be an odd, irreducible, continuous Galois representation, and let $N = N(\overline{\rho})$ be its minimal level. Suppose that $\overline{\rho}\,|_{G_{\mathbb{Q}_p}} = \psi_1 \oplus \psi_2$, such that ψ_1 is tamely ramified, and ψ_2 is unramified. Then $\psi_1\,|_{I_p} = \overline{\varepsilon}_p^{k-1}$ for some $2 \leq k \leq p-1$. By Serre's modularity conjecture, there exists a cuspidal eigenform $f \in \mathrm{H}^0(X_1(N), \omega^k)$, such that $\overline{\rho}_f \simeq \overline{\rho}$. Denote $\mathfrak{m}_{\overline{\rho}}$ to be the corresponding maximal ideal in $\mathbb{T}$. The restriction map $\pi : \mathrm{H}^0(X_1(N), \omega^k) \to \mathrm{H}^0(SS, \omega^k)$ has kernel $\mathrm{H}^0(X_1(N), \omega^{k-p-1})$, which is 0 for $2 \leq k \leq p-2$. For $k = p-1$, f does not lie in the kernel since by our assumption the Galois representation is non-trivial. Therefore, we obtain a maximal ideal $\mathfrak{m}_{D,\overline{\rho}} \subset \mathbb{T}_D$ given by $\pi(f)$.

Theorem 6.6 (multiplicity one)**.** *The space $\mathrm{H}^0(SS, \omega^k)[\mathfrak{m}_{D,\overline{\rho}}]$ has dimension at most 1.*

Proof. 1. For $2 \leq k \leq p-2$, B induces an isomorphism $B : \mathrm{H}^0(SS, \omega^k)[\mathfrak{m}_{D,\overline{\rho}}] \simeq \mathrm{H}^0(SS, \omega^{k+p+1})[\mathfrak{m}_{D,\overline{\rho}\otimes\overline{\varepsilon}_p}]$. Consider the Hecke equivariant short exact sequence:

$$0 \to \mathrm{H}^0(X_1(N), \omega^{k+2}) \xrightarrow{\mathrm{Ha}} \mathrm{H}^0(X_1(N), \omega^{k+p+1}) \to \mathrm{H}^0(SS, \omega^{k+p+1}) \to 0.$$

Since $\overline{\rho} \otimes \overline{\varepsilon}_p \mid_{I_p} = \overline{\varepsilon}_p^k \oplus \overline{\varepsilon}_p$, by weight part of Serre's conjecture[Edi92], the minimal weight that contains eigenform with Galois representation $\overline{\rho}\otimes\overline{\varepsilon}_p$ is $p+1+k$. Therefore, $\mathrm{H}^0(X_1(N), \omega^{k+p+1})_{\mathfrak{m}_{\overline{\rho}\otimes\overline{\varepsilon}_p}} \simeq \mathrm{H}^0(SS, \omega^{k+p+1})_{\mathfrak{m}_{D,\overline{\rho}\otimes\overline{\varepsilon}_p}}$, and it suffices to show that $\dim_{\mathbb{F}_p}\mathrm{H}^0(X_1(N), \omega^{k+p+1})[\mathfrak{m}_{\overline{\rho}\otimes\overline{\varepsilon}_p}] \leq 1$. The Hecke operators U_l for $l \mid N$ are also contained in $\mathbb{T}$ by [Bao26][Lemma 2.4.2], therefore, by considering q-expansion, we know that if

$$\dim_{\mathbb{F}_p}\mathrm{H}^0(X_1(N), \omega^{k+p+1})[\mathfrak{m}_{\overline{\rho}\otimes\overline{\varepsilon}_p}] \geq 2,$$

then there exists $0 \neq g \in \mathrm{H}^0(X_1(N), \omega^{k+p+1})[\mathfrak{m}_{\overline{\rho}\otimes\overline{\varepsilon}_p}]$ whose q-expansion has the shape $g = \sum_n a_n q^{np}$, which means $g \in \ker(\theta)$. But g is not divisible by Hasse invariant, and $p \nmid k+p+1$, however by [Kat77] such g doesn't exist.

2. For $k = p-1$, V_p induces an isomorphism $V_p : \mathrm{H}^0(SS, \omega^{p-1})[\mathfrak{m}_{D,\overline{\rho}}] \simeq \mathrm{H}^0(SS, \omega^{p(p-1)})[\mathfrak{m}_{D,\overline{\rho}}]$, and B induces an isomorphism

$$B^{-(p-3)} : \mathrm{H}^0(SS, \omega^{pk})[\mathfrak{m}_{D,\overline{\rho}}] \simeq \mathrm{H}^0(SS, \omega^{p+3})[\mathfrak{m}_{D,\overline{\rho}\otimes\overline{\varepsilon}_p^2}].$$

Since $\overline{\rho} \otimes \overline{\varepsilon}_p^4 \mid_{I_p} = \overline{\varepsilon}_p \oplus \overline{\varepsilon}_p^2$, by weight part of Serre's conjecture[Edi92], the minimal weight that contains eigenform with Galois representation $\overline{\rho} \otimes \overline{\varepsilon}_p^2$ is $p+3$, and the remaining arguments are same as above.

□

Theorem 6.7. *There exists an eigenform $f \in \mathrm{H}^0(X_0(p), \omega_{p-k})$ such that the Galois representation $\overline{\rho}_f$ associated to f is isomorphic to $\overline{\rho}$. Moreover,* $\dim_{\mathbb{F}_p}\mathrm{H}^0(X_0(p), \omega_{p-k})[\mathfrak{m}_{\overline{\rho}}] = 1$.

Proof. Let SS be the reduced closed subscheme corresponding to the supersingular points on $X_0(p)$, there is a short exact sequence:

$$0 \to \omega_{p-k} \xrightarrow{s\mapsto(s|_{X_0(p)^F}, s|_{X_0(p)^V})} \omega_{p-k} \mid_{X_0(p)^F} \oplus \omega_{p-k} \mid_{X_0(p)^V} \xrightarrow{(s_1,s_2)\mapsto s_1|_{SS} - s_2|_{SS}} \omega_{p-k} \mid_{SS} \to 0.$$

By 6.3, this induces a short exact sequence:

$$0 \to \mathrm{H}^0(X_0(p), \omega_{p-k})[\mathfrak{m}_{\overline{\rho}}] \to \mathrm{H}^0(X, \omega^k)[\mathfrak{m}_{\overline{\rho}}] \oplus \mathrm{H}^0(X, \omega^{p+1-k})[\mathfrak{m}_{\overline{\rho}\otimes\overline{\varepsilon}_p^{-(p-k)}}] \to \mathrm{H}^0(SS, \omega_{p-k})[\mathfrak{m}_{\overline{\rho}}].$$

The restriction maps induce

$$\mathrm{H}^0(SS, \omega^k)[\mathfrak{m}_{\overline{\rho}}] \simeq \mathrm{H}^0(SS, \omega_{p-k})[\mathfrak{m}_{\overline{\rho}}] \simeq \mathrm{H}^0(SS, \omega^{p+1-k})[\mathfrak{m}_{\overline{\rho}\otimes\overline{\varepsilon}_p^{-(p-k)}}].$$

By the existence of companion forms[Gro90], we know that

$$\dim_{\mathbb{F}_p} \mathrm{H}^0(X, \omega^{p+1-k})[\mathfrak{m}_{\overline{\rho}\otimes\bar{\varepsilon}_p^{-(p-k)}}] \geq 1,$$

and by considering q-expansions as in 6.6 we have

$$\dim_{\mathbb{F}_p} \mathrm{H}^0(X, \omega^{k})[\mathfrak{m}_{\overline{\rho}}] = \dim_{\mathbb{F}_p} \mathrm{H}^0(X, \omega^{p+1-k})[\mathfrak{m}_{\overline{\rho}\otimes\bar{\varepsilon}_p^{-(p-k)}}] = 1.$$

The restriction map $\mathrm{H}^0(X, \omega^k)[\mathfrak{m}_{\overline{\rho}}] \to \mathrm{H}^0(SS, \omega^k)[\mathfrak{m}_{\overline{\rho}}]$ has kernel $\mathrm{H}^0(X, \omega^{k-p+1})[\mathfrak{m}_{\overline{\rho}}] = 0$(for $2 \leq k \leq p-2$ this is true because $k-p+1 < 0$, for $k = p-1$ this is true because $\mathrm{H}^0(X, \mathcal{O}) = \mathbb{F}_p$ only contains trivial eigensystem). Since we also have $\dim_{\mathbb{F}_p} \mathrm{H}^0(SS, \omega_{p-k})[\mathfrak{m}_{\overline{\rho}}] = 1$ by the multiplicity one theorem 6.6, the map $\mathrm{H}^0(X, \omega^k)[\mathfrak{m}_{\overline{\rho}}] \oplus \mathrm{H}^0(X, \omega^{p+1-k})[\mathfrak{m}_{\overline{\rho}\otimes\bar{\varepsilon}_p^{-(p-k)}}] \to \mathrm{H}^0(SS, \omega_{p-k})[\mathfrak{m}_{\overline{\rho}}]$ is surjective. Therefore, $\dim_{\mathbb{F}_p} \mathrm{H}^0(X_0(p), \omega_{p-k})[\mathfrak{m}_{\overline{\rho}}] = 1$. □

Remark 6.8. There should be an alternative proof of the modularity theorem based on an analysis of the Cousin complexes using integral p-adic Hodge theory, for example via integral Sen theory. I would be very interested in exploring this approach, which should be better suited to generalization, since the multiplicity-one result does not hold for more general Shimura varieties.

6.3 Galois representation valued in the Hecke algebra

In this section we construct Galois representation valued in the weight one Hecke algebra, and prove the corresponding local-global compatibility result.

Let $\mathbb{T}^{\mathrm{abs}} = \mathbb{Z}_p[(\mathbb{Z}/N\mathbb{Z})^\times][U_p, T_l, l \nmid Np]$ be the abstract Hecke algebra, and let $\mathbb{T}_a = \mathrm{Im}(\mathbb{T}^{\mathrm{abs}} \to \mathrm{End}_{D(\mathbb{Z}_p)}(\mathrm{R}\Gamma(\mathcal{X}_0(p), \omega_a)))$ for $0 \leq a \leq p-2$, where $D(\mathbb{Z}_p)$ is the derived category of $\mathbb{Z}_p$-modules. Let R be a complete topological $\mathbb{Z}_p$-algebra, for a p-adic weight $\kappa : \mathbb{Z}_p^\times \to R^\times$, denote $\mathbb{T}_\kappa^{\mathrm{ord}} = \mathrm{Im}(\mathbb{T}_R^{\mathrm{abs}} \to \mathrm{End}(e(U_p)\mathrm{H}^0(\mathfrak{X}_R^{\mathrm{ord}}, \omega^\kappa)))$, where $\omega^\kappa = \omega^{\kappa^{\mathrm{un}}} \otimes_{\Lambda,\kappa} R$ is a line bundle on $\mathfrak{X}_R^{\mathrm{ord}}$.

Let $\mathfrak{m} \subset \mathbb{T}^{\mathrm{abs}}$ be an non-Eisenstein maximal ideal, our goal in this section is to construct a Galois representation

$$\rho_{\mathfrak{m}} : G_{\mathbb{Q}} \to \mathrm{GL}_2((\mathbb{T}_a)_{\mathfrak{m}}).$$

Lemma 6.9. *Let R be a complete $\mathbb{Z}_p$-algebra, and $\kappa : \mathbb{Z}_p^\times \to R^\times$ be a continuous character. Then:*

1. $\mathrm{H}^0(\mathfrak{X}_\Lambda^{\mathrm{ord}}, \omega^{\kappa^{\mathrm{un}}})\hat{\otimes}_{\Lambda,\kappa} R \simeq \mathrm{H}^0(\mathfrak{X}_R^{\mathrm{ord}}, \omega^\kappa)$,
2. *If R is also a finite Λ-algebra, then* $\mathrm{H}_c^1(\mathfrak{X}_\Lambda^{\mathrm{ord}}, \omega^{\kappa^{\mathrm{un}}})\hat{\otimes}_{\Lambda,\kappa} R \simeq \mathrm{H}_c^1(\mathfrak{X}_R^{\mathrm{ord}}, \omega^\kappa)$.
3. *There is a perfect pairing* $e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}) \times e(F)\mathrm{H}_c^1(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_a}(-D)) \to \mathbb{Z}_p$

Proof. 1 follows from the affiness of $\mathfrak{X}^{\mathrm{ord}}$. For 2, since $\mathfrak{X}^{\mathrm{ord}}$ is affine, the functor that maps a finite Λ-algebra R to $\mathrm{H}_c^1(\mathfrak{X}^{\mathrm{ord}}, \omega^{\mathrm{un}} \otimes_\Lambda R)$ is exact due to the vanishing of H^0. Therefore, taking a finite free presentation of R gives the desired

result. For 3, [BP22, Theorem 4.18] constructed a perfect pairing between finite projective Λ-modules([BP22, Theorem 4.15])

$$e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa^{\mathrm{un}}})\times e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{2-\kappa^{\mathrm{un}}}(-D))\to\Lambda.$$

Since $\kappa_a+\kappa_{-a}=2$, the result follows from base change and using 12. □

On $\mathrm{R}\Gamma(\mathcal{X}_0(p),\omega_a)$, there is a T_l-action as constructed in 6.1, and we denote the induced action on $\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega_a)$, $\mathrm{H}^1_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p),\omega_a)$, and $\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega_a)$ also by T_l. Consider the T_l action on $\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_{-a}})=\lim_n\mathrm{H}^0(X_n^{\mathrm{ord}},\omega^{1-p^{n-1}a})$ by taking inverse limit of T_l on finite level, and similarly we have the T_l action on $\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_a})=\lim_n\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{1+ap^{n-1}})$. Therefore by 6.3, there are commutative diagrams:

$$\begin{array}{ccc}\mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega_a)&\xrightarrow{\sim}&\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_{-a}})\\ {\scriptstyle T_l}\downarrow&&\downarrow{\scriptstyle T_l}\\ \mathrm{H}^0(\mathfrak{X}_0(p)^{F,\mathrm{ord}},\omega_a)&\xrightarrow{\sim}&\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_{-a}}),\end{array}\qquad\begin{array}{ccc}\mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega_a)&\xrightarrow{\sim}&\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_a})\\ {\scriptstyle T_l}\downarrow&&\downarrow{\scriptstyle [l]^{-a}T_l}\\ \mathrm{H}^1_c(\mathfrak{X}_0(p)^{V,\mathrm{ord}},\omega_a)&\xrightarrow{\sim}&\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_a})\end{array}$$

Lemma 6.10. *There is a commutative diagram:*

$$\begin{array}{ccc}e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_a})&\xrightarrow{\sim}&e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_{-a}}(-D))^*\\ \downarrow{\scriptstyle T_l}&&\downarrow{\scriptstyle [l]^aT_l^*}\\ e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_a})&\xrightarrow{\sim}&e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_{-a}}(-D))^*,\end{array}$$

where the horizontal isomorphisms are induced by the perfect pairing, and T_l^* is the dual of T_l.

Proof. For $n\geq 1$, over X_n we have $\mathrm{H}^1(\mathcal{X},\omega^k)/p^n=\mathrm{H}^1(X_n,\omega^k)$, and $\mathrm{H}^0(\mathcal{X},\omega^k)/p^n=\mathrm{H}^0(X_n,\omega^k)$ for $k\geq 2$. There is a classicality theorem coming from modulo p^n of [BP22, Theorem 4.15]:

1. If $k\leq -1$, then $e(F)\mathrm{H}^1_c(X_n^{\mathrm{ord}},\omega^k)=e(T_p)\mathrm{H}^1(X_n,\omega^k)$,
2. If $k\geq 3$, then $e(U_p)\mathrm{H}^0(X_n^{\mathrm{ord}},\omega^k)=e(T_p)\mathrm{H}^0(X_n,\omega^k)$.

The duality isomorphism is induced by the diagram:

$$\begin{array}{ccc}e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_a})&\xrightarrow{\quad\sim\quad}&e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}},\omega^{\kappa_{-a}}(-D))^*\\ \downarrow\sim&&\downarrow\sim\\ \lim_n e(F)\mathrm{H}^1_c(X_n^{\mathrm{ord}},\omega^{1+p^{n-1}a})&\xrightarrow{\quad\sim\quad}&\lim_n e(U_p)\mathrm{H}^0(X_n^{\mathrm{ord}},\omega^{1-p^{n-1}a}(-D))^*\\ \downarrow\sim&&\sim\uparrow\\ \lim_n e(T_p)\mathrm{H}^1(X_n,\omega^{1+p^{n-1}a-p^{n-1}(p-1)})&\xrightarrow{\sim}&\lim_n e(T_p)\mathrm{H}^0(X_n,\omega^{1-p^{n-1}a+p^{n-1}(p-1)}(-D))^*,\end{array}$$

where the bottom horizontal isomorphism is given by the usual Serre duality, and the lower vertical isomorphisms are given by the classicality theorem above(at least for n large). Therefore, it suffices to show that the following diagram commutes:

$$\begin{array}{ccc} \mathrm{H}^1(X_n, \omega^{1+p^{n-1}a-p^{n-1}(p-1)}) & \xrightarrow{SD} & \mathrm{H}^0(X_n, \omega^{1-p^{n-1}a+p^{n-1}(p-1)}(-D))^* \\ \downarrow{\scriptstyle T_l} & & \downarrow{\scriptstyle l^{p^{n-1}a}T_l^*} \\ \mathrm{H}^1(X_n, \omega^{1+p^{n-1}a-p^{n-1}(p-1)}) & \xrightarrow{SD} & \mathrm{H}^0(X_n, \omega^{1-p^{n-1}a+p^{n-1}(p-1)}(-D))^*. \end{array}$$

The proof of this is the same as [Edi92, Proposition 7.3]. □

Proposition 6.11. *For any $0 \leq a \leq p-2$, there is a surjective $\mathbb{T}^{\mathrm{abs}}_{\mathfrak{m}}$-algebra morphism $(\mathbb{T}^{\mathrm{ord}}_{\kappa_{-a}})_{\mathfrak{m}} \twoheadrightarrow (\mathbb{T}_a)_{\mathfrak{m}}$ (they can be 0).*

Proof. It suffices to check that the natural map $(\mathbb{T}^{\mathrm{ord}}_{\kappa_{-a}})_{\mathfrak{m}} \dashrightarrow (\mathbb{T}_a)_{\mathfrak{m}}$ is well defined. For $0 \leq a \leq p-3$, 5.26 and 5.16 shows that

$$\begin{aligned} \mathrm{R}\Gamma(\mathcal{X}_0(p), \omega_a) &= \left[e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}) \longrightarrow e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_a})\right] \\ &= \left[e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}) \longrightarrow e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}(-D))^*\right]. \end{aligned}$$

By 6.10, the T_l-action on $\mathrm{R}\Gamma(\mathcal{X}_0(p), \omega_a)$ is compatible with the (T_l, T_l^*)-action on degree $0, 1$, i.e:

$$\begin{array}{ccc} \left[e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}) \longrightarrow e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}(-D))^*\right] \\ \downarrow{\scriptstyle T_l} \qquad\qquad \downarrow{\scriptstyle T_l^*} \\ \left[e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}) \longrightarrow e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}(-D))^*\right] \end{array}$$

Since $\mathfrak{m}$ is non-Eisenstein, $e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}})_{\mathfrak{m}} = e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}(-D))_{\mathfrak{m}}$. Therefore, if $t \in \mathbb{T}^{\mathrm{abs}}_{\mathfrak{m}}$ such that $t \mapsto 0 \in (\mathbb{T}^{\mathrm{ord}}_{\kappa_{-a}})_{\mathfrak{m}}$, then the action of t^* on $e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}}(-D))^*_{\mathfrak{m}} = e(U_p)\mathrm{H}^0(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{-a}})^*_{\mathfrak{m}}$ is 0, and this gives the required map $(\mathbb{T}^{\mathrm{ord}}_{\kappa_{-a}})_{\mathfrak{m}} \twoheadrightarrow (\mathbb{T}_a)_{\mathfrak{m}}$. For the case $a = p-2$, by5.26 $e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{p-2}}) \hookrightarrow e(U_p)\mathrm{H}^1_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p), \omega_{p-2})$ is a full rank lattice, therefore the dual $e(U_p)\mathrm{H}^1_{\mathfrak{X}_0(p)^V}(\mathfrak{X}_0(p), \omega_{p-2})^* \hookrightarrow e(F)\mathrm{H}^1_c(\mathfrak{X}^{\mathrm{ord}}, \omega^{\kappa_{p-2}})^*$ is injective, and the rest of the arguments are the same. □

Similarly, we have the following by considering U_p^t and the corresponding Cousin complex.

Proposition 6.12. *For any $0 \leq a \leq p-2$, there is a surjection $\mathbb{T}^{\mathrm{ord}}_{\kappa_a, \mathfrak{m}} \twoheadrightarrow \mathbb{T}_{a,\mathfrak{m}}$, which maps $U_p^t = \langle p \rangle^{-1} U_p$ to $U_p^t = U_p^{-1}$, and maps $T_l \cdot [l]^{-a}$ to T_l.*

Let $\mathbb{T}_\Lambda^{\mathrm{ord}} = \mathrm{Im}(\mathbb{T}_\Lambda^{\mathrm{abs}} \to \mathrm{End}(e(U_p)\mathrm{H}^0(\mathfrak{X}_\Lambda^{\mathrm{ord}}, \omega^{\kappa^{\mathrm{un}}})))$ be the Λ-adic ordinary Hecke algebra. We will construct a Λ-adic ordinary Galois representation $\rho_{\Lambda,\mathfrak{m}} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{T}_{\Lambda,\mathfrak{m}}^{\mathrm{ord}})$ in the next section, and we can construct the required Galois representation $\rho : G_{\mathbb{Q}} \to \mathrm{GL}_2((\mathbb{T}_a)_{\mathfrak{m}})$ through the surjections $\mathbb{T}_{\Lambda,\mathfrak{m}}^{\mathrm{ord}} \twoheadrightarrow \mathbb{T}_{\Lambda,\mathfrak{m}}^{\mathrm{ord}} \otimes_{\Lambda,\kappa_{-a}} \mathbb{Z}_p \twoheadrightarrow \mathbb{T}_{\kappa_{-a},\mathfrak{m}}^{\mathrm{ord}} \twoheadrightarrow \mathbb{T}_{a,\mathfrak{m}}$.

6.4 Λ-adic ordinary Galois representation

Let $\overline{\rho} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{F}}_p)$ be a continuous, odd, irreducible Galois representation. Let us suppose that

$$\overline{\rho}|_{G_p} = \begin{pmatrix} \chi_1 & * \\ 0 & \chi_2 \end{pmatrix},$$

where $\chi_1, \chi_2 : G_p \to \overline{\mathbb{F}}_p^\times$ are continuous characters, $\chi_2 = \lambda(\alpha)$ is unramified, and $\chi_1 = \bar{\varepsilon}_p^k \otimes \lambda(\beta)$ for some $1 \le k \le p-2$. Let $\mathfrak{m} \subset \mathbb{T}^{\mathrm{abs}}$ be the corresponding maximal ideal generated by $\langle T_l - \mathrm{tr}\,\overline{\rho}(\mathrm{Frob}_l), l \nmid Np, U_p - \alpha \rangle$, and the corresponding Nebentypus character. Denote $M_\Lambda^{\mathrm{ord}} = e(U_p)\mathrm{H}^0(\mathfrak{X}_\Lambda^{\mathrm{ord}}, \omega^{\kappa^{\mathrm{un}}})$, which is a finite free Λ-module by [BP22, Theorem 4.15]. Let $\mathbb{T}_k^{\mathrm{ord}} = \mathrm{Im}(\mathbb{T}^{\mathrm{abs}} \to \mathrm{End}(e(U_p)\mathrm{H}^0(\mathcal{X}, \omega^k)))$, be the classical weight k ordinary Hecke algebra.

Proposition 6.13. *The natural map $\mathbb{T}_\Lambda^{\mathrm{ord}} \to \prod_{k\ge 3} \mathbb{T}_k^{\mathrm{ord}}$ is injective, and $\mathbb{T}_\Lambda^{\mathrm{ord}}$ is reduced.*

Proof. M_Λ^{ord} is a free Λ-module, and the control theorems [BP22, Theorem 4.15] implies that $M_\Lambda^{\mathrm{ord}} \otimes_{\Lambda,k} \mathbb{Z}_p \simeq e(U_p)\mathrm{H}^0(\mathcal{X}, \omega^k)$ is a free $\mathbb{Z}_p$-module with the same rank n. Suppose $t \in \mathbb{T}_\Lambda^{\mathrm{ord}}$ lies in the kernel, take a basis of M_Λ^{ord} and write $t = (t_{i,j})_{1\le i,j\le n}$ as a matrix in $\mathrm{End}(M_\Lambda^{\mathrm{ord}}) \simeq M_{n\times n}(\Lambda)$. Then the image of t in $\mathrm{End}(e(U_p)\mathrm{H}^0(\mathcal{X}, \omega^k)) = \mathrm{End}(M_\Lambda^{\mathrm{ord}} \otimes_{\Lambda,k} \mathbb{Z}_p) \simeq M_{n\times n}(\mathbb{Z}_p)$ is 0, which means $t_{i,j} \in \cap_{k\ge 3} \ker(k : \Lambda \to \mathbb{Z}_p) = 0$. We show that by classical theory $\mathbb{T}_k^{\mathrm{ord}}$ are reduced for $k \ge 2$, and the reducedness of $\mathbb{T}_\Lambda^{\mathrm{ord}}$ follows. By classical theory of modular forms, the Hecke operators $T_l, l \nmid N$ act semisimply on $e(U_p)\mathrm{H}^0(\mathcal{X}_{\mathbb{Q}_p}, \omega^k)$, and therefore decompose into eigenspaces after extending the coefficient. The U_p-action satisfies $U_p^2 - T_pU_p + p^{k-1}\langle p\rangle = 0$, therefore, for $k \ge 2$, there is at most one eigenvalue with p-adic valuation 0, hence the U_p-action on the ordinary part is also semisimple. Let $\mathbb{T}_{\mathbb{C},k}^{\mathrm{ord}}$ be the Hecke algebra of $M_k(\Gamma_1(N), \mathbb{C})^{\mathrm{ord}}$, then freeness of $e(U_p)\mathrm{H}^0(\mathcal{X}, \omega^k)$ again gives an embedding $\mathbb{T}_k^{\mathrm{ord}} \hookrightarrow \mathbb{T}_{\mathbb{C},k}^{\mathrm{ord}} = \prod_{f \text{ eigenform}} \mathbb{C}$, and therefore $\mathbb{T}_k^{\mathrm{ord}}$ is reduced. □

Put $A = \mathbb{T}_{\Lambda,\mathfrak{m}}^{\mathrm{ord}}$ and $A_k = \mathbb{T}_{k,\mathfrak{m}}^{\mathrm{ord}}$. The image of $A \hookrightarrow \prod_{k\ge 3} A_k$ is closed, since A is compact and the target is Hausdorff.

By classical theory, for each such k there is a continuous two-dimensional pseudocharacter

$$t_k : G_{\mathbb{Q},Np} \longrightarrow A_k$$

satisfying $t_k(\mathrm{Frob}_\ell) = T_\ell$ for $\ell \nmid Np$. Taking their product gives a continuous pseudocharacter

$$\widetilde{t} : G_{\mathbb{Q},Np} \longrightarrow \prod_{k\ge 3} A_k.$$

Its values at Frobenius elements lie in the image of A. By Chebotarev density and closedness of this image, $\widetilde{t}$ therefore factors through a continuous pseudocharacter

$$t : G_{\mathbb{Q},Np} \longrightarrow A.$$

Its reduction modulo $\mathfrak{m}$ is $\operatorname{tr}\bar{\rho}$. Since $\bar{\rho}$ is absolutely irreducible, the Nyssen–Rouquier theorem gives a continuous representation

$$\rho : G_{\mathbb{Q},Np} \longrightarrow \mathrm{GL}_2(A)$$

with $\operatorname{tr}\rho = t$. In particular, $\operatorname{tr}\rho(\mathrm{Frob}_\ell) = T_\ell$ for every $\ell \nmid Np$.

Let ϵ_p denote the p-adic cyclotomic character and put $\epsilon^{\mathrm{univ}} = \chi^{\mathrm{univ}} \circ \epsilon_p$, where $\chi^{\mathrm{univ}} : \mathbb{Z}_p^\times \to \Lambda^\times$ is the universal character that maps z to $[z]$. Using

$$\det\rho(g) = \frac{t(g)^2 - t(g^2)}{2},$$

we see that the weight-k specialization of $\det\rho(\mathrm{Frob}_\ell)$ is $\ell^{k-1}\langle\ell\rangle_k$. Injectivity of the specialization map therefore gives

$$\det\rho(\mathrm{Frob}_\ell) = \chi^{\mathrm{univ}}(\ell)\ell^{-1}\langle\ell\rangle = \epsilon^{\mathrm{univ}}(\mathrm{Frob}_\ell)\epsilon_p(\mathrm{Frob}_\ell)^{-1}\langle\ell\rangle$$

for every $\ell \nmid Np$.

Theorem 6.14. *Let $\rho : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{T}^{\mathrm{ord}}_{\Lambda,\mathfrak{m}})$ be the Galois representation constructed above, then*

$$\rho|_{G_p} \simeq \begin{pmatrix} \epsilon^{\mathrm{univ}} \cdot \epsilon_p^{-1} \cdot \lambda(\frac{\langle p\rangle}{U_p}) & * \\ 0 & \lambda(U_p) \end{pmatrix}$$

.

Proof. We use the strategy established in [Ger10], but our case is much simpler since we assumed $\chi_1|_{I_p}$ is non-trivial. Let $\psi_1 = \epsilon^{\mathrm{univ}} \cdot \epsilon_p^{-1}|_{I_p} : I_p \to \Lambda^\times$, and let $\psi_2 : I_p \to \Lambda^\times$ to be the trivial character. Consider the following framed deformation problem. Let $\mathcal{C}_\Lambda$ denote the category of Artinian local Λ-algebras with residue field $\mathbb{F}$. Define

$$D_\Lambda^{\square,\mathrm{ord}} : \mathcal{C}_\Lambda \longrightarrow \mathrm{Sets},$$
$$A \longmapsto \left\{ \begin{array}{l} \text{continuous lifts } \rho_A : G_p \to \mathrm{GL}_2(A) \text{ of } \bar{\rho}, \\ \text{admitting a } G_p\text{-stable rank-one direct summand } L_A \subset A^2, \\ \text{such that } I_p \text{ acts on } L_A \text{ through } \psi_{1,A} \\ \text{and on } A^2/L_A \text{ through } \psi_{2,A} \end{array} \right\},$$

where $\psi_{i,A}$ is the composite $I_p \xrightarrow{\psi_i} \Lambda^\times \to A^\times$. Under our hypothesis that $\chi_1|_{I_p} \neq 1$, this moduli problem is pro-representable [Sno18, §4] by $R_\Lambda^{\square,\mathrm{ord}}$. Let

$R^{\square}_{\Lambda}$ be the universal framed deformation ring. By classical theory, we have the following commutative diagram:

$$\begin{array}{ccc} \mathbb{T}^{\mathrm{ord}}_{\Lambda} & \hookrightarrow & \prod_{k\geq 3}\mathbb{T}^{\mathrm{ord}}_{k} \\ \uparrow & \nearrow & \uparrow \\ R^{\square}_{\Lambda} & \twoheadrightarrow & R^{\square,\mathrm{ord}}_{\Lambda} \end{array}$$

Therefore there exists a factorization given by the dash arrow, which means that

$$\rho|_{G_p} \simeq \begin{pmatrix} \epsilon^{\mathrm{univ}} \cdot \epsilon_p^{-1} \cdot \lambda(\frac{\langle p\rangle}{U_p}) & * \\ 0 & \lambda(U_p) \end{pmatrix}$$

. □

Corollary 6.15. *Suppose* $1 \leq a \leq p-2$.

1. *If* $k = p-a-1$, *then there is a continuous Galois representation* $\rho : G_{\mathbb{Q},Np} \to \mathrm{GL}_2(\mathbb{T}^{\mathrm{ord}}_{\kappa_{-a},\mathfrak{m}})$, *satisfying*

$$\mathrm{tr}\,\rho(\mathrm{Frob}_l) = T_l,\ \det\rho(\mathrm{Frob}_l) = [l]^{-a}\langle l\rangle$$

for every $l \nmid Np$, *and*

$$\rho|_{G_p} \simeq \begin{pmatrix} [\overline{\varepsilon}_p]^{-a} \cdot \lambda(\frac{\langle p\rangle}{U_p}) & * \\ 0 & \lambda(U_p) \end{pmatrix}$$

.

2. *If* $k = a$, *then there is a continuous Galois representation* $\rho : G_{\mathbb{Q},Np} \to \mathrm{GL}_2(\mathbb{T}^{\mathrm{ord}}_{\kappa_a,\mathfrak{m}})$, *satisfying*

$$\mathrm{tr}\,\rho(\mathrm{Frob}_l) = T_l,\ \det\rho(\mathrm{Frob}_l) = [l]^{a}\langle l\rangle$$

for every $l \nmid Np$, *and*

$$\rho|_{G_p} \simeq \begin{pmatrix} [\overline{\varepsilon}_p]^{a} \cdot \lambda(\frac{\langle p\rangle}{U_p}) & * \\ 0 & \lambda(U_p) \end{pmatrix}$$

.

Proof. This follows from 6.14 by passing through the surjections $\mathbb{T}^{\mathrm{ord}}_{\Lambda,\mathfrak{m}} \twoheadrightarrow \mathbb{T}^{\mathrm{ord}}_{\kappa_{-a}}$, and $\mathbb{T}^{\mathrm{ord}}_{\Lambda,\mathfrak{m}} \twoheadrightarrow \mathbb{T}^{\mathrm{ord}}_{\kappa_a}$. □

6.5 Local Global Compatibility for Galois representation valued in Hecke algebra

The main theorem of this chapter is the following.

Theorem 6.16. *For $1 \leq a \leq p-2$, the Galois representation $\rho : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{T}_{a,\mathfrak{m}})$ constructed in 6.3 is split at p, that is to say:*

$$\rho|_{G_p} \simeq [\overline{\varepsilon}_p]^{-a}\lambda(\langle p\rangle/U_p) \oplus \lambda(U_p).$$

Proof. By 6.11 and 6.15,

$$\rho|_{G_p} \simeq \begin{pmatrix} [\overline{\varepsilon}_p]^{-a} \cdot \lambda(\frac{\langle p\rangle}{U_p}) & * \\ 0 & \lambda(U_p) \end{pmatrix}.$$

On the other hand, by 6.12 and 6.15,

$$\rho|_{G_p} \simeq \begin{pmatrix} [\overline{\varepsilon}_p]^{a} \cdot \lambda(U_p) & * \\ 0 & \lambda(\frac{\langle p\rangle}{U_p}) \end{pmatrix} \otimes [\overline{\varepsilon}_p]^{-a} \simeq \begin{pmatrix} \lambda(U_p) & * \\ 0 & [\overline{\varepsilon}_p]^{-a}\lambda(\frac{\langle p\rangle}{U_p}) \end{pmatrix}.$$

Therefore, there are two eigenlines with different characters, and

$$\rho|_{G_p} \simeq [\overline{\varepsilon}_p]^{-a}\lambda(\langle p\rangle/U_p) \oplus \lambda(U_p).$$

□

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